

MTH4107/MTH4207: Introduction to Probability

Rosemary J. Harris

School of Mathematical Sciences



Notes corresponding to undergraduate lecture course

· Autumn 2021 ·

CONTENTS

0	Prologue	1
0.1	Motivation	1
0.2	This Course	3
0.3	Further Resources	4
0.4	Acknowledgements	5
1	Sample Spaces and Events	6
1.1	Framework	6
1.2	Basic Set Theory	8
1.3	More on Events	11
1.4	Further Exercises	12
2	Properties of Probabilities	14
2.1	Kolmogorov's Axioms	14
2.2	Deductions from the Axioms	16
2.3	Inclusion-Exclusion Formulae	19
2.4	Equally-Likely Outcomes	22
2.5	Further Exercises	24
3	Sampling	25
3.1	Basics for Sampling	25
3.2	Ordered Sampling with Replacement	26
3.3	Ordered Sampling without Replacement	27
3.4	Unordered Sampling without Replacement	29
3.5	Sampling in Practice	31
3.6	Further Exercises	33
4	Conditional Probability	35
4.1	Introducing Extra Information	35
4.2	Implications of Extra Information	37
4.3	The Multiplication Rule	39
4.4	Ordered Sampling Revisited	40
4.5	Further Exercises	41

5	Independence	43
5.1	Independence for Two Events – Basic Definition	43
5.2	Independence for Two Events – More Details	44
5.3	Independence for Three or More Events	46
5.4	Conditional Independence	48
5.5	Further Exercises	50
6	Total Probability and Bayes’ Theorem	52
6.1	Law of Total Probability	52
6.2	Total Probability for Conditional Probabilities	54
6.3	Bayes’ Theorem	56
6.4	From Axioms to Applications	58
6.5	Further Exercises	59
7	Interlude (and Self-Study Ideas)	61
7.1	Looking Back and Looking Forward	61
7.2	Tips for Reading the Lecture Notes	61
7.3	Tips for Doing Examples/Exercises	62
8	Introduction to Random Variables	64
8.1	Concept of a Random Variable	64
8.2	Distributions of Discrete Random Variables	66
8.3	Properties of the Probability Mass Function	69
8.4	Further Exercises	72
9	Expectation and Variance	74
9.1	Expected Value	74
9.2	Expectation of a Function of a Random Variable	76
9.3	Moments and Variance	78
9.4	Useful Properties of Expectation and Variance	81
9.5	Further Exercises	82
10	Special Discrete Random Variables	84
10.1	Bernoulli Distribution	84
10.2	Binomial Distribution	85
10.3	Geometric Distribution	87
10.4	Poisson Distribution	88
10.5	Distributions in Practice	91
10.6	Further Exercises	93
11	Several Random Variables	95
11.1	Joint and Marginal Distributions	95
11.2	Expectations in the Multivariate Context	97
11.3	Independence for Random Variables	99
11.4	Binomial Distribution Revisited	101
11.5	Further Exercises	102

12 Covariance and Conditional Expectation	104
12.1 Covariance and Correlation	104
12.2 Conditional Expectation	106
12.3 Law of Total Probability for Expectations	108
12.4 Further Exercises	110
13 Epilogue (and Exams etc.)	111
13.1 Tips for Revision and Exams	111
13.2 Probability in Perspective	112
13.3 Bonus Further Exercise	112
A Errata	114

CHAPTER 0

Prologue

0.1 Motivation

Before starting the module properly, we will spend a bit of time thinking about what probability is and why it is important to study. As a warm-up, I invite you to think about the following question.

Exercise 0.1: Birthday matching

How likely is it that two people in your tutorial group share a birthday?

[You may well have seen a calculation similar to this before; if not, then try to guess what the answer might be and perhaps discuss it with your class in the first tutorial.]

In general terms, probability theory is about “chance”; it helps to describe situations where there is some randomness, i.e., events we cannot predict with certainty. Such situations could be truly random (arguably like tossing a coin) or they may simply be beyond our knowledge (like the birthdays). Of course, probability is about more than just description – we want to be able to *quantify* the randomness (loosely speaking, “give it a number”). Indeed, for Exercise 0.1 the best answer would be not just something vague like “very low” but a fraction or decimal with value depending on the size of your group. [You should certainly be able to do this calculation by the end of Chapter 3. In fact, most people have rather poor intuition for the birthday problem and similar probability questions so you might find that your initial guess was a long way from the true answer.]

Introductory probability courses are typically full of examples involving birthdays, dice, coins, playing cards, etc. and I’m afraid you will see plenty of them here too. They enable us to clearly demonstrate the mathematical framework of sample spaces and events (as introduced in Chapter 1) but you should not assume that the applications of probability are limited to such artificial scenarios. To

name just a few more “real world” examples: forecasting the weather, predicting financial markets, and modelling the spread of diseases all rely crucially on probability theory. At QMUL you will encounter probability in many later courses, e.g., if you take “Actuarial Mathematics” modules you will be concerned with the probabilities of life and death, together with their implications for life insurance.

Notice that in the above paragraphs we have not actually defined probability. In fact, the question of what probability is does not have an entirely satisfactory answer. We need to associate a number to an event which will measure the likelihood of it occurring but what does this really mean? Well, you can think of the required number as some kind of limiting frequency – an informal definition is given by the following procedure:

- Repeat an experiment (say, roll a die¹) N times;
- Let A denote an event (say, the die shows an even number);
- Suppose the event comes up m times (among the N repetitions of the experiment);
- Then, in the limit of very large values of N , the ratio m/N gives the probability of the event A .

In Chapter 2 we will give a more precise mathematical definition which, roughly speaking, defines probability in terms of the properties it should have. This is called the “axiomatic approach to probability” and we will see how many important results can be derived from the basic axioms. Probability theory is thus a beautiful demonstration of pure maths (prepare for some proofs!) as well as an important tool in applied maths.

In concluding this introductory section, let me raise a few more thought-provoking questions to illustrate some important concepts we will meet during the course.

Exercise 0.2: Are you smarter than a pigeon?

On the mean streets of London a street performer engages you in a socially-distanced card game. He shows you three playing cards of which one is the ace of hearts. You will win a mystery prize if you pick this card. No prize is associated with the other cards.

The performer shuffles the cards and holds them up so he can see what they are but you cannot. You are asked to pick a card by pointing at it. As part of the game, he now shows you one of the two unpicked cards which he knows is not the ace. Now only the card you picked and one remaining card are left unrevealed. You are asked if you would like to switch your choice. *Should you switch?*

¹For the avoidance of doubt, “die” is the singular of “dice”.

[The above exercise shows the danger of assuming that all outcomes are equally likely – we will discuss this more carefully later.]

Exercise 0.3: Innocent or guilty?

Professor Damson has been discovered dead on the top floor of the Maths Building (making rather a mess of the nice carpet) with the murder weapon left by her body. A suspect's fingerprints match those found on the murder weapon. The probability of a match occurring by chance is around 1 in 50,000. Is the suspect more likely to be innocent or more likely to be guilty?

[This “prosecutor’s fallacy” scenario emphasizes how important it is to work out exactly what information we are given and what we want to know. I encourage you to think carefully about it now and return to it in due course; to answer properly requires conditional probability (which we will introduce in Chapter 4) and Bayes’ Theorem (Chapter 6).]

Exercise 0.4: Choices

Would you rather...

- ...be given £5, or toss a coin winning £10 if it comes up heads?
- ...be given £5000, or toss a coin winning £10000 if it comes up heads?
- ...be given £1, or toss a coin 10 times winning £1000 if it comes up heads every time?

[This last exercise is not entirely a mathematical one and there is no right or wrong answer for each part. However some tools from probability can help to explain the various choices. In each case the average amount we expect to win and the degree of variation in our gain are relevant. Properties of so-called random variables (the focus of the second half of this course) can be used to describe the choices quantitatively. Of course there are lots of extra ingredients which will influence an individual’s decision (for instance how useful a particular sum of money is to them, and how much they enjoy the excitement of taking a risk). One attempt to model some of these extra factors mathematically is the idea of utility functions from game theory but this is beyond the scope of the present module.]

0.2 This Course

These notes define the examinable content for the course – *everything here is examinable unless explicitly stated otherwise*. Each chapter corresponds to one week’s material and the notes should always be available by the start of the relevant week so you can read them in advance of the teaching sessions. To help you with learning, important new **probability words** are generally printed in **bold**. You

must be sure to understand what these mean, both in everyday language (could you explain them to your grandmother or your next-door neighbour?) and in terms of the associated mathematical formalism. The notes also contain a number of “*examples*” and “*exercises*”. For the former, you will find full details of the working which you should study carefully to see not just how to get correct answers but how to present your solutions (an important skill for university-level mathematics). For the latter, only selected answers are in the notes although many solutions will be presented in lectures and tutorials (see below). A good way to check your own understanding would be to read the text and try the associated exercises – please ask if you have any difficulty getting the right answers. I intend the unstarred exercises to correspond roughly to the Key Objectives for the course (i.e., everyone should be able to do them); single-starred exercises should be attempted by those aiming for a high grade and double-starred exercises are for those seeking a real challenge (beyond the syllabus).

The content of the notes will be presented in timetabled lectures (typically one hour per section) in which the main points will be emphasized and solutions to some of the exercises given. These lectures will feature various “real-time” written demonstrations and you are strongly recommended to follow along and fill in the gaps in the slides or annotate these notes – *for most people, the practice of writing out maths (rather than just looking at a screen) helps the material to be absorbed and the necessary notation to become automatic*. In addition, there are tutorials which will include more detailed discussion of the coursework and provide a further opportunity for queries/misunderstandings to be addressed. The QMplus page contains full information about course practicalities (timetable, coursework, assessment details, etc.) and important announcements/corrections will also be posted there.

0.3 Further Resources

The course is designed to be fairly self-contained and does not follow any one textbook. However, if you would like further background reading (and many more practice exercises!) then the following are recommended.

- Sheldon Ross, *A First Course in Probability* (Pearson, 2020):
 - Cited in these notes as [Ros20];
 - Probably closest to the general treatment of this course.
- David F. Anderson, Timo Seppäläinen, and Benedek Valkó, *Introduction to Probability* (Cambridge, 2018):

- Cited in these notes as [ASV18];
- Slightly more rigorous but very readable.
- Henk Tijms, *Understanding Probability* (Cambridge, 2012):
 - Cited in these notes as [Tij12];
 - Content covered in this module is contained in Chapters 7–9, while parts of Chapters 1–6 nicely motivate the theoretical ideas with examples.

[All of these are available in physical and electronic form from the QMUL library.]

0.4 Acknowledgements

Large sections of these notes are based on those of previous lecturers, Dr Robert Johnson and Dr Wolfram Just, and I am greatly indebted to both of them. However, the responsibility for errors and typos remains mine; if you find any, please contact me at rosemary.harris@qmul.ac.uk.

CHAPTER 1

Sample Spaces and Events

1.1 Framework

The general setting is that we perform an **experiment** and record an **outcome**. Outcomes must be precisely specified, mutually exclusive and cover all possibilities.

Definition 1.1. *The **sample space** is the set of all possible outcomes for the experiment. It is denoted by $\mathcal{S}$ (or Ω).*

Definition 1.2. *An **event** is a subset of the sample space. The event **occurs** if the actual outcome is an element of this subset.*

To see how the framework is applied in practice, let's consider some examples. We start with a very simple one.

Example 1.1: Die roll

Suppose your experiment is to roll an ordinary six-sided die and record the number showing as the outcome.

- (a) Use set notation to write down the sample space.
- (b) Denote by A the event “You roll an even number” and write it as a subset of the sample space.

Solution:

- (a) The sample space is the set containing the integers 1 to 6 (inclusive) so we write

$$\mathcal{S} = \{1, 2, 3, 4, 5, 6\}.$$

- (b) The event corresponding to the rolling of an even number is $A = \{2, 4, 6\}$.

The situation is more complicated when the outcome is not just an observation of a single thing. In choosing your notation you then need to think about whether order matters.

Example 1.2: Tossing thrice

You toss a coin three times and, for each toss, observe whether you get a Head or whether you get a Tail.

- (a) Use set notation to write down the sample space.
- (b) State the following events as subsets of the sample space:
 - “Exactly one Head is seen”;
 - “The second toss is a Head”.

Solution:

- (a) Denoting a Head with h and a Tail with t , we can write the sample space as

$$\mathcal{S} = \{hhh, hht, hth, htt, thh, tht, tth, ttt\}$$

where, for example, htt means the first toss is a Head, the second is a Tail, and the third is also a Tail. Note that here order matters: htt is not the same as tth so we have to include both. In general, if we need to include information about order, we can either write an ordered list as above or use round brackets and commas, e.g., (h, t, t) . However, this is not the only way to write the sample space for this example. If you are a (lazy) experimenter recording outcomes you might realise that you only need to note which tosses are, say, Heads and you have all the information about what happened. So, adding a tilde (“squiggle”) to the $\mathcal{S}$ to indicate we’ve changed notation, we could write

$$\tilde{\mathcal{S}} = \{\{1, 2, 3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1\}, \{2\}, \{3\}, \{\}\}$$

where we record the set of tosses which are Heads so the outcome $\{1, 3\}$ corresponds to hth . Note that the curly braces indicate that order doesn’t matter; $\{3, 1\}$ would mean exactly the same thing as $\{1, 3\}$. [Here we are representing the outcomes within the sample space as sets; you may recall that in set notation the order of elements is unimportant – for a brief review of set theory see the next section.]

- (b) Using the first notation for the sample space and denoting the events “Exactly one Head is seen” and “The second toss is a Head” by B and C respectively, we have

$$B = \{htt, tht, tth\} \quad \text{and} \quad C = \{hhh, hht, thh, tht\}.$$

Denoting the same events by $\tilde{B}$ and $\tilde{C}$ in the second notation, we have,

$$\tilde{B} = \{\{1\}, \{2\}, \{3\}\} \quad \text{and} \quad \tilde{C} = \{\{2\}, \{1, 2\}, \{2, 3\}, \{1, 2, 3\}\}.$$

[Normally you should avoid using two different notations within the same solution but, if for some reason you need to, you should distinguish them clearly, e.g., by $\mathcal{S}$ and $\tilde{\mathcal{S}}$, $\mathcal{S}_1$ and $\mathcal{S}_2$, or $\mathcal{S}$ and $\mathcal{S}'$; in the last case, the prime is not to be confused with the set complement for which we will use a c superscript. Notice too that the number of elements is always invariant under a change of notation and this can sometimes provide a way to check your answers are sensible.]

Exercise 1.3: Two die rolls

You throw an ordinary six-sided die and write down the number showing. Then you throw it again and write down the number showing.

- (a) Write down the sample space for this experiment.
- (b) How many elements does the sample space contain?
- (c) Write in words some possible events corresponding to this experiment and then state them as subsets of the sample space.

In this course we will mainly be interested in experiments with discrete outcomes and finite sample spaces (especially when proving rigorous results). However, we will also encounter some cases where the sample space contains a countably infinite number of elements. [The extension to continuous outcomes will be left for future courses.]

Exercise 1.4: Exam strategy

Suppose your experiment is to take an exam repeatedly until you pass.¹ Ignoring complications such as finite lifetimes, which of the following could represent the sample space?

- (a) $\mathcal{S}_1 = \{P, FP, FFP, FFFP, \dots\}$,
- (b) $\mathcal{S}_2 = \{1, 2, 3, 4, \dots\}$,
- (c) $\mathcal{S}_3 = \{0, 1, 2, 3, \dots\}$,
- (d) $\mathcal{S}_4 = \{FFFP, FFP, FP, P\}$.

Definition 1.3. An event E is a **simple event** (or **elementary event**) if it consists of a single element of the sample space $\mathcal{S}$.

Exercise 1.5: Simple events

Identify some simple events from the examples in this section.

1.2 Basic Set Theory

We use extensively the terminology and notation of basic set theory; if you are taking the module “Numbers, Sets and Functions”, you should find some helpful overlap in material. Informally a set is an unordered collection of well-defined distinct objects. For instance, $\{b, -2.4, \square\}$ is a set. However, we do not regard $\{2, 3, 3\}$ as a set because it contains repeated elements (not distinct).² We stress

¹This is not a good strategy at QMUL – you only get two attempts at each exam!

²Some mathematicians allow a set to be written with repeated elements with the convention that, e.g., $\{1, 1, 2\}$ is the same set as $\{1, 2\}$. We will *never* do that in this course – the sample space is a set where the elements are mutually exclusive outcomes (which cannot repeat each other!) so it makes absolute sense to forbid repetition.

that order does not matter so, e.g., $\{2, 3, 5, 1\}$ and $\{3, 5, 2, 1\}$ are the same set. Indeed, two sets A and B are equal, $A = B$, if they contain precisely the same elements.

A set can be specified in various ways:

- By listing all the objects in it between braces $\{, \}$ and separated by commas, e.g., $\{1, 2, 3, 4\}$.
- By listing enough elements to determine a pattern (usually for infinite sets), e.g., $\{2, 4, 6, 8, \dots\}$ which is the set of positive even integers. [A set which can be written as a comma-separated list is said to be countable.]
- By giving a rule, e.g., $\{x : x \text{ is an even integer}\}$ which we read as “the set of all x such that x is an even integer”.

If A is a set we write $x \in A$ to mean that the object x is in the set A and we say that x is an **element** of A . If x is not an element of A then we write $x \notin A$. For a finite set, the **size** (or **cardinality**) is just the number of elements; if $A = \{a_1, a_2, \dots, a_n\}$, we write $|A| = n$. [Do not confuse the size of a set with the absolute value of a number.]

We now summarize some facts which will be useful in the rest of this course. Let A and B both be sets.

- $A \cup B$ (“ A union B ”) is the set of elements in A or B (or both):

$$A \cup B = \{x : x \in A \text{ or } x \in B\}. \quad (1.1)$$

- $A \cap B$ (“ A intersection B ”) is the set of elements in both A and B :³

$$A \cap B = \{x : x \in A \text{ and } x \in B\}. \quad (1.2)$$

- $A \setminus B$ (“ A take away B ”) is the set of elements in A but not in B :

$$A \setminus B = \{x : x \in A \text{ and } x \notin B\}. \quad (1.3)$$

- $A \triangle B$ (“symmetric difference of A and B ”) is the set of elements in either A or B but not both:

$$A \triangle B = (A \setminus B) \cup (B \setminus A). \quad (1.4)$$

³Some books, including [Ros20, ASV18, Tij12], use AB without a “cap” in the middle to denote the intersection of two events.

- If all the elements of A are contained in the set B , we say that A is a subset of B and we write $A \subseteq B$.
- If all sets are subsets of some fixed set $\mathcal{S}$, then A^c (“the complement of A ”) is the set of all elements of $\mathcal{S}$ which are not elements of A :

$$A^c = \mathcal{S} \setminus A. \quad (1.5)$$

- We say two sets A and B are **disjoint** (or **mutually exclusive**) if they have no element in common, i.e., $A \cap B = \{\}$. The empty set $\{\}$ is often denoted by $\emptyset$.

Exercise 1.6: Venn diagrams

Draw Venn diagrams to illustrate the bullet points above.

Exercise 1.7: Symmetric difference

Suppose that A and B are subsets of $\mathcal{S}$. Express $A \triangle B$ in terms of intersections, unions, and complements.

Exercise 1.8: Disjoint decomposition

Suppose that A and B are subsets of $\mathcal{S}$.

- Show that $A \cap B$ and $A \cap B^c$ are disjoint.
- Express A as a union of two disjoint sets.
- Express $A \cup B$ as a union of three mutually exclusive sets.

[These kind of tricks will be important later on.]

Exercise 1.9: Set examples (based on [ASV18])

Consider the following three sets:

$$A = \{2, 3, 4, 5, 6, 7, 8\}, \quad B = \{1, 3, 5, 7\}, \quad C = \{2, 4, 5, 8\}.$$

- Can the set $D = \{2, 4, 8\}$ be expressed in terms of A , B , and C using intersections, unions, and complements?
- Can the set $E = \{4, 5\}$ be expressed in terms of A , B , and C using intersections, unions, and complements?

It is very useful to remember the following two identities which are known as De Morgan’s laws:

$$(A \cup B)^c = A^c \cap B^c, \quad (1.6)$$

$$(A \cap B)^c = A^c \cup B^c. \quad (1.7)$$

*Exercise 1.10: *De Morgan's laws*

- (a) You may have seen a proof of De Morgan's laws elsewhere. Show that, in fact, you can derive (1.7) from (1.6) so you only really need to remember one of them.
- (b) Write down (or look up) the generalization of De Morgan's laws to n sets.

1.3 More on Events

The previous section was about set theory in general – we now return to the implications for events. Let A and B denote events (i.e., subsets of the sample space $\mathcal{S}$).

- If A is an event then A^c contains the elements of the sample space which are not contained in A , i.e., A^c is the event that “ A does not occur”.
- If A and B are events then the event E_1 “ A and B both occur” consists of all elements of both A and B , i.e., $E_1 = A \cap B$.
- The event E_2 “at least one of A or B occurs” consists of all elements in A or B , i.e., $E_2 = A \cup B$.
- The event E_3 “ A occurs but B does not” consists of all elements in A but not in B , i.e., $E_3 = A \setminus B$.
- The event E_4 “exactly one of A or B occurs” consists of all elements in A or B but not in both, i.e., $E_4 = A \triangle B$.

Example 1.11: Die roll (revisited)

You roll a die as in Example 1.1. Let A be the event that an even number occurs and B be the event that the outcome is a prime number. Express the following events in terms of A and B and write them explicitly as subsets of the sample space:

F_1 : “The outcome is an even number or a prime”;

F_2 : “The outcome is either an even number or a prime (i.e., not both)”.

Solution:

We have $A = \{2, 4, 6\}$ (see Example 1.1) and $B = \{2, 3, 5\}$. Using these, the required events are

$$F_1 = A \cup B = \{2, 3, 4, 5, 6\},$$

and

$$F_2 = A \triangle B = \{3, 4, 5, 6\}.$$

Exercise 1.12: Lecture attendance

Four students, Alisha, Bilal, Chloe and Daniel, are supposed to be attending a lecture course. In the last lecture of the semester their attendance is recorded.

- (a) Write down the sample space, explaining your notation carefully.
- (b) Write down the event “Exactly three of them attend the last lecture” as a set.
- (c) Write down the event “Alisha attends the last lecture but Daniel does not” as a set.

Exercise 1.13: Rugby squad

One player is chosen at random from a squad of world cup rugby players. Let C be the event “the player chosen is the captain”, F be the event “the player chosen is a forward”, and I be the event “the player chosen is injured”.

- (a) Express the following in symbols:
 - (i) The event “an injured forward is chosen”;
 - (ii) The event “the chosen player is a forward but is not the captain”;
 - (iii) The statement “none of the forwards are injured”.
- (b) Express the following in words:
 - (i) The event $F^c \cup I^c$;
 - (ii) The statement $|I^c| < 15$.

1.4 Further Exercises

Exercise 1.14: Real-life probability

Find an article from the news illustrating a scenario where probability might be important. Try to identify the experiment, the possible outcomes, and some events which might be of interest.

Exercise 1.15: Horse racing

A race takes place between three horses Adobe, Brandy, and Chopin. It is possible that one or more of them may fall and so fail to complete the race. The finishing horses are recorded in the order in which they finish.

- (a) Write down the sample space, explaining your notation.
- (b) Write down the event “The race is won by Adobe” as a set.
- (c) Write down the event “Brandy falls” as a set.
- (d) Write down the event “All horses complete the race” as a set.

Exercise 1.16: Rules for stopping

You toss an ordinary coin repeatedly, recording which side it lands on each toss. You do this until you have seen either two Heads or three Tails in total and then you stop.

- (a) Write down the sample space.

- (b) Write down the event “you toss the coin exactly four times” as a subset of the sample space.

I perform the same experiment but I do not stop until I have seen either seven Heads or eight Tails in total. Let E_i be the event “I toss the coin exactly i times”.

- (c) For which i is it the case that $E_i = \emptyset$?

CHAPTER 2

Properties of Probabilities

2.1 Kolmogorov's Axioms

We want to assign a numerical value to an event which reflects the chance that it occurs. To be more precise, probability is a concept/recipe (or, in formal terms, a function) $\mathbb{P}$ which assigns a (real) number $\mathbb{P}(A)$ to each event A .

Sometimes we have some intuition about how probabilities should be assigned. For example, if we toss a *fair* coin we should have probability $1/2$ of seeing Heads and probability $1/2$ of seeing Tails. This is an example of the *special case* where all outcomes are equally likely and the probability of an event A is just the ratio of the number of outcomes in A to the total number of outcomes in $\mathcal{S}$. We will explore this situation more in Section 2.4 but note that it does not constitute a general recipe for probability – one cannot just assume equally-likely outcomes without good reason! For example, what if we toss a *biased* coin? What probabilities would be reasonable in that case? Can they be any real numbers?

The formal approach is to regard probability as a mathematical construction satisfying certain **axioms**.¹

Definition 2.1 (Kolmogorov's axioms for probability). *Probability is a function $\mathbb{P}$ which assigns to each event A a real number $\mathbb{P}(A)$ such that:*

- (a) *For every event A we have $\mathbb{P}(A) \geq 0$,*
- (b) *$\mathbb{P}(\mathcal{S}) = 1$,*
- (c) *If $A_1, A_2, \dots, A_n$ are n pairwise disjoint events ($A_k \cap A_\ell = \emptyset$ for all $k \neq \ell$)*

¹Roughly speaking, an axiom is a statement which is assumed to be true and then can be used to make other logical deductions.

then

$$\mathbb{P}(A_1 \cup A_2 \cup \cdots \cup A_n) = \mathbb{P}(A_1) + \mathbb{P}(A_2) + \cdots + \mathbb{P}(A_n) = \sum_{k=1}^n \mathbb{P}(A_k).$$

Remarks:

- The function $\mathbb{P}$ is sometimes called a **probability measure**.
- **Pairwise disjoint** in Definition 2.1(c) simply means that every pair of events is disjoint. [Equivalently, we can use the term **mutually exclusive**.]
- Definition 2.1(c) is here stated for a finite number of events. In fact, there is a version for a countably infinite number of events as well; we will use that implicitly in Chapter 8 but we do not state it formally here as we would need to first clarify the notion of an infinite sum.² Further subtleties occur if $\mathcal{S}$ is not countable; this situation is left to more advanced courses.

Exercise 2.1: Fair coin

Show that the probabilities associated with a single toss of a fair coin satisfy Kolmogorov's Axioms. [Hint: We will check this for the whole class of experiments with equally-likely outcomes in Section 2.4.]

Example 2.2: Biased coin

Consider a single toss of a biased coin where the outcome is still Heads (h) or Tails (t) but the probabilities of the possible events are given by:

$$\begin{aligned} \mathbb{P}(\emptyset) &= 0 && \text{[Seeing nothing is not possible.],} \\ \mathbb{P}(\{h\}) &= \frac{1}{3}, \\ \mathbb{P}(\{t\}) &= \frac{2}{3}, \\ \mathbb{P}(\{h, t\}) &= 1 && \text{[We must see either a Head or a Tail.]} \end{aligned}$$

Show that this defines a probability measure, i.e., that the given probabilities satisfy Kolmogorov's Axioms.

Solution:

All four events (i.e., all four subsets of the sample space $\{h, t\}$) have probabilities greater than or equal to zero so Definition 2.1(a) is obviously satisfied. We also straightforwardly have $\mathbb{P}(\mathcal{S}) = \mathbb{P}(\{h, t\}) = 1$ so Definition 2.1(b) is also satisfied. For Definition 2.1(c) we

²A concept many of you will meet in the module "Calculus II".

need to check all possible unions of pairwise disjoint events:

$$\begin{aligned}\mathbb{P}(\{h\} \cup \{t\}) &= \mathbb{P}(\{h, t\}) = 1 = \frac{1}{3} + \frac{2}{3} = \mathbb{P}(\{h\}) + \mathbb{P}(\{t\}), \\ \mathbb{P}(\emptyset \cup \{h\}) &= \mathbb{P}(\{h\}) = \frac{1}{3} = 0 + \frac{1}{3} = \mathbb{P}(\emptyset) + \mathbb{P}(\{h\}), \\ \mathbb{P}(\emptyset \cup \{t\}) &= \mathbb{P}(\{t\}) = \frac{2}{3} = 0 + \frac{2}{3} = \mathbb{P}(\emptyset) + \mathbb{P}(\{t\}), \\ \mathbb{P}(\emptyset \cup \{h, t\}) &= \mathbb{P}(\{h, t\}) = 1 = 0 + 1 = \mathbb{P}(\emptyset) + \mathbb{P}(\{h, t\}), \\ \mathbb{P}(\emptyset \cup \{h\} \cup \{t\}) &= \mathbb{P}(\{h, t\}) = 1 = 0 + \frac{1}{3} + \frac{2}{3} = \mathbb{P}(\emptyset) + \mathbb{P}(\{h\}) + \mathbb{P}(\{t\}).\end{aligned}$$

Hence Definition 2.1(c) is satisfied and we have a probability measure as required.

Note that for simple events you may occasionally see notation like $\mathbb{P}(h)$ as a shorthand for $\mathbb{P}(\{h\})$ but it is better to include the curly braces as a reminder that events are always sets.

Exercise 2.3: A more complicated construction

Let $\mathcal{S} = \{1, 2, 3, 4, 5, 6, 7, 8\}$ and for $A \subseteq \mathcal{S}$ define a probability measure by

$$\mathbb{P}(A) = \frac{1}{12} (|A \cap \{1, 2, 3, 4\}| + 2|A \cap \{5, 6, 7, 8\}|).$$

- (a) Verify that this satisfies the axioms for probability.
- (b) Give an example of a physical situation which this probability measure could describe.

2.2 Deductions from the Axioms

Starting from the axioms we can deduce various properties. Hopefully, these will agree with our intuition about probability – if they did not then this would suggest that we had not made a good choice of axioms! The proofs of all of these are simple deductions from the axioms.

Proposition 2.2. *If A is an event then*

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A).$$

This statement makes perfect sense: if $\mathbb{P}(A)$ is the probability of the event A then the probability of the complementary event A^c should be $1 - \mathbb{P}(A)$. [This can be very useful in calculations; sometimes it is much easier to calculate $\mathbb{P}(A^c)$ than $\mathbb{P}(A)$.] Although the proposition may be obvious, we want to provide a formal proof starting from Definition 2.1 to provide evidence that our axioms are consistent with the real world and to demonstrate the structure of probability theory.

Proof:

Let A be any event. Then we can set $A_1 = A$ and $A_2 = A^c$. By definition of the complement, $A_1 \cap A_2 = \emptyset$ and so we can apply Definition 2.1(c), with $n = 2$, to get

$$\mathbb{P}(A_1 \cup A_2) = \mathbb{P}(A_1) + \mathbb{P}(A_2) = \mathbb{P}(A) + \mathbb{P}(A^c). \quad (2.1)$$

Now, again by definition of the complement, $A_1 \cup A_2 = \mathcal{S}$ so, using Definition 2.1(b),

$$1 = \mathbb{P}(\mathcal{S}) = \mathbb{P}(A_1 \cup A_2). \quad (2.2)$$

Hence, combining (2.1) and (2.2), we have

$$1 = \mathbb{P}(A) + \mathbb{P}(A^c)$$

and rearranging this gives the desired result. $\square$

We can use what we have just proved to deduce further results which are called here “corollaries” as they are straightforward consequences of the preceding proposition.

Corollary 2.3.

$$\mathbb{P}(\emptyset) = 0.$$

This statement makes perfect sense as well. The probability of “no outcome” is zero. [We already had this in Example 2.2 but we now see that it *has* to be true.]

Proof:

By definition of the complement, $\mathcal{S}^c = \mathcal{S} \setminus \mathcal{S} = \emptyset$. Hence, by Proposition 2.2,

$$\mathbb{P}(\emptyset) = \mathbb{P}(\mathcal{S}^c) = 1 - \mathbb{P}(\mathcal{S}).$$

Then, using Definition 2.1(b), we have

$$\mathbb{P}(\emptyset) = 1 - 1 = 0,$$

as required. $\square$

Corollary 2.4. *If A is an event then $\mathbb{P}(A) \leq 1$.*

Again the statement agrees with our intuition. Probabilities are always smaller than or equal to one.

Proof:

By Proposition 2.2,

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A).$$

But A^c is also an event so, by Definition 2.1(a),

$$0 \leq \mathbb{P}(A^c) = 1 - \mathbb{P}(A)$$

and hence

$$\mathbb{P}(A) \leq 1,$$

as required. $\square$

The following statements are less obvious consequences of Definition 2.1 and the statements we have shown so far. Thus we call them again propositions.

Proposition 2.5. *If A and B are events and $A \subseteq B$ then*

$$\mathbb{P}(A) \leq \mathbb{P}(B).$$

This statement looks sensible as well. If an event B contains all the outcomes of an event A then the probability of the former must be at least as big as that of the latter.

Proof:

Consider the events $A_1 = A$ and $A_2 = B \setminus A$, with $A \subseteq B$. Then $A_1 \cap A_2 = \emptyset$ (i.e., the two events are pairwise disjoint) and $A_1 \cup A_2 = B$. [A Venn diagram may help you to see what is going on here.] So, by Definition 2.1(c), with $n = 2$,

$$\mathbb{P}(B) = \mathbb{P}(A_1 \cup A_2) = \mathbb{P}(A_1) + \mathbb{P}(A_2) = \mathbb{P}(A) + \mathbb{P}(B \setminus A).$$

Since $B \setminus A$ is also an event, Definition 2.1(a) tells us that

$$\mathbb{P}(B) - \mathbb{P}(A) = \mathbb{P}(B \setminus A) \geq 0,$$

and the required statement follows by rearrangement. $\square$

Proposition 2.6. *If $A = \{a_1, a_2, \dots, a_n\}$ is a finite event then*

$$\mathbb{P}(A) = \mathbb{P}(\{a_1\}) + \mathbb{P}(\{a_2\}) + \dots + \mathbb{P}(\{a_n\}) = \sum_{i=1}^n \mathbb{P}(\{a_i\}).$$

This statement is quite remarkable. The probability of a (finite) event is the sum of the probabilities of the corresponding simple events. [Again, you may occasionally see $\mathbb{P}(a_i)$ for $\mathbb{P}(\{a_i\})$ although writing the former should really be avoided.]

Proof:

Denote the simple events by $A_i = \{a_i\}$, $i = 1, \dots, n$. These events are pairwise

disjoint and $A_1 \cup A_2 \cup \cdots \cup A_n = A$. Hence by Definition 2.1(c),

$$\begin{aligned}\mathbb{P}(A) &= \mathbb{P}(A_1 \cup A_2 \cup \cdots \cup A_n) \\ &= \mathbb{P}(A_1) + \mathbb{P}(A_2) + \cdots + \mathbb{P}(A_n) \\ &= \mathbb{P}(\{a_1\}) + \mathbb{P}(\{a_2\}) + \cdots + \mathbb{P}(\{a_n\}),\end{aligned}$$

as required. $\square$

Remarks:

- Note that combining Proposition 2.6 with Definition 2.1(b) leads to the obvious fact that the sum of the individual probabilities for all outcomes in the sample space is unity. In other words, this agrees with our intuition that “probabilities sum to one”.
- Notice that most of the proofs in this section involve expressing events of interest as the union of disjoint events so that Definition 2.1(c) can be applied. This is a powerful general strategy.

Exercise 2.4: Rugby squad (revisited)

Consider the set-up of Exercise 1.13. Suppose that 50% of the squad are forwards, 25% are injured and 10% are injured forwards and that the player chosen is equally likely to be any member of the squad. Calculate the probability that the chosen player is a forward who is not injured.

Exercise 2.5: Friendly discussion

Let A and B be events with $\mathbb{P}(A) = 1/2$, $\mathbb{P}(B) = 1/4$, $\mathbb{P}(A \cap B) = 1/10$. Your friend says that $\mathbb{P}(A \cup B) = 3/4$. Explain carefully whether or not they are correct.

2.3 Inclusion-Exclusion Formulae

We now move on to what was once described by the Italian-American mathematician Gian-Carlo Rota as “One of the most useful principles of enumeration in discrete probability and combinatorial theory” [Rot64].

Proposition 2.7 (Inclusion-exclusion for two events). *For any two events A and B we have*

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

This statement is not entirely obvious. For general events the probability of the event “ A or B ” is not the sum of the probabilities of events A and B ; in fact, because of some “double counting” one needs to correct by the probability of the event “ A and B ”.

Proof:

Consider the three events $E_1 = A \setminus B$, $E_2 = A \cap B$ and $E_3 = B \setminus A$. The events are pairwise disjoint and $E_1 \cup E_2 \cup E_3 = A \cup B$. [This should remind you of Exercise 1.8.] Hence, by Definition 2.1(c) with $n = 3$,

$$\mathbb{P}(A \cup B) = \mathbb{P}(E_1) + \mathbb{P}(E_2) + \mathbb{P}(E_3).$$

Furthermore $E_1 \cup E_2 = A$ and $E_2 \cup E_3 = B$. Thus Definition 2.1(c) with $n = 2$ yields

$$\mathbb{P}(A) = \mathbb{P}(E_1) + \mathbb{P}(E_2),$$

$$\mathbb{P}(B) = \mathbb{P}(E_2) + \mathbb{P}(E_3).$$

Since $\mathbb{P}(A \cap B) = \mathbb{P}(E_2)$, we finally have

$$\begin{aligned} \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B) &= \mathbb{P}(E_1) + \mathbb{P}(E_2) + \mathbb{P}(E_2) + \mathbb{P}(E_3) - \mathbb{P}(E_2) \\ &= \mathbb{P}(E_1) + \mathbb{P}(E_2) + \mathbb{P}(E_3) \\ &= \mathbb{P}(A \cup B), \end{aligned}$$

as required. □

Although normally written in the form above, the inclusion-exclusion formula can obviously be rearranged to express any one of $\mathbb{P}(A)$, $\mathbb{P}(B)$, $\mathbb{P}(A \cup B)$, $\mathbb{P}(A \cap B)$ in terms of the other three.

Example 2.6: A tale of two cities

A bus operator operates two lines each connecting the city Xinji with Yiwu. Line 1 runs a bus A from Xinji to Ulanqab and a bus B from Ulanqab to Yiwu, so that passengers have to change at Ulanqab. Line 2 runs a bus C from Xinji to Weihui and a bus D from Weihui to Yiwu, so that passengers have to change at Weihui. Buses are running with probabilities $\mathbb{P}(A) = 0.9$, $\mathbb{P}(B) = 0.8$, $\mathbb{P}(C) = 0.7$, $\mathbb{P}(D) = 0.8$. Furthermore the company ensures that at least three of the four buses are running.

Using the inclusion-exclusion principle, or otherwise, compute the probability that I can travel via line 1 from Xinji to Yiwu. [2018 exam question (part)]

Solution:

The probability of being able to travel via line 1 is $\mathbb{P}(A \cap B)$. Using the inclusion-exclusion formula we can write this as

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cup B)$$

but this only helps if we know $\mathbb{P}(A \cup B)$. The trick here is to remember that the union of A and B means at least one of A and B occurs. Since at least three of the four buses are

running, at least one of the two on line 1 must be running and hence $\mathbb{P}(A \cup B) = 1$. The probability of being able to travel via line 1 is thus

$$\mathbb{P}(A \cap B) = 0.9 + 0.8 - 1 = 0.7.$$

Proposition 2.8 (Inclusion-exclusion for three events). *For any three events A , B , and C we have*

$$\mathbb{P}(A \cup B \cup C) = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(A \cap B) - \mathbb{P}(A \cap C) - \mathbb{P}(B \cap C) + \mathbb{P}(A \cap B \cap C).$$

As for two events, there exists an “intuitive” argument but that is not a proof.

Proof:

Essentially we will apply Proposition 2.7 three times. Let $D = A \cup B$ so that $A \cup B \cup C = C \cup D$. Then

$$\begin{aligned} \mathbb{P}(A \cup B \cup C) &= \mathbb{P}(C \cup D) \\ &= \mathbb{P}(C) + \mathbb{P}(D) - \mathbb{P}(C \cap D) \quad [\text{by Proposition 2.7}] \\ &= \mathbb{P}(C) + \mathbb{P}(A \cup B) - \mathbb{P}(C \cap D) \\ &= \mathbb{P}(C) + \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B) - \mathbb{P}(C \cap D) \quad [\text{by Proposition 2.7}]. \end{aligned} \tag{2.3}$$

Now $C \cap D = C \cap (A \cup B) = (C \cap A) \cup (C \cap B)$ so that

$$\begin{aligned} \mathbb{P}(C \cap D) &= \mathbb{P}((C \cap A) \cup (C \cap B)) \\ &= \mathbb{P}(C \cap A) + \mathbb{P}(C \cap B) - \mathbb{P}((C \cap A) \cap (C \cap B)) \quad [\text{by Proposition 2.7}] \\ &= \mathbb{P}(C \cap A) + \mathbb{P}(C \cap B) - \mathbb{P}(A \cap B \cap C). \end{aligned} \tag{2.4}$$

Finally, substituting (2.4) in (2.3) yields

$$\mathbb{P}(A \cup B \cup C) = \mathbb{P}(C) + \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B) - \mathbb{P}(C \cap A) - \mathbb{P}(C \cap B) + \mathbb{P}(A \cap B \cap C),$$

as required. □

Exercise 2.7: Three events

Suppose that the probabilities for each of the three events A , B , and C are $1/3$, i.e.

$$\mathbb{P}(A) = \mathbb{P}(B) = \mathbb{P}(C) = \frac{1}{3}.$$

Furthermore, assume that the probabilities for each of the events “ A and B ”, “ A and C ”,

and “ B and C ” are $1/10$, i.e.,

$$\mathbb{P}(A \cap B) = \mathbb{P}(A \cap C) = \mathbb{P}(B \cap C) = \frac{1}{10}.$$

What can be said about the probability of the event that none of A , B , or C occur?

*Exercise 2.8: **Inclusion-exclusion for more events*

- (a) Derive the inclusion-exclusion formula for four events.
- (b) Now try to write down a general formula for n events. How could you prove it?

Remark: Notice how the few simple axioms of Section 2.1 led to a plethora of results in Sections 2.2 and 2.3 – such structure is essentially what mathematics is about. In the next section, we will see how the general framework applies to the special case of equally-likely outcomes.

2.4 Equally-Likely Outcomes

As mentioned at the beginning of Section 2.1, in some situations it is reasonable to say that the probability of an event A is the ratio of the number of outcomes in A to the total number of outcomes in $\mathcal{S}$, i.e., to define probability by

$$\mathbb{P}(A) = \frac{|A|}{|\mathcal{S}|}. \quad (2.5)$$

For example, if we roll a fair die then the sample space is $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$ and the probability of the event A “the number shown is smaller than 3” is given by

$$\mathbb{P}(A) = \frac{|\{1, 2\}|}{|\{1, 2, 3, 4, 5, 6\}|} = \frac{2}{6} = \frac{1}{3}.$$

We emphasize again that we are using here that every outcome in the sample space is equally likely (we say “we pick an outcome at random”) and this *special case* should not be assumed without justification; for example, it wouldn’t apply to a biased die. We also run into difficulties if $\mathcal{S}$ is infinite. If $\mathcal{S} = \mathbb{N}$ (the set of positive integers) as in Exercise 1.4, then there is no reasonable way to choose an element of $\mathcal{S}$ with all outcomes equally likely; there are, however, ways to choose so that every positive integer has some chance of occurring.

Example 2.9: Checking the axioms

Suppose that the sample space $\mathcal{S}$ is finite. Show that defining probabilities according to (2.5) satisfies Kolmogorov’s Axioms.

Solution:

We need to check all three parts of Definition 2.1.

(a) If A is an event then, by definition, it is a set (a subset of $\mathcal{S}$) with $|A| \geq 0$ and

$$\mathbb{P}(A) = \frac{|A|}{|\mathcal{S}|} \geq \frac{0}{|\mathcal{S}|} = 0.$$

(b) We straightforwardly have

$$\mathbb{P}(\mathcal{S}) = \frac{|\mathcal{S}|}{|\mathcal{S}|} = 1.$$

(c) If $A_1, A_2, \dots, A_n$ are pairwise disjoint subsets of $\mathcal{S}$ then

$$|A_1 \cup A_2 \cup \dots \cup A_n| = |A_1| + |A_2| + \dots + |A_n|,$$

so

$$\begin{aligned} \mathbb{P}(A_1 \cup A_2 \cup \dots \cup A_n) &= \frac{|A_1| + |A_2| + \dots + |A_n|}{|\mathcal{S}|} \\ &= \frac{|A_1|}{|\mathcal{S}|} + \frac{|A_2|}{|\mathcal{S}|} + \dots + \frac{|A_n|}{|\mathcal{S}|} \\ &= \mathbb{P}(A_1) + \mathbb{P}(A_2) + \dots + \mathbb{P}(A_n). \end{aligned}$$

Hence all three axioms are satisfied.

Since Definition 2.1 is fulfilled, all the other results in Sections 2.2 and 2.3 also hold. In particular, the inclusion-exclusion formula for two events reads

$$\frac{|A \cup B|}{|\mathcal{S}|} = \frac{|A|}{|\mathcal{S}|} + \frac{|B|}{|\mathcal{S}|} - \frac{|A \cap B|}{|\mathcal{S}|}$$

which implies

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

This last statement is a statement about the sizes of finite sets which can be proved in other ways as well. This cross-fertilization of ideas between different branches of mathematics is an important part of higher-level study (and research!).

Note that in this set-up of equally-likely outcomes, calculating probabilities becomes counting! In the next chapter we will see some combinatorial arguments for finding $|A|$ and $|\mathcal{S}|$ in different situations.

Exercise 2.10: Getting out of jail

In the game of Monopoly one way to get out of jail is to throw “a double”. What is the probability that when you roll two fair six-sided dice they both show the same number?

Exercise 2.11: Two Heads are better than one

You toss a fair coin five times. What is the probability that you see at least two Heads?

2.5 Further Exercises

Exercise 2.12: Probability calculations

Let A and B be events with $\mathbb{P}(A) = 1/2$, $\mathbb{P}(B) = 1/4$, $\mathbb{P}(A \cap B) = 1/10$. Calculate the following probabilities:

- (a) $\mathbb{P}(B^c)$,
- (b) $\mathbb{P}(A \cup B)$,
- (c) $\mathbb{P}(A \cap B^c)$.

Exercise 2.13: Zeros

Let A and B denote two events with $\mathbb{P}(A \cup B) = 0$. Show that $\mathbb{P}(A) = 0$ and $\mathbb{P}(B) = 0$.

Exercise 2.14: More identities

Suppose that A and B are events.

- (a) Using the inclusion-exclusion principle, or otherwise, show that

$$\mathbb{P}(A \cap B) \geq \mathbb{P}(A) + \mathbb{P}(B) - 1.$$

- (b) Using the events $C = A \triangle B$ and $D = A \cap B$, or otherwise, show that

$$\mathbb{P}(A \triangle B) = \mathbb{P}(A) + \mathbb{P}(B) - 2\mathbb{P}(A \cap B).$$

Make sure that each step in your proofs is justified by a definition, axiom or result from the notes (or is a simple manipulation).

CHAPTER 3

Sampling

3.1 Basics for Sampling

Throughout this chapter we focus again on the special case where all elements of the sample space are equally likely, as in Section 2.4. In this situation, calculating probability essentially boils down to counting the number of ways of making some selection. Specifically, we are often interested in finding how many ways there are of choosing r things from a collection of n things. This is called **sampling from an n -element set**. The number of ways of doing this depends on exactly what we mean by selection: is the order important and is repetition allowed? We will see three distinct cases in these notes and illustrate them with examples.

Before getting into the details, it is worth mentioning a fundamental idea that we use implicitly: the so-called “basic principle of counting” [Ros20]: if there are m possible outcomes of experiment 1 and n possible outcomes of experiment 2, then there are $m \times n$ outcomes of the two experiments together. For example, if there are four main courses on a cafeteria menu and three desserts then there are a total of twelve different meals which can be chosen. This principle can be straightforwardly generalized to more than two experiments.

Let us also emphasize some notation which will be important in the following sections. As we already know, a set $\{a_1, a_2, \dots, a_n\}$ is an *unordered* collection of n *distinct* objects. In contrast, an n -tuple $(b_1, b_2, \dots, b_n)$ is an *ordered* collection of n objects which are *not necessarily distinct* – you can think of this as a coordinate in an n -dimensional space. A 2-tuple is a “pair”, e.g., $(1, 2)$; a 3-tuple is a “triple”, e.g., $(1, 2, 1)$, a 4-tuple is a “quadruple”, e.g., $(1, 2, 1, 3)$, and so on.

3.2 Ordered Sampling with Replacement

Example 3.1: Words

How many ordered three-letter strings can be made from the letters A, B, C, D, E if repetition is allowed? [We will call these “words” although they don’t have to be real words in English, for example BDB, ACE, CAB, ABC, DDD , are all allowable possibilities.]

Solution:

This is an ordered selection of three things from a collection of five things with repetition allowed – you can think of taking Scrabble tiles from a bag containing (one each of) A, B, C, D, E , and replacing each letter after you have written it down. There are five choices for the first letter. For each of these, there are five choices for the second letter and, for each of these, there are five choices for the third letter. Hence, using the basic principle of counting, there are a total of $5 \times 5 \times 5 = 5^3 = 125$ possible words.

In formal terms, we let the set of letters be $U = \{A, B, C, D, E\}$. Then the experiment is to pick three letters in order and each outcome can be written as a triple, e.g., (A, C, E) . The sample space of all words is the set of all such triples, i.e., $\mathcal{S} = \{(s_1, s_2, s_3) : s_i \in U\}$ and we have $|\mathcal{S}| = 5^3$.

The above example illustrates the general principle of this section:

- If we make an ordered selection of r things from a set $U = \{u_1, u_2, \dots, u_n\}$ with replacement (i.e., we allow repetition, an element can be selected more than once) then the sample space is the set of all r -tuples consisting of elements of U . That is

$$\mathcal{S} = \{(s_1, s_2, \dots, s_r) : s_i \in U\}.$$

- If $|U| = n$ there are n choices for u_1 ; for each of these, there are n choices for u_2 , and so on. Hence, we have

$$|\mathcal{S}| = |U|^r = n^r. \quad (3.1)$$

To determine the probability of a given event, in the framework of equally-likely outcomes, we also need to calculate the cardinality of that event. This can often be done by straightforwardly applying the basic principle of counting, in a similar way to the calculation of $|\mathcal{S}|$. Some examples should illustrate the point.

Example 3.2: More words

Consider the experiment of Example 3.1 and suppose that all outcomes are equally likely – there is no bias in the selection of letters from the bag. Find the probability of the following events:

E_1 : “A randomly-chosen word contains no vowels”;

E_2 : “A randomly-chosen word begins and ends with the same letter”.

Solution:

If the word contains no vowels then it must be made up solely of the letters B , C , and D . Hence there are three choices for the first letter, three choices for the second letter, and three choices for the third letter. It follows that

$$\mathbb{P}(E_1) = \frac{|E_1|}{|\mathcal{S}|} = \frac{3 \times 3 \times 3}{125} = \frac{27}{125}.$$

If the word begins and ends with the same letter then there are five choices for the first letter and five choices for the second letter but only one choice for the third because it must be the same as the first. It follows that

$$\mathbb{P}(E_2) = \frac{|E_2|}{|\mathcal{S}|} = \frac{5 \times 5 \times 1}{125} = \frac{25}{125} = \frac{1}{5}.$$

Exercise 3.3: Bank PIN

When I open a bank account I am allocated a 4-digit personal identification number (which may begin with one or more zeros) at random.

- (a) What is the cardinality of the sample space for this experiment?
- (b) By computing the cardinality of each of these events find the probability that:
 - (i) Every digit of my number is even;
 - (ii) My number is palindromic (reads the same forwards as backwards);
 - (iii) No digit of my number exceeds 7;
 - (iv) The largest digit in my number is exactly 7.

3.3 Ordered Sampling without Replacement

Example 3.4: Yet more words

Consider again the experiment of Example 3.1 but now suppose that repetition of letters is *not* allowed – you can still think of taking three letters from the bag of five but now without replacement. How many three-letter words exist in this case?

Solution:

There are still five choices for the first letter but now there are only four choices for the second letter (only four letters are “left in the bag”) and only three choices for the third letter (only three letters “left in the bag”). Hence there are $5 \times 4 \times 3 = 60$ different words.

Again this example illustrates a general principle:

- If we make an ordered selection of r things from a set $U = \{u_1, u_2, \dots, u_n\}$ without replacement (i.e., repetition is not allowed, each element can be

selected only once) then the sample space is the set of all ordered r -tuples of *distinct* elements of U . That is

$$\mathcal{S} = \{(s_1, s_2, \dots, s_r) : s_i \in U \text{ with } s_i \neq s_j \text{ for all } i \neq j\}.$$

- To find the cardinality of $\mathcal{S}$, notice that if $|U| = n$ there are n choices for s_1 ; for each of these choices there are $n - 1$ choices for s_2 ; for each of these there are $n - 2$ choices for s_3 , and so on. Hence,

$$\begin{aligned} |\mathcal{S}| &= n \times (n - 1) \times (n - 2) \times \cdots \times (n - r + 1) \\ &= \frac{n \times (n - 1) \times (n - 2) \times \cdots \times (n - r + 1) \times (n - r) \times (n - r - 1) \times \cdots \times 2 \times 1}{(n - r) \times (n - r - 1) \times \cdots \times 2 \times 1} \\ &= \frac{n!}{(n - r)!} \end{aligned} \tag{3.2}$$

where $k! = k \times (k - 1) \times \cdots \times 2 \times 1$ (known as “ k factorial”) and we have the convention $0! = 1$.

Example 3.5: Letter permutations

How many permutations (rearrangements) of the letters A, B, C, D, E exist? In other words, how many five-letter “anagrams” can we make from these letters without repetition?

Solution:

Considering how many choices there are for each letter, the number of permutations is $5 \times 4 \times 3 \times 2 \times 1 = 5! = 120$.

In general, a permutation is an ordered sample of n things without replacement from the set $U = \{u_1, u_2, \dots, u_n\}$ of n things. Hence, we can find the number of permutations using (3.2) with $r = n$: there are $n!/0! = n!$ of them. This will be an important result when we move on to unordered sampling in the next section.

Sometimes we’re interested in events with no repetition as a subset of a sample space where repetition is allowed. In this case, we need to combine the methods of this section and the last.

Example 3.6: Complicated words

Suppose we do the experiment of Examples 3.1 and 3.2, i.e., we sample letters *with* replacement. What is the probability that a randomly-chosen word has no repeated letters?

Solution:

The sample space of all possible words has $|\mathcal{S}| = 125$ (Example 3.1). Let E_3 be the event that a word has no repeated letters. Then $|E_3| = 60$ (Example 3.4) so

$$\mathbb{P}(E_3) = \frac{|E_3|}{|\mathcal{S}|} = \frac{60}{125} = \frac{12}{25}.$$

*Exercise 3.7: *Bank PIN (revisited)*

Consider the bank account example of Exercise 3.3. Find the probability that:

- (a) The number has at least one repeated digit;
- (b) The digits in the number are in strictly increasing order.

3.4 Unordered Sampling without Replacement

Example 3.8: Letters again

How many ways are there to choose three letters from five without replacement if the order doesn't matter? [You can think of drawing letters from a bag in a Scrabble-type game where the important thing is just what letters you get, not what order you get them.]

Solution:

We already know that there are $5!/2! = 60$ ways to make an ordered selection without replacement (see Example 3.4) but that is overcounting because now we want, e.g., *ECB*, *EBC*, *BEC*, *BCE*, *CBE*, *CEB* to all count as the *same* outcome. For each choice of three letters there are $3!=6$ permutations (see Example 3.5) so the overcounting is by a factor of six and there are $60/6 = 10$ ways to make an unordered selection without replacement.

Once again, the above example illustrates a general idea:

- If we make an unordered selection of r things from a set $U = \{u_1, u_2, \dots, u_n\}$ without replacement then we obtain a subset of U of size r with, by definition, distinct elements.
- The corresponding sample space is the set of all subsets of r elements of U :

$$\mathcal{S} = \{A \subseteq U : |A| = r\}.$$

- An *ordered* sample is obtained by taking an element of this sample space $\mathcal{S}$ and putting its elements in order. Each element of the sample space can be ordered in $r!$ ways and so, if $|U| = n$, then (using the formula (3.2) for ordered selections without replacement) we must have that

$$r! \times |\mathcal{S}| = \frac{n!}{(n-r)!},$$

and so

$$|\mathcal{S}| = \frac{n!}{(n-r)!r!}. \quad (3.3)$$

Remark: We normally use the notation

$$\binom{n}{r} = \frac{n!}{(n-r)!r!}.$$

Here $\binom{n}{r}$ (read as “ n choose r ”) is called a **binomial coefficient**. By convention $\binom{n}{r} = 0$ when $r > n$, which makes sense since we can’t choose more than r things from n without replacement. Binomial coefficients appear in many other places, for example, you may have encountered them in the binomial theorem:

$$\begin{aligned}(a+b)^n &= \binom{n}{0}a^n b^0 + \binom{n}{1}a^{n-1}b^1 + \binom{n}{2}a^{n-2}b^2 + \cdots + \binom{n}{n-1}a^1b^{n-1} + \binom{n}{n}a^0b^n \\ &= \sum_{k=0}^n \binom{n}{k}a^k b^{n-k}.\end{aligned}$$

*Exercise 3.9: **Binomial theorem*

Verify the binomial theorem formula for $n = 2$ and $n = 3$. Then prove it for general n . [Hint: Use induction.]

Example 3.10: Course representation

How many ways are there to select four course representatives with two students from the 246 studying MTH4107, and two students from the 256 studying MTH4207?

Solution:

There are $\binom{246}{2}$ ways to choose the MTH4107 students and $\binom{256}{2}$ ways to choose the MTH4207 students. Hence, by the basic principle of counting,

$$\begin{aligned}\text{Number of selections} &= \binom{246}{2} \times \binom{256}{2} \\ &= \frac{246!}{244!2!} \times \frac{256!}{254!2!} \\ &= \frac{246 \times 245}{2} \times \frac{256 \times 255}{2} \\ &= 123 \times 245 \times 128 \times 255 \\ &= 983606400.\end{aligned}$$

Exercise 3.11: Lotto

An entry into the UK lottery consists of a list of six numbers. During the draw, six numbered balls are chosen at random and if a player matches all six (regardless of order) they win the jackpot.

- Until 2015, the balls were numbered 1 to 49 (inclusive). What was the probability of a particular entry winning?
- From 2015, the balls have been numbered 1 to 59 (inclusive). What is the probability of winning now?

- (c) *What is the probability of matching five balls out of six in the current set-up? Can you generalize to the probability of matching n balls?

3.5 Sampling in Practice

In the preceding sections we have shown the following.

Theorem 3.1. *The number of ways of selecting (sampling) r objects from an n -element set is*

- (a) *Ordered with replacement (repetition allowed): n^r ,*
- (b) *Ordered without replacement (no repetition): $\frac{n!}{(n-r)!}$,*
- (c) *Unordered without replacement (no repetition): $\binom{n}{r}$.*

Remarks:

- It is important when answering questions involving sampling that you read the question carefully and decide what sort of sampling is involved. Specifically, how many things are you selecting, what set are you selecting from, does the order matter, and is repetition allowed or not?
- Sometimes we can consider an experiment as either ordered or unordered sampling. In particular, if we are interested in an event where order doesn't matter, it is generally possible to consider a sample space of ordered outcomes and then count how many of these ordered outcomes are in our event. The important thing is to be consistent, i.e., to consider outcomes recorded in the same way when determining the cardinality of both sample space and event.
- We have not covered the case of unordered sampling with replacement (i.e., repetition allowed). This is rather more difficult to deal with but sometimes the previous point provides a workaround.

Some examples should serve as further illustration.

Example 3.12: Coins

Suppose we have ten coins, seven gold ones and three copper ones, and we pick four of these coins at random (i.e., such that all outcomes are equally likely). Let F_1 be the event that we pick four gold coins. Determine $\mathbb{P}(F_1)$ using both ordered and unordered sampling.

Solution:

Since we pick coins at random, $\mathbb{P}(F_1) = |F_1|/|\mathcal{S}|$. To calculate the probability we therefore need to determine the size of the sample space and the size of the event. The set U of objects contains seven different gold coins and three different copper coins, $U = \{c_1, c_2, c_3, g_1, g_2, \dots, g_6, g_7\}$. It is important to note that the coins are all different objects even if they may share the same colour – by definition the elements of a set must be distinct.

Let us first consider the experiment as ordered sampling without replacement. In this case, the outcomes are an ordered selection of four objects, i.e., a 4-tuple (s_1, s_2, s_3, s_4) with $s_k \in U$ and no repetition (i.e., $s_i \neq s_j$ for $i \neq j$). Using $n = 10$ and $r = 4$ in Theorem 3.1(b), the size of the sample space is

$$|\mathcal{S}| = \frac{10!}{(10-4)!} = \frac{10!}{6!} = 10 \times 9 \times 8 \times 7.$$

The event F_1 consists of ordered samples without replacement of four things from the set of seven gold coins, for instance, one outcome in F_1 is (g_5, g_2, g_6, g_3) . How many such outcomes are there? Well, again using Theorem 3.1(b), but now with $n = 7$ and $r = 4$, we have

$$|F_1| = \frac{7!}{(7-4)!} = \frac{7!}{3!} = 7 \times 6 \times 5 \times 4,$$

and hence

$$\mathbb{P}(F_1) = \frac{|F_1|}{|\mathcal{S}|} = \frac{7 \times 6 \times 5 \times 4}{10 \times 9 \times 8 \times 7} = \frac{1}{6}.$$

Now let us consider the experiment as unordered sampling – imagine revealing all the picked coins at once rather than one-by-one. The outcomes are now subsets of U with cardinality four (i.e., $\{s_1, s_2, s_3, s_4\}$ with $s_k \in U$) and to calculate the size of the sample space we need to use Theorem 3.1(c), with $n = 10$ and $r = 4$:

$$|\mathcal{S}| = \binom{10}{4} = \frac{10!}{6!4!} = \frac{10 \times 9 \times 8 \times 7}{4 \times 3 \times 2 \times 1}.$$

To be consistent, in this framework the outcomes in F_1 must also be considered as unordered: each outcome is a four-element subset of the set of seven gold coins, for instance, $\{g_1, g_3, g_6, g_7\}$. The number of such outcomes is given by Theorem 3.1(c), now with $n = 7$ and $r = 4$:

$$|F_1| = \binom{7}{4} = \frac{7!}{3!4!} = \frac{7 \times 6 \times 5 \times 4}{4 \times 3 \times 2 \times 1}.$$

Hence, finally we have,

$$\mathbb{P}(F_1) = \frac{|F_1|}{|\mathcal{S}|} = \frac{\binom{7}{4}}{\binom{10}{4}} = \frac{7 \times 6 \times 5 \times 4}{10 \times 9 \times 8 \times 7} = \frac{1}{6},$$

which, of course, is the same result as obtained using ordered sampling. [In the fortunate circumstance that you are able to do a question using two different methods, comparison of the results provides a good way to check your answer!]

Exercise 3.13: Coins again

Consider again the set-up of Example 3.12 and determine the probability of the following events:

- (a) The event, F_2 , that we pick two gold coins followed by two copper coins;
- (b) The event, F_3 , that we pick two gold and two copper coins in any order.

In each case, which of ordered and unordered sampling can be used?

Example 3.14: Poker dice

Suppose we roll five fair dice. What is the probability to roll “a pair” (i.e., for two of the dice to show the same number while the other three are all different)?

Solution:

We are sampling $r = 5$ objects from the set $U = \{1, 2, 3, 4, 5, 6\}$ of size $n = 6$. The experiment allows repetition so we have to consider ordered sampling with replacement. [Remember that we haven’t really covered how to treat unordered sampling with replacement!] Outcomes are thus 5-tuples and the size of the sample space is given by Theorem 3.1(a):

$$|\mathcal{S}| = 6^5.$$

Now let G be the event “we roll a pair”. We consider first those outcomes in the event G which have the pair as the first two entries, i.e., outcomes of the form (p, p, r_1, r_2, r_3) . There are obviously six choices for p ; for each of those there are five choices for r_1 ; for each of those there are four choices for r_2 ; for each of those there are three choices for r_3 . Hence by the basic principle of counting, there are $6 \times 5 \times 4 \times 3$ such outcomes. However, the pair doesn’t have to appear as the first two entries of our 5-tuple; we need to take other arrangements into account, e.g., (p, r_1, p, r_2, r_3) , (r_1, r_2, r_3, p, p) . Since we are effectively choosing two (distinct) dice from five to be the pair, there are $\binom{5}{2} = 10$ different “patterns”. For each of those patterns there are $6 \times 5 \times 4 \times 3$ outcomes. Hence, putting everything together, we have

$$|G| = 10 \times 6 \times 5 \times 4 \times 3,$$

and

$$\mathbb{P}(G) = \frac{|G|}{|\mathcal{S}|} = \frac{10 \times 6 \times 5 \times 4 \times 3}{6^5} = \frac{25}{54}.$$

3.6 Further Exercises

Exercise 3.15: Real-life sampling

Think of a real-life situation involving sampling and discuss which of the methods from this chapter would be appropriate to analyse it.

Exercise 3.16: More course representation

Look back at Example 3.10. Now suppose that four course representatives are chosen

at random from the 502 students. What is the probability that two are chosen from MTH4107 and two from MTH4207? [Hint: Look at Exercise 3.13(b).]

Exercise 3.17: Cricket squad

Each member of a squad of 18 cricketers is either a batsman or a bowler. The squad comprises 10 batsmen and 8 bowlers. An eccentric coach chooses a team by picking a random set of 11 players from the squad.

- (a) What is the probability that the team is made up of six batsmen and five bowlers?
- (b) What is the probability that the team contains fewer than three bowlers?

*Exercise 3.18: *Binomial identities*

Let $1 \leq r \leq n$. A subset of $\{1, 2, \dots, n\}$ of cardinality r is chosen at random.

- (a) Calculate the probability that 1 is an element of the chosen subset.
- (b) Without using your answer to part (a), calculate the probability that 1 is *not* an element of the chosen subset.
- (c) Deduce that

$$\binom{n}{r} = \binom{n-1}{r} + \binom{n-1}{r-1}.$$

CHAPTER 4

Conditional Probability

4.1 Introducing Extra Information

Additional information (a so-called “condition”) may change the probability ascribed to an event. This is important, for example, in medical testing/care and in pricing life insurance.

Exercise 4.1: The power of knowledge

Think of some more real-life examples where extra information would change the probability given to an event.

Example 4.2: Information about a die

Consider rolling a fair six-sided die once and recording the number showing.

- (a) Determine the probabilities of the following two events:
- A : “The number shown is odd”;
 B : “The number shown is smaller than four”.
- (b) Now suppose somebody tells us that the number shown is odd, i.e., we know that event A has happened (we say “event A is given”). What is now the probability of event B (given A)?

Solution:

- (a) Just as in Example 1.1 we can simply write the sample space as $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$ and we have $A = \{1, 3, 5\}$ and $B = \{1, 2, 3\}$. Hence, since the die is fair and outcomes are equally likely,

$$\mathbb{P}(A) = \frac{|A|}{|\mathcal{S}|} = \frac{3}{6} = \frac{1}{2},$$
$$\mathbb{P}(B) = \frac{|B|}{|\mathcal{S}|} = \frac{3}{6} = \frac{1}{2}.$$

(b) Again using the fact that outcomes are equally likely, we have

$$\mathbb{P}(B \text{ given } A) = \frac{|\{1, 3\}|}{|\{1, 3, 5\}|} = \frac{2}{3}.$$

Notice that we can write

$$\begin{aligned} \mathbb{P}(B \text{ given } A) &= \frac{|\{1, 3\}|}{|\{1, 3, 5\}|} \\ &= \frac{|\{1, 3\}|/|\mathcal{S}|}{|\{1, 3, 5\}|/|\mathcal{S}|} \\ &= \frac{|A \cap B|/|\mathcal{S}|}{|A|/|\mathcal{S}|} \\ &= \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)} \end{aligned}$$

This last expression gives a general definition of conditional probability (the probability of the event B conditioned on event A having occurring), which holds beyond the special case of equally-likely outcomes.

Definition 4.1. If E_1 and E_2 are events and $\mathbb{P}(E_1) \neq 0$ then the **conditional probability of E_2 given E_1** , usually denoted by $\mathbb{P}(E_2|E_1)$, is

$$\mathbb{P}(E_2|E_1) = \frac{\mathbb{P}(E_1 \cap E_2)}{\mathbb{P}(E_1)}.$$

Remarks:

- The notation $\mathbb{P}(E_2|E_1)$ is rather unfortunate since $E_2|E_1$ is not an event. Do not confuse the conditional probability $\mathbb{P}(E_2|E_1)$ with $\mathbb{P}(E_2 \setminus E_1)$, the probability for the event “ E_2 and not E_1 ”.
- Note that the definition does not require that E_2 happens after E_1 , only that we know about E_1 but not about E_2 . One way of thinking of this is to imagine that the experiment is performed secretly and the fact that E_1 occurred is revealed to you (without the full outcome being revealed). The conditional probability of E_2 given E_1 is the new probability of E_2 in these circumstances
- It is worth stressing again that this is a general definition of conditional probability; for the remainder of the chapter (and these notes) you should not assume equally-likely outcomes without good reason.

Exercise 4.3: Coloured pens

You have three pens coloured blue, red, and green. You pick one at random and then you pick another without replacement.

- (a) Find the conditional probability that the second pen is blue, given the first pen is red.
- (b) Find the conditional probability that the first pen is red, given the second pen is blue.

4.2 Implications of Extra Information

In the last section we introduced the idea of extra information; we here explore further its implications. Note, in particular, that conditional probability can be used to measure how the occurrence of some event influences the chance of another event occurring:

- If $\mathbb{P}(E_2|E_1) < \mathbb{P}(E_2)$ then E_1 occurring makes E_2 less probable;
- If $\mathbb{P}(E_2|E_1) > \mathbb{P}(E_2)$ then E_1 occurring makes E_2 more probable;
- If $\mathbb{P}(E_2|E_1) = \mathbb{P}(E_2)$ then the event E_1 has no impact on the probability of event E_2 and we say the events are **independent** (see Chapter 5).

We now illustrate this with a long example.

Example 4.4: More information about a die

Consider rolling a fair six-sided die twice.

- (a) Determine the probability of the following events:

A : “A ‘six’ occurs on the first roll”;

B : “A double is rolled”;

C : “At least one odd number is rolled”.

- (b) Now find the conditional probabilities for the following:

(i) Rolling a double assuming the first roll is a six.

(ii) Rolling a six first assuming a double is rolled.

(iii) Rolling at least one odd number assuming a double is rolled.

(iv) Rolling a double assuming at least one odd number is rolled.

Solution:

- (a) This situation can be treated as ordered sampling with replacement; we write the outcomes as ordered pairs and $|\mathcal{S}| = 36$ (see Exercise 1.3). Since outcomes are equally likely we can simply count the number of outcomes in each event to find

$$\mathbb{P}(A) = \frac{|A|}{|\mathcal{S}|} = \frac{6}{36} = \frac{1}{6},$$

$$\mathbb{P}(B) = \frac{|B|}{|\mathcal{S}|} = \frac{6}{36} = \frac{1}{6},$$

$$\mathbb{P}(C) = \frac{|C|}{|\mathcal{S}|} = \frac{27}{36} = \frac{3}{4}.$$

[If the last of these is not obvious, use the fact that there are 3×3 ways to get two even numbers, together with Proposition 2.2.]

- (b) (i) We have $A \cap B = \{(6, 6)\}$ so

$$\mathbb{P}(A \cap B) = \frac{|A \cap B|}{|\mathcal{S}|} = \frac{1}{36}$$

and hence

$$\mathbb{P}(B|A) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)} = \frac{1/36}{1/6} = \frac{1}{6}.$$

This fits with the intuition that rolling a six first does not change the probability of a double.

- (ii) Now we have

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \frac{1/36}{1/6} = \frac{1}{6}.$$

It is perhaps slightly less obvious that rolling a double does not change the probability of getting a six on the first roll. However, remember that conditional probability does not require the condition to happen “first”; indeed, here event B happens “after” event A has happened.

- (iii) Here $B \cap C = \{(1, 1), (3, 3), (5, 5)\}$ so

$$\mathbb{P}(B \cap C) = \frac{|B \cap C|}{|\mathcal{S}|} = \frac{3}{36} = \frac{1}{12}$$

and hence

$$\mathbb{P}(C|B) = \frac{\mathbb{P}(B \cap C)}{\mathbb{P}(B)} = \frac{1/12}{1/6} = \frac{1}{2}.$$

So rolling a double *reduces* the probability of having at least one odd number.

- (iv) Now we have

$$\mathbb{P}(B|C) = \frac{\mathbb{P}(B \cap C)}{\mathbb{P}(C)} = \frac{1/12}{3/4} = \frac{1}{9}.$$

Similarly, rolling at least one odd number *reduces* the probability of a double – this is probably not obvious at all!

Notice from the last example that for events E_1 and E_2 , the probability of E_1 given E_2 does not have to be equal to the probability of E_2 given E_1 ; in general these are different probabilities (a fact which is related to the discussion of Exercise 0.3).

Exercise 4.5: Conditional deductions

Let E_1 and E_2 be events with $\mathbb{P}(E_1) > 0$ and $\mathbb{P}(E_2) > 0$.

- (a) Prove that if $\mathbb{P}(E_1|E_2) > \mathbb{P}(E_1)$ then $\mathbb{P}(E_2|E_1) > \mathbb{P}(E_2)$.
- (b) Prove that $\mathbb{P}(E_1^c|E_2) = 1 - \mathbb{P}(E_1|E_2)$.
- (c) If $\mathbb{P}(E_1|E_2)$ is known, what can be deduced about $\mathbb{P}(E_1|E_2^c)$?

4.3 The Multiplication Rule

We have already seen how we can calculate the conditional probability $P(E_2|E_1)$ from knowledge of $\mathbb{P}(E_1 \cap E_2)$. However, by straightforwardly rearranging Definition 4.1, one can write

$$\mathbb{P}(E_1 \cap E_2) = \mathbb{P}(E_1)\mathbb{P}(E_2|E_1), \quad (4.1)$$

which is a useful formula for calculating $\mathbb{P}(E_1 \cap E_2)$ from knowledge of the conditional probability $P(E_2|E_1)$. In fact, one can generalize this to an arbitrary number of events as expressed in the following theorem.

Theorem 4.2. *Let $E_1, E_2, \dots, E_n$ be events (where $n \geq 2$), then*

$$\mathbb{P}(E_1 \cap E_2 \cap \dots \cap E_n) = \mathbb{P}(E_1) \times \mathbb{P}(E_2|E_1) \times \mathbb{P}(E_3|E_1 \cap E_2) \times \dots \times \mathbb{P}(E_n|E_1 \cap E_2 \cap \dots \cap E_{n-1})$$

provided that all of the conditional probabilities involved are defined.

Proof:

For a given number of events, one can easily prove the theorem statement by plugging in the definition of conditional probabilities and cancelling common factors in the numerator and denominator. To prove it in general we will use induction.

As the base case, let us take $n = 2$. This is precisely what we already have in (4.1) from direct rearrangement of Definition 4.1. Now, for the inductive step, we assume we have shown that the statement of the theorem holds for $n = k$, i.e.,

$$\mathbb{P}(E_1 \cap E_2 \cap \dots \cap E_k) = \mathbb{P}(E_1) \times \mathbb{P}(E_2|E_1) \times \mathbb{P}(E_3|E_1 \cap E_2) \times \dots \times \mathbb{P}(E_k|E_1 \cap E_2 \cap \dots \cap E_{k-1}) \quad (4.2)$$

and seek to prove the statement for $n = k + 1$. First note that we can write $E_1 \cap E_2 \cap \dots \cap E_{k+1}$ as $F_1 \cap F_2$ with $F_1 = E_1 \cap E_2 \cap \dots \cap E_k$ and $F_2 = E_{k+1}$. From Definition 4.1, we have $\mathbb{P}(F_1 \cap F_2) = \mathbb{P}(F_1)\mathbb{P}(F_2|F_1)$, i.e.,

$$\mathbb{P}(E_1 \cap E_2 \cap \dots \cap E_{k+1}) = \mathbb{P}(E_1 \cap E_2 \cap \dots \cap E_k) \times \mathbb{P}(E_{k+1}|E_1 \cap E_2 \cap \dots \cap E_k). \quad (4.3)$$

Substituting (4.2) into (4.3) yields

$$\begin{aligned} \mathbb{P}(E_1 \cap E_2 \cap \dots \cap E_{k+1}) &= \mathbb{P}(E_1) \times \mathbb{P}(E_2|E_1) \times \mathbb{P}(E_3|E_1 \cap E_2) \times \dots \\ &\quad \times \mathbb{P}(E_k|E_1 \cap E_2 \cap \dots \cap E_{k-1}) \times \mathbb{P}(E_{k+1}|E_1 \cap E_2 \cap \dots \cap E_k) \end{aligned}$$

which means the statement of the theorem also holds for $n = k + 1$ and hence, by the principle of induction, for all $n \geq 2$. $\square$

Exercise 4.6: Coins (revisited)

Reconsider the event F_1 in Example 3.12 (picking four gold coins from a set of seven gold and three copper coins). Demonstrate that one can also work out $\mathbb{P}(F_1)$ using conditional probability.

4.4 Ordered Sampling Revisited

The exercise at the end of the previous section already gave a hint that conditional probability provides an alternative approach to questions involving ordered sampling. We now show how this approach can be used to confirm our earlier results for the cardinality of the sample space in the cases of ordered sampling with and without replacement.

Consider again that we pick at random r things in order from a set U of size $n = |U|$. Let us denote a fixed outcome of this experiment by $(v_1, v_2, \dots, v_r)$, with $v_k \in U$, and the corresponding event that this particular outcome occurs by A , i.e., $A = \{(v_1, v_2, \dots, v_r)\}$. Notice that A is a simple event so, since all outcomes are equally likely, we have

$$\mathbb{P}(A) = \frac{1}{|\mathcal{S}|}.$$

Hence, if we can calculate $\mathbb{P}(A)$ by conditional probability, it will give us an expression for $|\mathcal{S}|$.

Let E_k denote the event that the k th pick gives v_k (with $v_k \in U$). The event A can then be written as $A = E_1 \cap E_2 \cap \dots \cap E_r$ and, by Theorem 4.2, the probability of this event is given as

$$\mathbb{P}(E_1 \cap E_2 \cap \dots \cap E_r) = \mathbb{P}(E_1) \times \mathbb{P}(E_2|E_1) \times \mathbb{P}(E_3|E_1 \cap E_2) \times \dots \times \mathbb{P}(E_r|E_1 \cap E_2 \cap \dots \cap E_{r-1}).$$

Note that this expression is valid whether we sample with or without replacement but the form of the conditional probabilities is different in each case.

Example 4.7: Ordered sampling without replacement (revisited)

Suppose we sample *without* replacement and $v_1, v_2, \dots, v_r$ are distinct. Show that $\mathbb{P}(A) = (n-r)!/n!$.

Solution:

In this case,

$$\mathbb{P}(E_1) = \frac{1}{n},$$

as we pick the element v_1 at random from the set U of size $|U| = n$. Similarly,

$$\mathbb{P}(E_2|E_1) = \frac{1}{n-1},$$

as we pick the element v_2 at random from the set $U \setminus \{v_1\}$ of size $n - 1$. In general,

$$\mathbb{P}(E_i | E_1 \cap E_2 \cap \cdots \cap E_{i-1}) = \frac{1}{n - i + 1},$$

as we pick the element v_i at random from the set $U \setminus \{v_1, v_2, \dots, v_{i-1}\}$ of size $n - i + 1$. Hence

$$\begin{aligned} \mathbb{P}(A) &= \mathbb{P}(E_1 \cap E_2 \cap \cdots \cap E_r) \\ &= \frac{1}{n} \times \frac{1}{n-1} \times \cdots \times \frac{1}{n-r+1} \\ &= \frac{(n-r)!}{n!}. \end{aligned}$$

Exercise 4.8: Ordered sampling with replacement (revisited)

Modify Example 4.7 to sampling *with* replacement. Show that $\mathbb{P}(A) = 1/n^r$ in this case.

Unsurprisingly, the results for $\mathbb{P}(A)$ in Example 4.7 and Exercise 4.8 agree with $1/|\mathcal{S}|$ when $|\mathcal{S}|$ is calculated from Theorem 3.1(b) and 3.1(a) respectively. However, you may find the equally-likely assumption more transparent with the method of this chapter.

4.5 Further Exercises

Exercise 4.9: More die rolling

A standard fair die is rolled twice.

- Find the probability that the sum of the two rolls is at least nine.
- Find the conditional probability that the first roll is a “four” given that the sum of the two rolls is at least nine.
- Find the conditional probability that the first roll is not a “four” given that the sum of the two rolls is at least nine.
- Find the conditional probability that the first roll is a “four” given that the sum of the two rolls is less than nine.
- Find the conditional probability that the sum of the two rolls is at least nine given that the first roll is a “four”.

Exercise 4.10: Travel difficulties

When you travel into university you notice whether your train is late and by how much and also whether you are able to get a seat on it. Let A be the event “the train is not late”, B be the event “the train is late but by not more than 15 minutes”, and C be the event “you are able to get a seat”. Suppose that $\mathbb{P}(A) = 1/2$, $\mathbb{P}(B) = 1/4$, $\mathbb{P}(C) = 1/3$ and $\mathbb{P}(A \cap C) = 1/4$.

- (a) Show that the conditional probability that the train is more than 15 minutes late given that the train is late is equal to $1/2$.
- (b) Show that the conditional probability that you get a seat given that the train is late is equal to $1/6$.

Exercise 4.11: Medical testing

Two treatments for a disease are tested on a group of 390 patients. Treatment A is given to 160 patients of whom 100 are men and 60 are women; 20 of these men and 40 of these women recover. Treatment B is given to 230 patients of whom 210 are men and 20 are women; 50 of these men and 15 of these women recover.

- (a) For which of A and B is there a higher probability that a patient chosen randomly from among those given that treatment recovers? Express this as an inequality between two conditional probabilities.
- (b) For which of A and B is there a higher probability that a man chosen randomly from among those given that treatment recovers? Express this as an inequality between two conditional probabilities.
- (c) For which of A and B is there a higher probability that a woman chosen randomly from among those given that treatment recovers? Express this as an inequality between two conditional probabilities.
- (d) Compare the inequality in part (a) with the inequalities in part (b) and (c). Are you surprised by the result?

CHAPTER 5

Independence

5.1 Independence for Two Events – Basic Definition

The examples we have seen so far tell us that the probability $\mathbb{P}(E_1 \cap E_2)$ being the product $\mathbb{P}(E_1) \times \mathbb{P}(E_2)$ is a very special case.

Exercise 5.1: Finding the unusual

Look back through all the previous examples in the notes and find cases where the probability of the intersection of two events is the product of their individual probabilities, i.e., $\mathbb{P}(E_1 \cap E_2) = \mathbb{P}(E_1)\mathbb{P}(E_2)$. Can you identify what these situations have in common?

Example 5.2: Yet more die rolling

Consider again rolling a fair six-sided die twice. Let A denote the event “the first roll shows an even number” and B denote the event “the number shown on the second roll is larger than four”. Determine the probabilities $\mathbb{P}(A)$, $\mathbb{P}(B)$, $\mathbb{P}(A \cap B)$, $\mathbb{P}(A|B)$, and $\mathbb{P}(B|A)$.

Solution:

Just as in Example 4.4, we can treat this situation as ordered sampling with replacement and write the outcomes as ordered pairs. We obviously have $|\mathcal{S}| = 36$ and, counting the number of possibilities for each die roll we have $|A| = 3 \times 6 = 18$ and $|B| = 6 \times 2 = 12$. Hence, since all outcomes are equally likely and (2.5) applies,

$$\mathbb{P}(A) = \frac{18}{36} = \frac{1}{2} \quad \text{and} \quad \mathbb{P}(B) = \frac{12}{36} = \frac{1}{3}.$$

For the event $A \cap B$ (“the first roll is even and the second roll is larger than four”), there are obviously three possibilities for the first roll and two for the second so $|A \cap B| = 3 \times 2 = 6$ and

$$\mathbb{P}(A \cap B) = \frac{6}{36} = \frac{1}{6}.$$

Notice here that

$$\mathbb{P}(A \cap B) = \frac{1}{6} = \frac{1}{2} \times \frac{1}{3} = \mathbb{P}(A)\mathbb{P}(B).$$

As you may already know, events with such a property are said to be independent. We also have that

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \frac{1/6}{1/3} = \frac{1}{2},$$

and

$$\mathbb{P}(B|A) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)} = \frac{1/6}{1/2} = \frac{1}{3}.$$

Note that $\mathbb{P}(A|B) = \mathbb{P}(A)$ and $\mathbb{P}(B|A) = \mathbb{P}(B)$, i.e. the conditional probabilities do not depend on the condition. We will further explore the connection between conditional probabilities and independence in the next section.

Definition 5.1. We say that the events E_1 and E_2 are (pairwise) **independent** if

$$\mathbb{P}(E_1 \cap E_2) = \mathbb{P}(E_1)\mathbb{P}(E_2).$$

[If this equation does not hold we say that the events are **dependent**.]

Remarks:

- Be careful not to assume independence without good reason. You may assume that two events E_1 and E_2 are independent in the following situations:
 - (i) They are clearly physically unrelated (e.g., they are associated with different tosses of a coin);
 - (ii) You calculate their probabilities and find that $\mathbb{P}(E_1 \cap E_2) = \mathbb{P}(E_1)\mathbb{P}(E_2)$ (i.e., you check Definition 5.1);
 - (iii) A question tells you that they are independent!
- Independence is *not* equivalent to being physically unrelated. Physically unrelated events are always independent [see (i) above] but, as we shall see, physically related events may or may not be independent.

Exercise 5.3: Different buses

Suppose there are two buses, the Alphabus and the Betabus, running from station P to station Q along two different routes. Consider the events

A : “The Alphabus is running”;

B : “The Betabus is running”.

Assuming that these events are independent and have probabilities $\mathbb{P}(A) = 9/10$ and $\mathbb{P}(B) = 4/5$, determine the probability that one can travel from P to Q by bus.

5.2 Independence for Two Events – More Details

As Example 5.2 already suggested, there is a connection between independence and conditional probability.

Theorem 5.2. *Let E_1 and E_2 be events with $\mathbb{P}(E_1) > 0$ and $\mathbb{P}(E_2) > 0$. The following are equivalent:*

- (a) E_1 and E_2 are independent,
- (b) $\mathbb{P}(E_1|E_2) = \mathbb{P}(E_1)$,
- (c) $\mathbb{P}(E_2|E_1) = \mathbb{P}(E_2)$.

Loosely speaking, this says that if E_1 and E_2 are independent then telling you that E_1 occurred does not change the probability that E_2 occurred, and vice versa.

Proof:

It is sufficient to show that (a) implies (b), (b) implies (c), and (c) implies (a).

- **(a) implies (b):** We start by assuming that (a) is true, i.e., we suppose E_1 and E_2 are independent so, from Definition 5.1,

$$\mathbb{P}(E_1 \cap E_2) = \mathbb{P}(E_1)\mathbb{P}(E_2).$$

Since $\mathbb{P}(E_2) \neq 0$, we can use Definition 4.1 to show

$$\mathbb{P}(E_1|E_2) = \frac{\mathbb{P}(E_1 \cap E_2)}{\mathbb{P}(E_2)} = \frac{\mathbb{P}(E_1)\mathbb{P}(E_2)}{\mathbb{P}(E_2)} = \mathbb{P}(E_1).$$

Hence (b) is also true.

- **(b) implies (c):** We start by assuming (b) is true, i.e., $\mathbb{P}(E_1|E_2) = \mathbb{P}(E_1)$. Then, by Definition 4.1,

$$\frac{\mathbb{P}(E_1 \cap E_2)}{\mathbb{P}(E_2)} = \mathbb{P}(E_1).$$

Since $\mathbb{P}(E_1) \neq 0$ (and $E_1 \cap E_2 = E_2 \cap E_1$), it follows that

$$\frac{\mathbb{P}(E_2 \cap E_1)}{\mathbb{P}(E_1)} = \mathbb{P}(E_2)$$

which means, again by Definition 4.1, $\mathbb{P}(E_2|E_1) = \mathbb{P}(E_2)$, i.e., (c) is also true.

- **(c) implies (a):** We start by assuming (c) is true, i.e., $\mathbb{P}(E_2|E_1) = \mathbb{P}(E_2)$. Then, once again by Definition 4.1,

$$\frac{\mathbb{P}(E_2 \cap E_1)}{\mathbb{P}(E_1)} = \mathbb{P}(E_2)$$

which implies $\mathbb{P}(E_2 \cap E_1) = \mathbb{P}(E_1)\mathbb{P}(E_2)$, i.e., E_1 and E_2 satisfy Definition 5.1 and are independent. Hence (a) is also true and the proof is complete. $\square$

Independence can also be disguised in other ways as illustrated by the following example.

Example 5.4: Another implication

Prove that the equality $\mathbb{P}(E_1 \cup E_2) = \mathbb{P}(E_1)\mathbb{P}(E_2^c) + \mathbb{P}(E_2)$ implies that E_1 and E_2 are independent.

Solution:

We start from

$$\mathbb{P}(E_1 \cup E_2) = \mathbb{P}(E_1)\mathbb{P}(E_2^c) + \mathbb{P}(E_2)$$

and use the inclusion-exclusion formula (Proposition 2.7) on the left-hand side and the obvious Proposition 2.2 on the right-hand side to obtain:

$$\mathbb{P}(E_1) + \mathbb{P}(E_2) - \mathbb{P}(E_1 \cap E_2) = \mathbb{P}(E_1)[1 - \mathbb{P}(E_2)] + \mathbb{P}(E_2).$$

Cancelling $\mathbb{P}(E_1) + \mathbb{P}(E_2)$ on the two sides, this yields

$$-\mathbb{P}(E_1 \cap E_2) = -\mathbb{P}(E_1)\mathbb{P}(E_2),$$

which is trivially equivalent to the usual condition for independence:

$$\mathbb{P}(E_1 \cap E_2) = \mathbb{P}(E_1)\mathbb{P}(E_2).$$

Hence the original equality implies that E_1 and E_2 are independent.

Exercise 5.5: More information about a die (revisited)

Look back at Example 4.4. Are the events A and B defined in that example independent? Are B and C ? Are A and C ? [You may use any of the equivalent results from this section.]

5.3 Independence for Three or More Events

Exercise 5.5 may prompt you to ask whether it is actually possible to find three events which are pairwise independent. You may also wonder whether just looking at pairs is enough to define independence in the case of more than two events. In this section we address such questions.

Example 5.6: Even more boring dice rolling

Consider (yet again) again rolling a fair six-sided die twice. Now look at the following events:

D : “The first roll shows an odd number”;

E : “The second roll shows an odd number”;

F : “The sum of the two rolls is an odd number”.

Determine whether these events are pairwise independent.

Solution:

Obviously $|\mathcal{S}| = 36$ and $|D| = |E| = 18$ so $\mathbb{P}(D) = \mathbb{P}(E) = 1/2$ (cf. Example 5.2). The cardinality of F is slightly less obvious but note that if the sum is odd, one roll must be odd and the other even; there are nine ordered pairs which are odd followed by even and nine ordered pairs which are even followed by odd. Hence, we also have $\mathbb{P}(F) = 18/36 = 1/2$. Turning our attention to independence, we now consider each pair of events in turn.

- D and E relate to different die rolls so we can argue that they are physically unrelated. Hence D and E are independent. This is easily confirmed since

$$\mathbb{P}(D \cap E) = \frac{|D \cap E|}{|\mathcal{S}|} = \frac{9}{36} = \frac{1}{4} = \frac{1}{2} \times \frac{1}{2} = \mathbb{P}(D)\mathbb{P}(E).$$

- The event $D \cap F$ contains all pairs which are odd followed by even so

$$\mathbb{P}(D \cap F) = \frac{|D \cap F|}{|\mathcal{S}|} = \frac{9}{36} = \frac{1}{4} = \frac{1}{2} \times \frac{1}{2} = \mathbb{P}(D)\mathbb{P}(F).$$

Hence D and F are independent.

- Similarly, the event $E \cap F$ contains all pairs which are even followed by odd so

$$\mathbb{P}(E \cap F) = \frac{|E \cap F|}{|\mathcal{S}|} = \frac{9}{36} = \frac{1}{4} = \frac{1}{2} \times \frac{1}{2} = \mathbb{P}(E)\mathbb{P}(F).$$

Hence E and F are independent.

Each pair of events is independent so we say that D , E , and F are pairwise independent. However, notice that the event $D \cap E \cap F$ is impossible since when both outcomes are odd the sum is even. Hence

$$\mathbb{P}(D \cap E \cap F) = \mathbb{P}(\emptyset) = 0 \neq \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \mathbb{P}(D)\mathbb{P}(E)\mathbb{P}(F).$$

In fact, for three or more events the notion of independence is slightly more subtle than for two events. For example, for three events we have the following definition.

Definition 5.3. Three events E_1 , E_2 , and E_3 are called *pairwise independent* if

$$\mathbb{P}(E_1 \cap E_2) = \mathbb{P}(E_1)\mathbb{P}(E_2),$$

$$\mathbb{P}(E_1 \cap E_3) = \mathbb{P}(E_1)\mathbb{P}(E_3),$$

$$\mathbb{P}(E_2 \cap E_3) = \mathbb{P}(E_2)\mathbb{P}(E_3).$$

The three events are called *mutually independent* if in addition

$$\mathbb{P}(E_1 \cap E_2 \cap E_3) = \mathbb{P}(E_1)\mathbb{P}(E_2)\mathbb{P}(E_3).$$

Armed with this definition, we see that although the events in Example 5.6 are pairwise independent they are *not* mutually independent.

Definition 5.3 can be generalized to four, five, and more events. The formal definition looks awkward but the basic idea is that, for mutual independence, the probability of the intersection of any finite subset of the events should factorize into the probabilities of the individual events.

Definition 5.4. We say that the events $E_1, E_2, \dots, E_n$ are **mutually independent** if for any $2 \leq t \leq n$ and $1 \leq i_1 < i_2 < \dots < i_t \leq n$ we have

$$\mathbb{P}(E_{i_1} \cap E_{i_2} \cap \dots \cap E_{i_t}) = \mathbb{P}(E_{i_1}) \times \mathbb{P}(E_{i_2}) \times \dots \times \mathbb{P}(E_{i_t}).$$

Remark: If the term “independent” is used without qualification for three or more events, it generally means mutually independent. However, to avoid ambiguity we shall try to be careful in the use of “pairwise independent” and “mutually independent” as appropriate.

Exercise 5.7: Tossing thrice again

Suppose you toss a fair coin three times and record the sequence of Heads/Tails, as in Example 1.2. Now consider the following events:

- A: “The first and the second toss show the same result.”;
- B: “The first and the last toss show different results.”;
- C: “The first toss is a Tail.”.

Determine whether these three events are pairwise independent and whether they are mutually independent.

5.4 Conditional Independence

It is also possible to consider independence of two events given we know that a third event happens. This more advanced concept is captured in the following definition.

Definition 5.5. Two events E_1 and E_2 are said to be **conditionally independent** given an event E_3 if

$$\mathbb{P}(E_1 \cap E_2 | E_3) = \mathbb{P}(E_1 | E_3) \mathbb{P}(E_2 | E_3).$$

Example 5.8: Magic coins

A magician has two coins: one is fair; the other has probability 3/4 of coming up Heads. She picks a coin at random and tosses it twice. Consider the following events:

- F: “The magician picks the fair coin.”;
- H_1 : “The first toss is a Head.”;
- H_2 : “The second toss is a Head.”.

- (a) Are H_1 and H_2 conditionally independent given F ?
- (b) Are H_1 and H_2 conditionally independent given F^c ?
- (c) Are H_1 and H_2 independent?

Solution:

- (a) Assuming we pick the fair coin, the probability of a Head on each toss is $1/2$, i.e.,

$$\mathbb{P}(H_1|F) = \frac{1}{2}, \quad \text{and} \quad \mathbb{P}(H_2|F) = \frac{1}{2}.$$

Furthermore, different tosses of the *same* coin are considered to be independent (see the discussion below Definition 5.1) so

$$\mathbb{P}(H_1 \cap H_2|F) = \mathbb{P}(H_1|F)\mathbb{P}(H_2|F) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4},$$

i.e., H_1 and H_2 are conditionally independent given F .

- (b) Assuming we pick the biased coin, the probability of a Head on each toss is $3/4$, i.e.,

$$\mathbb{P}(H_1|F^c) = \frac{3}{4}, \quad \text{and} \quad \mathbb{P}(H_2|F^c) = \frac{3}{4}.$$

Again, different tosses of the *same* coin are considered to be independent so, by construction of the experiment,

$$\mathbb{P}(H_1 \cap H_2|F^c) = \mathbb{P}(H_1|F^c)\mathbb{P}(H_2|F^c) = \frac{3}{4} \times \frac{3}{4} = \frac{9}{16},$$

i.e., H_1 and H_2 are conditionally independent given F^c .

- (c) To check Definition 5.1 and determine whether H_1 and H_2 are independent, we need to know $\mathbb{P}(H_1)$, $\mathbb{P}(H_2)$, and $\mathbb{P}(H_1 \cap H_2)$. To calculate the first of these note that

$$\begin{aligned} \mathbb{P}(H_1) &= \mathbb{P}(H_1 \cap F) + \mathbb{P}(H_1 \cap F^c) \\ &= \mathbb{P}(H_1|F)\mathbb{P}(F) + \mathbb{P}(H_1|F^c)\mathbb{P}(F^c), \end{aligned}$$

where in the first line we have employed Kolmogorov's third axiom, Definition 2.1(c), and in the second line we have used the multiplication rule, Theorem 4.2. [In fact, this is a taster of a general method which will appear in the next chapter.] Now, since the coin is picked at random, we obviously have $\mathbb{P}(F) = \mathbb{P}(F^c) = 1/2$. Hence, substituting numbers, we find

$$\mathbb{P}(H_1) = \frac{1}{2} \times \frac{1}{2} + \frac{3}{4} \times \frac{1}{2} = \frac{5}{8}.$$

Similarly, we have

$$\begin{aligned}
 \mathbb{P}(H_2) &= \mathbb{P}(H_2 \cap F) + \mathbb{P}(H_2 \cap F^c) \\
 &= \mathbb{P}(H_2|F)\mathbb{P}(F) + \mathbb{P}(H_2|F^c)\mathbb{P}(F^c) \\
 &= \frac{1}{2} \times \frac{1}{2} + \frac{3}{4} \times \frac{1}{2} \\
 &= \frac{5}{8},
 \end{aligned}$$

and

$$\begin{aligned}
 \mathbb{P}(H_1 \cap H_2) &= \mathbb{P}(H_1 \cap H_2 \cap F) + \mathbb{P}(H_1 \cap H_2 \cap F^c) \\
 &= \mathbb{P}(H_1 \cap H_2|F)\mathbb{P}(F) + \mathbb{P}(H_1 \cap H_2|F^c)\mathbb{P}(F^c) \\
 &= \frac{1}{4} \times \frac{1}{2} + \frac{9}{16} \times \frac{1}{2} \\
 &= \frac{13}{32}.
 \end{aligned}$$

Since

$$\mathbb{P}(H_1 \cap H_2) = \frac{26}{64} \neq \frac{5}{8} \times \frac{5}{8} = \mathbb{P}(H_1)\mathbb{P}(H_2),$$

we finally conclude that the events H_1 and H_2 are *not* independent. This is intuitively reasonable, if we see a Head on the first toss we can reason that the magician is more likely to be using the biased coin, and this will affect the probability of a Head on the second toss.

Exercise 5.9: Are you bored of dice yet?

You roll a fair six-sided die twice. Let A be the event that the first roll is odd, B be the event that you roll at least one “six”, and C be the event that the sum of the rolls is seven.

- (a) Are the events A and B independent?
- (b) Are the events A , B , and C mutually independent?
- (c) Are the events A and B conditionally independent given C ?

*Exercise 5.10: *Challenging events*

- (a) Find an example with two events, A and B , which are independent but *not* conditionally independent with respect to another event C .
- (b) Find an example with two events, D and E , which are conditionally independent with respect to an event F but *not* with respect to F^c .

5.5 Further Exercises

Exercise 5.11: Integer selection

A positive integer from the set $\{1, 2, 3, \dots, 36\}$ is chosen at random with all choices equally

likely. Let E be the event “the chosen number is even”, O be the event “the chosen number is odd”, Q be the event “the chosen number is a perfect square”, and D_k be the event “the chosen number is a multiple of k ”. Carefully justify your answers to the following.

- (a) Are the events E and O independent?
- (b) Are the events E and Q independent?
- (c) Are the events O and Q independent?
- (d) Are the events D_3 and D_4 independent?
- (e) Are the events D_4 and D_6 independent?

Exercise 5.12: Top card

The top card of a thoroughly shuffled standard deck of playing cards is turned over. Let A be the event “the card is an ace”, R be the event “the card belongs to a red suit ($\diamond$ or $\heartsuit$)”, and M be the event “the card belongs to a major suit ($\heartsuit$ or $\spadesuit$)”. Show that the events A , R , and M are mutually independent.

*Exercise 5.13: *Practice with proof*

Prove that if A , B , and C are mutually independent events, A and $B \cup C$ are independent.

CHAPTER 6

Total Probability and Bayes' Theorem

6.1 Law of Total Probability

We saw in Example 5.8 that conditional probabilities can be used to compute the “total” probability of an event. To state this more formally, we need the idea of a partition which we illustrate first with another example.

Example 6.1: Tossing thrice (revisited)

A coin is tossed three times and the sequence of Heads/Tails is recorded just as in Example 1.2 (and Exercise 5.7). Consider the three events:

E_1 : “The first toss is a Head”;

E_2 : “The first toss is a Tail and the second toss is a Head”;

E_3 : “The first and second tosses are Tails”.

State these events in set notation and consider how they relate to the sample space.

Solution:

With the same (obvious) notation as in Example 1.2, each outcome is a list of Heads (h) and Tails (t) in the order in which they are seen. We have

$$E_1 = \{htt, hht, hth, htt\},$$

$$E_2 = \{thh, tht\},$$

$$E_3 = \{tth, ttt\}.$$

Notice that every outcome in the sample space appears in exactly one of these sets. The three events are pairwise disjoint ($E_i \cap E_j = \emptyset$ for $i \neq j$) and $\mathcal{S} = E_1 \cup E_2 \cup E_3$. Loosely speaking, the three events “split” the sample space into three parts; more formally we say that E_1 , E_2 , and E_3 partition the sample space.

Definition 6.1. The events $E_1, E_2, \dots, E_n$ **partition** $\mathcal{S}$ if they are pairwise disjoint (i.e., $E_k \cap E_\ell = \emptyset$ if $k \neq \ell$) and $E_1 \cup E_2 \cup \dots \cup E_n = \mathcal{S}$. We can also say that the set $\{E_1, E_2, \dots, E_n\}$ is a **partition** of $\mathcal{S}$.

Remarks:

- Some books explicitly require $E_1, E_2, \dots, E_n$ to be non-empty sets; we will not insist on that here, although in practice it will usually be true.
- Understanding the definition of a partition is important in seeing how to calculate the (total) probability of an event A from the conditional probabilities $\mathbb{P}(A|E_k)$ (i.e., the probabilities under certain constraints) and the so-called marginal probabilities $\mathbb{P}(E_k)$.

Theorem 6.2 (Law of total probability). *Suppose that $E_1, E_2, \dots, E_n$ partition $\mathcal{S}$ with $\mathbb{P}(E_k) > 0$ for $k = 1, 2, \dots, n$. Then for any event A we have*

$$\begin{aligned}\mathbb{P}(A) &= \mathbb{P}(A|E_1)\mathbb{P}(E_1) + \mathbb{P}(A|E_2)\mathbb{P}(E_2) + \dots + \mathbb{P}(A|E_n)\mathbb{P}(E_n) \\ &= \sum_{k=1}^n \mathbb{P}(A|E_k)\mathbb{P}(E_k).\end{aligned}$$

Proof:

Let $A_k = A \cap E_k$, for $k = 1, 2, \dots, n$. Note that, by Definition 6.1, the sets $E_1, E_2, \dots, E_n$ are pairwise disjoint and $E_1 \cup E_2 \cup \dots \cup E_n = \mathcal{S}$. Since $A_k \subseteq E_k$ the events $A_1, A_2, \dots, A_n$ are also pairwise disjoint and, furthermore,

$$A_1 \cup A_2 \cup \dots \cup A_n = A \cap (E_1 \cup E_2 \cup \dots \cup E_n) = A \cap \mathcal{S} = A.$$

Hence, by Definition 2.1(c),

$$\mathbb{P}(A) = \mathbb{P}(A_1) + \mathbb{P}(A_2) + \dots + \mathbb{P}(A_n). \quad (6.1)$$

Now, since $\mathbb{P}(E_k) > 0$, we also have (for $k = 1, 2, \dots, n$)

$$\mathbb{P}(A_k) = \mathbb{P}(A \cap E_k) = \frac{\mathbb{P}(A \cap E_k)}{\mathbb{P}(E_k)} \mathbb{P}(E_k) = \mathbb{P}(A|E_k)\mathbb{P}(E_k). \quad (6.2)$$

Substituting (6.2) in (6.1) yields the statement of the theorem. $\square$

In fact, we have already seen an example of the use of Theorem 6.2 in Example 5.8: by definition F and F^c partition $\mathcal{S}$. More generally, the approach is very widely applicable but for different problems one needs to think carefully about what partition to use. The technique is called **conditioning**.

Exercise 6.2: Mind the gap

In a recent survey [YouGov, 11th–16th June 2020], 1088 adults were asked (amongst other questions) if they thought Watford counted as part of London. The following excerpt

from the results shows the number of survey participants in different age categories and the percentage of them saying that they did consider Watford as part of London.

	Age			
	18-24	25-49	50-64	65+
Number of participants	124	544	247	173
Percentage saying Watford is in London	31	34	15	19

What is the probability a randomly-chosen participant thinks Watford is in London?

6.2 Total Probability for Conditional Probabilities

There is an analogue of Theorem 6.2 for conditional probabilities.

Theorem 6.3. *If $E_1, E_2, \dots, E_n$ partition $\mathcal{S}$ with $\mathbb{P}(E_k) > 0$ for $k = 1, 2, \dots, n$, then for events A and B with $\mathbb{P}(B \cap E_k) > 0$ for $k = 1, 2, \dots, n$, we have*

$$\begin{aligned}\mathbb{P}(A|B) &= \mathbb{P}(A|B \cap E_1)\mathbb{P}(E_1|B) + \mathbb{P}(A|B \cap E_2)\mathbb{P}(E_2|B) + \cdots + \mathbb{P}(A|B \cap E_n)\mathbb{P}(E_n|B) \\ &= \sum_{k=1}^n \mathbb{P}(A|B \cap E_k)\mathbb{P}(E_k|B).\end{aligned}$$

Proof:

The idea is to use the definition of conditional probability together with the result we proved in the previous section. Specifically, we start from Definition 4.1

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)},$$

and apply Theorem 6.2 to $\mathbb{P}(A \cap B)$ to yield

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B|E_1)\mathbb{P}(E_1) + \mathbb{P}(A \cap B|E_2)\mathbb{P}(E_2) + \cdots + \mathbb{P}(A \cap B|E_n)\mathbb{P}(E_n)}{\mathbb{P}(B)}.$$

(6.3)

Now, for $k = 1, 2, \dots, n$, we have

$$\begin{aligned}\frac{1}{\mathbb{P}(B)}\mathbb{P}(A \cap B|E_k)\mathbb{P}(E_k) &= \frac{1}{\mathbb{P}(B)} \frac{\mathbb{P}(A \cap B \cap E_k)}{\mathbb{P}(E_k)}\mathbb{P}(E_k) && \text{[by Definition 4.1]} \\ &= \frac{1}{\mathbb{P}(B)}\mathbb{P}(A \cap B \cap E_k) \frac{\mathbb{P}(B \cap E_k)}{\mathbb{P}(B \cap E_k)} && \text{[using } \mathbb{P}(B \cap E_k) > 0\text{]} \\ &= \frac{\mathbb{P}(A \cap B \cap E_k)}{\mathbb{P}(B \cap E_k)} \frac{\mathbb{P}(B \cap E_k)}{\mathbb{P}(B)} \\ &= \mathbb{P}(A|B \cap E_k)\mathbb{P}(E_k|B) && \text{[by Definition 4.1].}\end{aligned}$$

(6.4)

Substituting (6.4) in (6.3) yields the statement of the theorem. $\square$

Example 6.3: Magic coins (revisited)

Consider again the set-up of the magician in Example 5.8.

- (a) Supposing there is a Head on the first toss, determine the probability that the coin is fair.
- (b) Use the result from (a) together with Theorem 6.3 to find the probability of getting a Head on the second toss given there is a Head on the first toss.

Solution:

- (a) We already have $\mathbb{P}(F) = 1/2$, $\mathbb{P}(H_1|F) = 1/2$, and $\mathbb{P}(H_1) = 5/8$ (see Example 5.8). We expect $\mathbb{P}(F|H_1)$ to be different to $\mathbb{P}(H_1|F)$; to calculate the former, we start with the definition of conditional probability. Using the results we already know, we have¹

$$\begin{aligned}
 \mathbb{P}(F|H_1) &= \frac{\mathbb{P}(F \cap H_1)}{\mathbb{P}(H_1)} && \text{[by Definition 4.1]} \\
 &= \frac{\mathbb{P}(H_1|F)\mathbb{P}(F)}{\mathbb{P}(H_1)} && \text{[by Theorem 4.2]} \\
 &= \frac{(1/2) \times (1/2)}{5/8} \\
 &= \frac{2}{5}.
 \end{aligned}$$

- (b) Using Theorem 6.3 with the partition $\{F, F^c\}$ gives

$$\mathbb{P}(H_2|H_1) = \mathbb{P}(H_2|H_1 \cap F)\mathbb{P}(F|H_1) + \mathbb{P}(H_2|H_1 \cap F^c)\mathbb{P}(F^c|H_1). \quad (6.5)$$

We have $\mathbb{P}(F|H_1) = 2/5$ [from (a)] and $\mathbb{P}(F^c|H_1) = 1 - \mathbb{P}(F|H_1) = 3/5$ [see Exercise 4.5]. The other two conditional probabilities on the right-hand side of (6.5) look more complicated at first sight. However, the property of conditional independence discussed in Example 5.8, leads to considerable simplification. Starting once again with the definition of conditional probability, we find

$$\begin{aligned}
 \mathbb{P}(H_2|H_1 \cap F) &= \frac{\mathbb{P}(H_2 \cap H_1 \cap F)}{\mathbb{P}(H_1 \cap F)} && \text{[by Definition 4.1]} \\
 &= \frac{\mathbb{P}(H_2 \cap H_1|F)\mathbb{P}(F)}{\mathbb{P}(H_1|F)\mathbb{P}(F)} && \text{[by Theorem 4.2]} \\
 &= \frac{\mathbb{P}(H_2 \cap H_1|F)}{\mathbb{P}(H_1|F)} \\
 &= \frac{\mathbb{P}(H_2|F)\mathbb{P}(H_1|F)}{\mathbb{P}(H_1|F)} && \text{[by conditional independence of } H_1 \text{ and } H_2 \text{ given } F] \\
 &= \mathbb{P}(H_2|F) \\
 &= \frac{1}{2}.
 \end{aligned}$$

¹We'll see this method again in the next section.

Similarly, by the conditional independence of H_1 and H_2 given F^c , we have

$$\mathbb{P}(H_2|H_1 \cap F^c) = \mathbb{P}(H_2|F^c) = \frac{3}{4}.$$

Hence, putting everything together, we conclude

$$\mathbb{P}(H_2|H_1) = \frac{1}{2} \times \frac{2}{5} + \frac{3}{4} \times \frac{3}{5} = \frac{13}{20}.$$

Exercise 6.4: Magic coins (re-revisited)

Show that the answer to Example 6.3(b) is consistent with the analysis in Example 5.8(c).

6.3 Bayes' Theorem

As Example 6.3 reminded us, $\mathbb{P}(A|B)$ and $\mathbb{P}(B|A)$ are different conditional probabilities. However, as seen in that example, we can determine one from the other *if* we also know $\mathbb{P}(A)$ and $\mathbb{P}(B)$. The theorem is attributed to Thomas Bayes (1702–1761) although not actually published until after his death [Bay63].

Theorem 6.4 (Bayes' theorem). *If A and B are events with $\mathbb{P}(A), \mathbb{P}(B) > 0$, then*

$$\mathbb{P}(B|A) = \frac{\mathbb{P}(A|B)\mathbb{P}(B)}{\mathbb{P}(A)}.$$

Proof:

Starting again from Definition 4.1 (and using that $\mathbb{P}(A), \mathbb{P}(B) > 0$) we have

$$\begin{aligned} \mathbb{P}(B|A) &= \frac{\mathbb{P}(B \cap A)}{\mathbb{P}(A)} \\ &= \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)} \frac{\mathbb{P}(B)}{\mathbb{P}(B)} \\ &= \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} \frac{\mathbb{P}(B)}{\mathbb{P}(A)} \\ &= \mathbb{P}(A|B) \frac{\mathbb{P}(B)}{\mathbb{P}(A)}. \end{aligned}$$

[Instead of multiplying numerator and denominator by $\mathbb{P}(B)$ in the second line, one could use the multiplication rule (Theorem 4.2) for $\mathbb{P}(A \cap B)$.] $\square$

Remarks:

- Bayes' theorem has *many* practical applications.
- We often need to use Theorem 6.2 (law of total probability) to calculate the probability in the denominator of Theorem 6.4.

Example 6.5: Medical test

Suppose there is a disease which 0.1% of the population suffers from. A test for the disease has a 99% chance of giving a positive result for someone with the disease, and only a 0.5% chance of giving a positive result for someone without the disease (a “false positive”). What is the probability that a randomly-chosen person who tests positive actually has the disease?

Solution:

Let us define the events:

D : “The selected person has the disease”;

P : “The test for the selected person is positive”.

We know

$$\mathbb{P}(D) = \frac{1}{1000}, \quad \mathbb{P}(P|D) = \frac{99}{100}, \quad \text{and} \quad \mathbb{P}(P|D^c) = \frac{5}{1000}.$$

We want to compute $\mathbb{P}(D|P)$ so, using Bayes' theorem (Theorem 6.4), we write

$$\mathbb{P}(D|P) = \frac{\mathbb{P}(P|D)\mathbb{P}(D)}{\mathbb{P}(P)}.$$

To calculate $\mathbb{P}(P)$ we can use Theorem 6.2 with the partition $\{D, D^c\}$:

$$\begin{aligned} \mathbb{P}(P) &= \mathbb{P}(P|D)\mathbb{P}(D) + \mathbb{P}(P|D^c)\mathbb{P}(D^c) \\ &= \mathbb{P}(P|D)\mathbb{P}(D) + \mathbb{P}(P|D^c)(1 - \mathbb{P}(D)) \quad [\text{using Proposition 2.2}] \\ &= \frac{99}{100} \times \frac{1}{1000} + \frac{5}{1000} \times \left(1 - \frac{1}{1000}\right) \\ &= \frac{99}{100} \times \frac{1}{1000} + \frac{5}{1000} \times \frac{999}{1000}. \end{aligned}$$

Hence, we find

$$\begin{aligned} \mathbb{P}(D|P) &= \frac{(99/100) \times (1/1000)}{(99/100) \times (1/1000) + (5/1000) \times (999/1000)} \\ &= \frac{990}{5985} \\ &= \frac{22}{133} \\ &= 0.1654 \quad (\text{to 4 decimal places}). \end{aligned}$$

So assuming the test is positive, there is only about 17% chance that the person has the disease. In other words, about 83% of positive tests are false positives. Does this mean the test is useless or is there anything one can do about the problem?

*Exercise 6.6: *Double medical test*

Consider the medical testing scenario of Example 6.5. Supposing a person tests positive in two separate tests, what is the probability that they actually have the disease? State clearly any assumptions you make.

6.4 From Axioms to Applications

The law of total probability and Bayes' theorem are crucially important: not only are they needed for many exam questions but they have wide-reaching applications in real-life. At this point in the course it is worth pausing to see how far we have come. After introducing the language of sets and events, we started in what may have seemed quite an abstract way with Kolmogorov's axioms (Definition 2.1) specifying the properties that probability should have. These simple axioms and the definition of conditional probability (Definition 4.1) are the basic ingredients which have allowed us to build up to proving the more complex results in the present chapter. This illustrates both the beauty and the power of the axiomatic approach to probability. We are also finally in a position to revisit a burning question from Chapter 0...

Example 6.7: Innocent or guilty (revisited)

Look back at Exercise 0.3. Based on the evidence there, a prosecution lawyer argues that there is a 1 in 50,000 chance of the suspect being innocent.

- (a) Why is such an argument flawed?
- (b) Suppose that London has a population of 10 million and the murderer is assumed to be one of these. If there is no evidence against the suspect apart from the fingerprint match then it is reasonable to regard the suspect as a randomly-chosen citizen. Under this assumption, what is the probability the suspect is innocent?
- (c) How does the argument change if one knows that the suspect is one of only 100 people who had access to the building at the time Professor Damson was killed?

Solution:

- (a) Let us write I for the event “the suspect is innocent” and F for the event “the fingerprints of the suspect match those at the crime scene”. The prosecutor notes that $\mathbb{P}(F|I) = 1/50000$ and deduces that $\mathbb{P}(I|F) = 1/50000$. This is nonsense. In general there is no reason why the two conditional probabilities $\mathbb{P}(F|I)$ and $\mathbb{P}(I|F)$ should be equal or even close to equal.
- (b) By Bayes' theorem combined with the law of total probability (using partition $\{I, I^c\}$),

$$\mathbb{P}(I|F) = \frac{\mathbb{P}(F|I)\mathbb{P}(I)}{\mathbb{P}(F)} = \frac{\mathbb{P}(F|I)\mathbb{P}(I)}{\mathbb{P}(F|I)\mathbb{P}(I) + \mathbb{P}(F|I^c)\mathbb{P}(I^c)}.$$

Now, we are told $\mathbb{P}(F|I)$ and it is reasonable to assume that $\mathbb{P}(F|I^c) = 1$ since if the suspect is guilty the fingerprints should certainly match.² The quantity $\mathbb{P}(I)$ in Bayes' theorem is the probability that the suspect is innocent in the absence of any evidence

²In fact, you could set $\mathbb{P}(F|I^c)$ to be any reasonably large probability without changing the general conclusions.

at all. We are told that the suspect should be regarded as a randomly-chosen citizen from a city of 10 million people so $\mathbb{P}(I^c) = 1/10000000$ and $\mathbb{P}(I) = 1 - 1/10000000 = 9999999/10000000$. This gives

$$\begin{aligned}\mathbb{P}(I|F) &= \frac{(1/50000) \times (9999999/10000000)}{(1/50000) \times (9999999/10000000) + 1 \times (1/100000000)} \\ &= \frac{9999999}{9999999 + 50000} \\ &= 0.9950 \quad (\text{to 4 decimal places}).\end{aligned}$$

Hence there is about a 99.5% chance that the suspect is innocent.

- (c) This new information will decrease our initial value of $\mathbb{P}(I)$ which, remember, is the probability that our suspect is innocent before we consider the fingerprint evidence. From the information given it is now reasonable to treat the suspect as randomly-chosen from among the 100 people with access to the building. Hence we take $\mathbb{P}(I^c) = 1/100$ and $\mathbb{P}(I) = 99/100$ which gives

$$\begin{aligned}\mathbb{P}(I|F) &= \frac{(1/50000) \times (99/100)}{(1/50000) \times (99/100) + 1 \times (1/100)} \\ &= \frac{99}{99 + 50000} \\ &= 0.0020 \quad (\text{to 4 decimal places}).\end{aligned}$$

Hence there is now only about a 0.2% chance that the suspect is innocent.

The above example illustrates the so-called “prosecutor’s fallacy” which is not just of academic interest – it has been associated with several high-profile miscarriages of justice.

Exercise 6.8: Wrongful conviction

Find and discuss some real-life examples where a suspect has been convicted on the basis of faulty probabilistic arguments.

6.5 Further Exercises

Exercise 6.9: Football team

Two important members of a football team are injured. Suppose that their recoveries before the match are independent events and each recovers with probability p . If both are able to play then the team has probability $2/3$ of winning the match, if only one of them plays then the probability of winning is $5/12$ and if neither play the probability of winning is $1/6$. Show that the condition $p > 2/3$ guarantees that the match is won with probability greater than $1/2$.

Exercise 6.10: General partitioning

Which of the following partition $\mathcal{S}$ when A and B are arbitrary events? Justify your answers.

- (a) The four events A, A^c, B, B^c ,
- (b) The two events $A, B \setminus A$,
- (c) The four events $A \setminus B, B \setminus A, A \cap B, (A \cup B)^c$,
- (d) The three events $A \cap B, A \triangle B, A^c \cap B^c$,
- (e) The three events $A, B, (A \cup B)^c$.

Exercise 6.11: Lost key

Mimi and Rodolfo are looking for a key in the dark. Suppose that the key may be under the table, behind the bookshelf or in the corridor, and has a $1/3$ chance of being in each of these places. Mimi searches under the table; if the key is there she has a $3/5$ chance of finding it. Rodolfo searches behind the bookshelf; if the key is there he has a $1/5$ chance of finding it.

- (a) Calculate the probability that the key is found.
- (b) Suppose that the key is found. Calculate the conditional probability that it is found by Rodolfo.
- (c) Suppose that the key is not found. Calculate the conditional probability that it is in the corridor.

[Answers: $4/15$, $1/4$, $5/11$]

*Exercise 6.12: *Are you smarter than a pigeon? (revisited)*

Consider the version of the Monty Hall problem presented in Exercise 0.2. Let the cards be numbered 1 to 3 with your initial pick being card 1. Assume that in the case where the ace is card 1, the street performer reveals card 2 with probability p , and card 3 with probability $1 - p$. Otherwise, the card (out of 2 and 3) which is not the ace is always revealed.

- (a) Assuming that you do not switch your choice, compute the probability of winning, conditioned on the performer showing you card 2. Do the same conditioned on the performer showing you card 3.
- (b) Assuming that you do switch once the card has been revealed, compute the probability of winning, conditioned on the performer showing you card 2. Again, do the same conditioned on the performer showing you card 3. Check that regardless of the value of p , and regardless of which card is revealed, deciding to switch is always at least as good as deciding not to switch.
- (c) Use the law of total probability to calculate the probability of winning in both cases (a) and (b). Explain briefly why your result makes sense.

CHAPTER 7

Interlude (and Self-Study Ideas)

7.1 Looking Back and Looking Forward

In several of the situations seen in the first part of the course, we've been interested in numerical values associated with the outcome of an experiment, e.g., the sum of the numbers from two rolls of a die (Exercise 4.9) or the number of Heads when tossing a coin (Exercise 2.11). This leads naturally onto the idea of a random variable which will be the subject of the second half of the course. In order to understand the following chapters, it is crucial that you have a good grasp of the basic structure and definitions we have seen so far. You are thus encouraged to spend some time in self-study and review – it is in your own interests to address any lingering difficulties before going forward.

7.2 Tips for Reading the Lecture Notes

The lecture notes define the examinable content for the course so now would be a good moment to reread Chapters 0 to 6 carefully. As you do that, you may find the following helpful.

- *Concentrate on understanding, not memorizing.* In general, the more you actually understand, the less you need to learn. For an open-book exam, memorizing the notes word-by-word is especially pointless!
- *Read actively, not passively.* Highlight your notes or annotate them as you read and ask yourself questions as if you're an annoying lecturer! For example, when you read a definition, see if you can think of cases which satisfy it and cases which don't. In a proof, check you understand how to get from every line to the next; even better, cover the proof up and see if you can

work it out on your own, uncovering to give yourself a clue where necessary.

- *Watch the accompanying recordings.* If you need further help/explanation on a particular topic then try rewatching the associated recording; again you should do this actively, pausing to check which bits you do or don't understand and then following up on queries as appropriate.
- *Do the examples and exercises.* As you read, you should try to redo the examples/exercises which are already solved and attempt the "Further Exercises" if you have not done so. Remember that the starred exercises are somewhat harder so could be skipped on a first pass of the notes; come back to them later if you're aiming for a high mark. For more advice on examples/exercises, see the next section...

7.3 Tips for Doing Examples/Exercises

Perhaps the most important word in the title of this section is "doing". University study is not a spectator sport; to learn effectively you need to be actively developing your own "mathematical muscles". This means that it is *not* enough to read the solution to an exercise/example or to watch a video – you must try and do it for yourself. Some specific suggestions now follow:

- *As you read a question, highlight the important words/concepts.* For instance, are you told that two events are independent or that something is chosen "at random"? Usually such words are not there by accident but give important clues as to how to proceed.
- *Identify what you know and what you're trying to find.* If a problem is phrased entirely in words, you will usually need to establish some notation before you can do this (e.g., to define events) – often there are many valid notational choices but you need to be clear and consistent.
- *Think about the main tools that might help get from what you know to what you're trying to find.* If you suspect a particular theorem or definition will be useful, make sure you have the exact statement to hand.
- *In a written solution, show all your working.* For a proof (typically a question which says "prove", "show", or "derive") you should try to justify *every* step; for a more applied calculation you should indicate at least the main methods you are using (e.g., "by inclusion-exclusion" or "using Proposition 2.7").

- *Consider how to check your answer.* This is a really important skill since in the real world (and in exams!) there are rarely solutions to consult. For instance, you can ask yourself whether a calculated probability is plausible and whether there might be another way to do the same question.

CHAPTER 8

Introduction to Random Variables

8.1 Concept of a Random Variable

In many real-life experiments, you may be chiefly concerned with some numerical quantity not with the actual outcome. For example, you might care about the sum of the numbers on two dice when playing Monopoly, the number of questions right in a multiple-choice quiz,¹ or the percentage of the electorate voting for a particular candidate. Loosely speaking, a random variable is a “machine” which takes as input an outcome in the sample space and gives as output a single number. More formally, a random variable is a *function*.

Definition 8.1. A *random variable* is a function from $\mathcal{S}$ to $\mathbb{R}$.

Remarks:

- If $\mathcal{S}$ is uncountable then this definition is, in fact, not quite correct. It turns out that some functions are too complicated to regard as random variables (just as some sets are too complicated to regard as events). This subtlety is well beyond the scope of this module and will not concern us at all.
- Random variables are usually denoted by capital letters but should not be confused with events. To aid the distinction, in this course we *generally* use letters from towards the beginning of the alphabet for events and letters from towards the end for random variables.

Exercise 8.1: Real-life random variables

Think of some more examples of random variables in real life. Can you find two different random variables associated with the same experiment?

¹Of course, you *should* care also about understanding which questions you got wrong.

Random variables and events are different concepts but, as we now discuss, events can be described in terms of random variables. If X is a random variable then $\mathbb{P}(X)$ makes no sense as X is not an event. The set of all outcomes $\omega \in \mathcal{S}$ such that $X(\omega) = x$ is, however, an event. Note that we use a lowercase letter (sometimes labelled with a subscript) to denote a particular value of a random variable. We use the shorthand “ $X = x$ ” for the set $\{\omega \in \mathcal{S} : X(\omega) = x\}$. Hence, for example, $\mathbb{P}(X = 2)$ does make sense; it is the probability that the random variable X takes the value two. Similarly, “ $X \leq x$ ” denotes the set $\{\omega \in \mathcal{S} : X(\omega) \leq x\}$ so we can write things like $\mathbb{P}(X \leq 6)$.

Another type of event involves the relationship between the values of different random variables for the same experiment. For example, if Y and Z are both random variables on the same sample space (i.e., functions with the same domain), then “ $Y > 2Z$ ” is shorthand for the set $\{\omega \in \mathcal{S} : Y(\omega) > 2Z(\omega)\}$; in other words, $\mathbb{P}(Y > 2Z)$ is the probability of the event that the value of the random variable Y is more than double the value of the random variable Z . There will be more detailed analysis of cases where several random variables are of interest in Chapters 11 and 12.

Example 8.2: Sum of two dice

Suppose that we roll two fair six-sided dice and record the numbers showing as an ordered pair. Let X denote the sum of the two numbers.

- Describe X as a function, identifying its domain and range.
- Evaluate each of the following or explain why it does not make sense: $X((5, 2))$, $X((6, 4))$, $X((4, 6))$, $X((2, 2))$, $X(5, 6)$, $X(\emptyset)$.
- Determine the following probabilities: $\mathbb{P}(X = 5)$, $\mathbb{P}(X = 3)$, $\mathbb{P}(X = 1)$, $\mathbb{P}(X \leq 2)$, $\mathbb{P}(X \leq 12)$.

Solution:

The sample space is the set of all ordered pairs with elements which are integers between 1 and 6 (inclusive), i.e., $\mathcal{S} = \{(j, k) : j, k \in \{1, 2, 3, 4, 5, 6\}\}$, and $|\mathcal{S}| = 36$ (cf. Exercise 1.3, amongst others).

- To get the sum of the two dice, then for an outcome (j, k) we must write down $j + k$. This recipe is a function $(j, k) \mapsto j + k$ from $\mathcal{S}$, to the set of integers $\mathbb{Z}$ (or if you prefer the set of natural numbers $\mathbb{N}$). With this definition, we see that the domain is $\mathcal{S}$ and the range, i.e., the set of values that X can actually take is $\{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$.
- The first four function values are easily obtained:

$$X((5, 2)) = 7, \quad X((6, 4)) = 10 \quad X((4, 6)) = 10, \quad \text{and} \quad X((2, 2)) = 4.$$

[Note that the function X , in common with most random variables, is not injective.] However, rather sneakily, $X(5, 6)$ does not make sense as the input must be a pair

(an element of $\mathcal{S}$), not just two numbers. Similarly, $X(\emptyset)$ does not make sense as the empty set is not an element of $\mathcal{S}$ here.

- (c) The event “ $X = 5$ ” contains all outcomes (pairs) such that the sum of the two rolls is five, i.e., it is the set $\{(1, 4), (2, 3), (3, 2), (4, 1)\}$. Since the cardinality of this set is four and all outcomes are equally likely, we have $\mathbb{P}(X = 5) = 4/36 = 1/9$. The other probabilities are similarly determined:

$$\begin{aligned}\mathbb{P}(X = 3) &= \mathbb{P}(\{(1, 2), (2, 1)\}) = \frac{2}{36} = \frac{1}{18}, \\ \mathbb{P}(X = 1) &= \mathbb{P}(\emptyset) = 0, \\ \mathbb{P}(X \leq 2) &= \mathbb{P}(X = 2) = \mathbb{P}(\{(1, 1)\}) = \frac{1}{36}, \\ \mathbb{P}(X \leq 12) &= \mathbb{P}(\mathcal{S}) = 1.\end{aligned}$$

Exercise 8.3: Head count

Suppose we toss a fair coin three times and denote outcomes in the sample space $\mathcal{S}$ by listing, in order, the observed Heads (h) and Tails (t).

- (a) Let X be the random variable counting the number of Heads, and Y be the random variable counting the number of Tails.
- (i) State the range of the functions X and Y , i.e., list the values they can take.
 - (ii) Evaluate $X(hht)$ and $Y(hht)$.
- (b) Let Z be another random variable defined as $Z = \max\{X, Y\}$.
- (i) State the range of the function Z , i.e., list the values it can take.
 - (ii) Evaluate $Z(hht)$.
- (c) Determine the probability that we see more Tails than Heads.

8.2 Distributions of Discrete Random Variables

Recall from the previous section that we denote a random variable by an uppercase letter and a particular value of the random variable by a lowercase letter (sometimes labelled with a subscript). We can classify random variables according to the set of values that they take.

Definition 8.2. A random variable X is **discrete** if the set of values that X takes is either finite or countably infinite.

In this case we can label the possible values x_1, x_2, x_3 , etc. and use x_k for a generic value. In this course we only really consider such **discrete random variables**.

The case of **continuous random variables** is more complicated² but practically important – you will certainly encounter it in future probability/statistics courses.

Exercise 8.4: Classifying real-life random variables

Look back at the examples you suggested in Exercise 8.1 and identify whether each of them is a discrete or a continuous random variable.

Now let us turn to the central question of how to describe the probability distribution of a discrete random variable. We need to associate probabilities to the events of the random variable taking each possible value; this information is encoded in the so-called **probability mass function**.

Definition 8.3. *The **probability mass function** (p.m.f.) of a discrete random variable X is the function which given input x has output $\mathbb{P}(X = x)$:*

$$x \mapsto \mathbb{P}(X = x).$$

Remarks:

- The p.m.f. is sometimes denoted by p , i.e., we define $p(x) = \mathbb{P}(X = x)$; we must have $p(x_k) > 0$ if x_k is a possible value of the random variable.
- Do not confuse the lower case p , which is the “name” of a function, with the $\mathbb{P}$ for probability; $p(X = x)$ and $\mathbb{P}(x)$ are both wrong notation.
- In situations with more than one random variable we can label each p.m.f. with a subscript, e.g., $p_X(x) = \mathbb{P}(X = x)$ and $p_Y(y) = \mathbb{P}(Y = y)$.

In the next section we will return to general properties of the p.m.f.; for the present we note that it can be given either by a closed-form expression or by a table, as illustrated in the following examples and exercises.

Example 8.5: Sum of two dice (revisited)

Determine the probability mass function of the random variable X (sum of two dice rolls) from Example 8.2.

Solution:

The random variable X takes values in the set $\{2, 3, 4, \dots, 12\}$. Since the dice are fair, we can calculate probabilities from the cardinalities of the associated events just as we did

²If X is a continuous random variable, what can you say about $\mathbb{P}(X = x)$?

previously in Example 8.2:

$$\begin{aligned}
 \mathbb{P}(X = 2) &= \frac{|\{(1, 1)\}|}{36} = \frac{1}{36}, \\
 \mathbb{P}(X = 3) &= \frac{|\{(1, 2), (2, 1)\}|}{36} = \frac{2}{36} = \frac{1}{18}, \\
 \mathbb{P}(X = 4) &= \frac{|\{(1, 3), (2, 2), (3, 1)\}|}{36} = \frac{3}{36} = \frac{1}{12}, \\
 \mathbb{P}(X = 5) &= \frac{|\{(1, 4), (2, 3), (3, 2), (4, 1)\}|}{36} = \frac{4}{36} = \frac{1}{9}, \\
 \mathbb{P}(X = 6) &= \frac{|\{(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)\}|}{36} = \frac{5}{36}, \\
 \mathbb{P}(X = 7) &= \frac{|\{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}|}{36} = \frac{6}{36} = \frac{1}{6}, \\
 \mathbb{P}(X = 8) &= \frac{|\{(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)\}|}{36} = \frac{5}{36}, \\
 \mathbb{P}(X = 9) &= \frac{|\{(3, 6), (4, 5), (5, 4), (6, 3)\}|}{36} = \frac{4}{36} = \frac{1}{9}, \\
 \mathbb{P}(X = 10) &= \frac{|\{(4, 6), (5, 5), (6, 4)\}|}{36} = \frac{3}{36} = \frac{1}{12}, \\
 \mathbb{P}(X = 11) &= \frac{|\{(5, 6), (6, 5)\}|}{36} = \frac{2}{36} = \frac{1}{18}, \\
 \mathbb{P}(X = 12) &= \frac{|\{(6, 6)\}|}{36} = \frac{1}{36}.
 \end{aligned}$$

These results can simply be displayed in a table of the p.m.f.:

x	2	3	4	5	6	7	8	9	10	11	12
$\mathbb{P}(X = x)$	1/36	1/18	1/12	1/9	5/36	1/6	5/36	1/9	1/12	1/18	1/36

*Exercise 8.6: *Sum of two dice (re-revisited)*

Find a formula for the p.m.f. in Example 8.2. [Hint: It may help to rewrite all the probabilities with the same denominator.]

Example 8.7: Waiting for a Tail

Suppose you toss a fair coin until it comes up Tails. Determine the probability mass function of the random variable T which counts the number of tosses.

Solution:

Denoting as usual a Head by h and a Tail by t , the sample space can be written as $\{t, ht, hht, hhht, \dots\}$ where, e.g., $hhht$ means three Heads followed by a Tail. The random variable T takes values $1, 2, 3, 4, \dots$, i.e., values from the countably infinite set of natural numbers. It is easy to see that the function T is injective so to calculate the p.m.f., we have to calculate the probabilities of simple events. Since the coin is fair and different tosses are physically unrelated (so we can assume independence and multiply probabilities) we

have

$$\begin{aligned}\mathbb{P}(T = 1) &= \mathbb{P}(\{t\}) = \frac{1}{2}, \\ \mathbb{P}(T = 2) &= \mathbb{P}(\{ht\}) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}, \\ \mathbb{P}(T = 3) &= \mathbb{P}(\{hht\}) = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}, \\ &\vdots\end{aligned}$$

It is easy to spot the pattern and we can write the p.m.f. as a table

n	1	2	3	4	...
$\mathbb{P}(T = n)$	1/2	1/4	1/8	1/16	...

or as a compact formula

$$\mathbb{P}(T = n) = \begin{cases} \frac{1}{2^n} & \text{for } n \in \mathbb{N} \\ 0 & \text{otherwise.} \end{cases}$$

[Note that the “0 otherwise” line is sometimes not written in p.m.f. formulae, it being simply assumed that the probability is zero for values of the random variable which are not explicitly listed.]

Exercise 8.8: Balls in a bag

Four balls are randomly selected, without replacement, from a bag that contains 10 balls numbered from 1 to 10. Let X denote the largest number selected.

- List the values X takes.
- Calculate $\mathbb{P}(X = 5)$.
- Write a formula for the p.m.f. of X .
- Calculate $\mathbb{P}(X > 5)$.

8.3 Properties of the Probability Mass Function

Since the values assigned by the probability mass function are probabilities, they must obey Kolmogorov’s axioms. In particular, this means they must add up to one.

Proposition 8.4. *If X is a discrete random variable which takes values $x_1, x_2, x_3, \dots$, then*

$$\sum_k \mathbb{P}(X = x_k) = \mathbb{P}(X = x_1) + \mathbb{P}(X = x_2) + \mathbb{P}(X = x_3) + \dots = 1$$

where the sum is over all values which X takes (a finite or infinite set).

Proof:

The random variable X takes the values x_k (with $k = 1, 2, 3, \dots$); we let A_k be the event “ $X = x_k$ ”. The A_k ’s are pairwise disjoint which can easily be proved by contradiction. [If A_i and A_j were not disjoint for some $i \neq j$, then A_i and A_j would contain at least one common element, say ω , but that would mean that $X(\omega) = x_i$ and $X(\omega) = x_j$ which is impossible if $i \neq j$.] Furthermore $A_1 \cup A_2 \cup \dots = \mathcal{S}$ since, for any $\omega \in \mathcal{S}$, $X(\omega)$ takes some value. [$X(\omega) = x_i$ means $\omega \in A_i$ so there is no $\omega \in \mathcal{S}$ which is not in one of the A_k ’s.] In other words, the A_k ’s partition the sample space; together with Kolmogorov’s axioms, this yields

$$\begin{aligned}
 \sum_k \mathbb{P}(X = x_k) &= \mathbb{P}(X = x_1) + \mathbb{P}(X = x_2) + \mathbb{P}(X = x_3) + \dots \\
 &= \mathbb{P}(A_1) + \mathbb{P}(A_2) + \mathbb{P}(A_3) + \dots \\
 &= \mathbb{P}(A_1 \cup A_2 \cup A_3 \cup \dots) \quad [\text{using Definition 2.1(c)}] \\
 &= \mathbb{P}(\mathcal{S}) \\
 &= 1 \quad [\text{using Definition 2.1(b)}]
 \end{aligned}$$

and so the result is proved. □

Remarks:

- In the remainder of this course we will often use the Σ -notation for sums; this keeps things more compact but if it helps to see what’s going on, you can always write out terms “long hand”.
- We assume that any infinite sums we encounter here can be straightforwardly treated in a similar way to finite sums; you will learn much more about the subtleties of infinite series in the module Calculus II. In particular, we use the fact that Kolmogorov’s third axiom, Definition 2.1(c), also holds for a countably infinite number of events.
- Proposition 8.4 provides a good way to check that a calculated p.m.f. is at least plausible.

Example 8.9: Checking probability mass functions

Check that Proposition 8.4 holds for the p.m.f. of X in Example 8.5, and for the p.m.f. of T in Example 8.7

Solution:

For the p.m.f. of X , we have to check a finite sum:

$$\begin{aligned}
 \sum_{x=2}^{12} \mathbb{P}(X = x) &= \mathbb{P}(X = 2) + \mathbb{P}(X = 3) + \mathbb{P}(X = 4) + \mathbb{P}(X = 5) + \mathbb{P}(X = 6) + \mathbb{P}(X = 7) \\
 &\quad + \mathbb{P}(X = 8) + \mathbb{P}(X = 9) + \mathbb{P}(X = 10) + \mathbb{P}(X = 11) + \mathbb{P}(X = 12) \\
 &= \frac{1}{36} + \frac{2}{36} + \frac{3}{36} + \frac{4}{36} + \frac{5}{36} + \frac{6}{36} + \frac{7}{36} + \frac{6}{36} + \frac{5}{36} + \frac{4}{36} + \frac{3}{36} + \frac{2}{36} \\
 &= \frac{36}{36} \\
 &= 1.
 \end{aligned}$$

For the p.m.f. of T , we have to check an infinite sum:

$$\begin{aligned}
 \sum_{n=1}^{\infty} \mathbb{P}(T = n) &= \mathbb{P}(T = 1) + \mathbb{P}(T = 2) + \mathbb{P}(T = 3) + \mathbb{P}(T = 4) + \cdots \\
 &= \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots \\
 &= \frac{1}{2} \left(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots \right) \\
 &= \frac{1}{2} \times \frac{1}{1 - (1/2)} \\
 &= \frac{1/2}{1/2} \\
 &= 1
 \end{aligned}$$

[We have used here the formula for the sum of a geometric series, see Exercise 8.14.] Hence, in both cases Proposition 8.4 holds, as of course it must.

The fact that the events “ $X = x_k$ ” are pairwise disjoint means that we can find the probabilities of other events by summing values of the probability mass function. For example, if the random variable X takes values in the integers, then $\mathbb{P}(0 \leq X < 3) = \mathbb{P}(X = 0) + \mathbb{P}(X = 1) + \mathbb{P}(X = 2)$. We can also find the p.m.f. of another random variable, say Y , which is itself a function of X , by considering which values of X are mapped to which values of Y .

Exercise 8.10: From X to Y

A random variable X has the following probability mass function:

x	-2	-1	0	1	2
$\mathbb{P}(X = x)$	1/10	2/5	1/4	1/5	1/20

Let Y be a new random variable defined by $Y = X^2 + 4$.

- List the values Y takes.
- Find the probability mass function of Y .

We conclude this chapter by remarking that a closely related function to the probability mass function is the **cumulative distribution function** (c.d.f.), usually denoted by F , which given input x has output $\mathbb{P}(X \leq x)$. This plays an important role in more advanced probability theory.³

8.4 Further Exercises

Exercise 8.11: Probability practice

Calculate the following probabilities for the random variable X of Exercise 8.10: $\mathbb{P}(X = 2)$, $\mathbb{P}(X = 3)$, $\mathbb{P}(X \leq 1)$, $\mathbb{P}(X^2 < 2)$.

[Answers: $1/20$, 0 , $19/20$, $17/20$]

Exercise 8.12: Choosing your marbles

A bag contains six red marbles and two blue marbles. You choose five at random without replacement. Let B be the number of blue marbles you end up with and R be the number of red marbles you end up with. Find the probability mass function of B . Without doing any more calculations, write down the probability mass function of R .

Exercise 8.13: Unfair tossing

A coin which has probability p of coming up Heads is tossed three times. Let X be the number of Heads observed.

- List the values which the random variable X takes.
- Compute the probability mass function of X .
- Confirm the statement in Proposition 8.4 for the p.m.f. calculated in (b).

*Exercise 8.14: *Geometric series*

- Let $z \neq 1$ be a real number and n be a positive integer. Show that

$$1 + z + z^2 + z^3 + \cdots + z^{n-1} = \sum_{k=0}^{n-1} z^k = \frac{1 - z^n}{1 - z}.$$

[Hint: Define $S_n = 1 + z + z^2 + z^3 + \cdots + z^{n-1}$ and take the difference of S_n and zS_n .]

- Now assume that $|z| < 1$. By taking the limit $n \rightarrow \infty$, derive the sum of the geometric series

$$1 + z + z^2 + z^3 + \cdots = \sum_{k=0}^{\infty} z^k = \frac{1}{1 - z}.$$

- Suppose you toss the coin of Exercise 8.13 until either a Head appears or a total of n Tails has been seen. Let the random variable T be the number of tosses made.

³For continuous (non-discrete) random variables, one can still define the c.d.f. but the p.m.f. is replaced by a **probability density function** (p.d.f.).

Determine the probability mass function of T and use your result from part (a) to verify that Proposition 8.4 holds.

*Exercise 8.15: **Coin games*

- (a) Let the random variable Y_n be the number of Tails appearing when a fair coin is tossed n times. Determine the probability mass function of Y_n and hence deduce a closed-form expression for the sum

$$\sum_{k=0}^n \binom{n}{k}.$$

- (b) You play a game where you first choose a positive integer n and then toss a fair coin n times. You win a prize if you get exactly two Tails. How should you choose n to maximize your chances of winning? What is the probability of winning with an optimal choice of n ?

CHAPTER 9

Expectation and Variance

9.1 Expected Value

In this chapter we start thinking about how to characterize properties of the distributions of (discrete) random variables. Let us begin by imagining that we toss a fair coin 100 times. How many Heads should we expect to see? We'll return to this question when we introduce the binomial distribution in the next chapter but intuition probably already gives you a good idea of the answer. This thought experiment illustrates the idea of expected value or **expectation** which is defined as follows.

Definition 9.1. *If X is a discrete random variable which takes values $x_1, x_2, x_3, \dots$, then the **expectation** of X (or the expected value of X) is defined by*

$$\begin{aligned}\mathbb{E}(X) &= \sum_k x_k \mathbb{P}(X = x_k) \\ &= x_1 \mathbb{P}(X = x_1) + x_2 \mathbb{P}(X = x_2) + x_3 \mathbb{P}(X = x_3) + \dots\end{aligned}$$

Remarks:

- The expectation is sometimes called the **mean** and sometimes denoted by μ (the Greek letter mu).
- The sum again ranges over all the possible values of the random variable; there are some further subtleties in the case of infinite sums (largely beyond the scope of this course).
- The expected value does *not* have to be one of the possible values of the random variable.

Example 9.1: Expectation of a die

Let W be the number shown on rolling a fair six-sided die. Find $\mathbb{E}(W)$.

Solution:

The random variable W obviously has p.m.f. $\mathbb{P}(W = w) = 1/6$ for $w \in \{1, 2, 3, 4, 5, 6\}$ so

$$\begin{aligned}\mathbb{E}(W) &= \sum_{w=1}^6 w\mathbb{P}(W = w) \\ &= 1 \times \frac{1}{6} + 2 \times \frac{1}{6} + 3 \times \frac{1}{6} + 4 \times \frac{1}{6} + 5 \times \frac{1}{6} + 6 \times \frac{1}{6} \\ &= \frac{7}{2}.\end{aligned}$$

Exercise 9.2: Expectation of the sum of two dice

Let X be the sum of the numbers when two fair six-sided dice are rolled. Use the p.m.f. calculated in Example 8.5 to show that $\mathbb{E}(X) = 7$. Why does this result make sense in the light of Example 9.1? [We will see this more clearly in Chapter 11.]

Example 9.1 and Exercise 9.2 clearly illustrate that the expectation may or may not be one of the values the random variable can actually take: it is possible for the sum of the numbers on two dice to be seven (in fact the most likely value seen) but it is certainly not possible to roll 3.5 on a single die! What then can we say about the expectation in general? Is there any way to check our calculations? Well, you would (hopefully!) have been surprised if you had calculated $\mathbb{E}(W)$ for the single die as 7.1 or $\mathbb{E}(X)$ for the two dice as 1.5. This leads to the following proposition.

Proposition 9.2. *If $m \leq X(\omega) \leq M$ for all $\omega \in \mathcal{S}$, then*

$$m \leq \mathbb{E}(X) \leq M.$$

Proof:

If every value x_k ($k = 1, 2, 3, \dots$) that X takes is less than or equal to M , $x_k \leq M$, we have that $x_k\mathbb{P}(X = x_k) \leq M\mathbb{P}(X = x_k)$ [since probabilities are non-negative by Definition 2.1(a)] and, using also Proposition 8.4,

$$\mathbb{E}(X) = \sum_k x_k\mathbb{P}(X = x_k) \leq \sum_k M\mathbb{P}(X = x_k) = M \sum_k \mathbb{P}(X = x_k) = M.$$

Similarly, if every value that X takes is greater than or equal to m , $x_k \geq m$, we have that

$$\mathbb{E}(X) = \sum_k x_k\mathbb{P}(X = x_k) \geq m \sum_k \mathbb{P}(X = x_k) = m.$$

Hence, we have $m \leq \mathbb{E}(X) \leq M$ as required. $\square$

This proposition is admittedly less helpful in the case of a random variable taking values from a countably infinite set. Determining the expectation in those cases involves an infinite sum.

*Exercise 9.3: *Waiting in expectation*

Let T be the number of tosses of a fair coin up to (and including) the first time you see a Tail. Use the p.m.f. found in Example 8.7 to calculate $\mathbb{E}(T)$.

In general, it turns out that if a random variable can take infinitely many values, its expectation may be infinite or even not well defined.¹ From the point of view of the present course, this is a complication that need not trouble you (but see Exercise 9.17 for a non-examinable challenge).

9.2 Expectation of a Function of a Random Variable

We now turn our attention to finding the expected value of a function of a random variable. If we know the p.m.f. of a random variable X , how can we find the expectation of a function of X , say $f(X)$?

Example 9.4: From X to Y (revisited)

Find the expectation of the random variable $Y = X^2 + 4$ where X has the p.m.f. given in Exercise 8.10.

Solution:

Since we already calculated the p.m.f. of Y we can simply use that to determine $\mathbb{E}(Y)$:

$$\begin{aligned}\mathbb{E}(Y) &= 4 \times \mathbb{P}(Y = 4) + 5 \times \mathbb{P}(Y = 5) + 8 \times \mathbb{P}(Y = 8) \\ &= 4 \times \frac{1}{4} + 5 \times \frac{3}{5} + 8 \times \frac{3}{20} \\ &= \frac{26}{5}.\end{aligned}$$

Notice, however, that we can write this calculation in another way; with $f(X) = X^2 + 4$, we have

$$\begin{aligned}\mathbb{E}(Y) &= 4 \times \mathbb{P}(Y = 4) + 5 \times \mathbb{P}(Y = 5) + 8 \times \mathbb{P}(Y = 8) \\ &= 4 \times \mathbb{P}(X = 0) + 5 \times [\mathbb{P}(X = -1) + \mathbb{P}(X = 1)] + 8 \times [\mathbb{P}(X = -2) + \mathbb{P}(X = 2)] \\ &= 8 \times \mathbb{P}(X = -2) + 5 \times \mathbb{P}(X = -1) + 4 \times \mathbb{P}(X = 0) + 5 \times \mathbb{P}(X = 1) + 8 \times \mathbb{P}(X = 2) \\ &= f(-2)\mathbb{P}(X = -2) + f(-1)\mathbb{P}(X = -1) + f(0)\mathbb{P}(X = 0) + f(1)\mathbb{P}(X = 1) + f(2)\mathbb{P}(X = 2).\end{aligned}$$

You should check that using the values in the p.m.f. of X (see Exercise 8.10) again results in $\mathbb{E}(Y) = 26/5$.

¹This is related to the question of whether or not an infinite series converges which you can learn about in “Calculus II” and similar modules.

The above example illustrates a general principle which is stated in the next proposition.

Proposition 9.3. *If f is a real-valued function defined on the range of a discrete random variable X , then*

$$\begin{aligned}\mathbb{E}(f(X)) &= \sum_k f(x_k) \mathbb{P}(X = x_k) \\ &= f(x_1) \mathbb{P}(X = x_1) + f(x_2) \mathbb{P}(X = x_2) + f(x_3) \mathbb{P}(X = x_3) + \cdots\end{aligned}$$

where the sum ranges over all possible values x_k of X .

The proof is omitted; it is straightforward but requires some slightly cumbersome notation. Proposition 9.3 has many useful consequences.

Example 9.5: Useful expectations

Let X be a discrete random variable, taking the values $x_1, x_2, x_3, \dots$, and c be a constant (a real number). Show that:

- (a) $\mathbb{E}(X + c) = \mathbb{E}(X) + c$,
- (b) $\mathbb{E}(cX) = c\mathbb{E}(X)$.

[You may use without proof the series properties $\sum_k (a_k + b_k) = \sum_k a_k + \sum_k b_k$ and $\sum_k ca_k = c \sum_k a_k$.]

Solution:

- (a) From Proposition 9.3, we have

$$\begin{aligned}\mathbb{E}(X + c) &= \sum_k (x_k + c) \mathbb{P}(X = x_k) \\ &= \sum_k x_k \mathbb{P}(X = x_k) + \sum_k c \mathbb{P}(X = x_k) \\ &= \mathbb{E}(X) + c \sum_k \mathbb{P}(X = x_k) \quad [\text{using Definition 9.1}] \\ &= \mathbb{E}(X) + c \times 1 \quad [\text{using Proposition 8.4}] \\ &= \mathbb{E}(X) + c.\end{aligned}$$

- (b) Similarly, Proposition 9.3 yields

$$\begin{aligned}\mathbb{E}(cX) &= \sum_k cx_k \mathbb{P}(X = x_k) \\ &= c \sum_k x_k \mathbb{P}(X = x_k) \\ &= c\mathbb{E}(X) \quad [\text{using Definition 9.1}].\end{aligned}$$

Exercise 9.6: Profit margins

Suppose that the Great Expectations restaurant prepares four takeaway meals in advance each evening at a cost of £4 each. Each takeaway meal is sold for £9 but any unsold meal goes to waste. Let X denote the number of these meals sold in a given evening and Y denote the profit made on them (in pounds). The restaurant owner observes that $\mathbb{P}(X = 0) = 1/12$, $\mathbb{P}(X = 3) = 1/6$, $\mathbb{P}(X = 4) = 1/8$, and $\mathbb{E}(X) = 2$.

- (a) Find the p.m.f. of X .
- (b) Determine $\mathbb{E}(Y)$.

9.3 Moments and Variance

An important special case of the treatment in the previous section is expectations of X^n (where n is a natural number); these expectations are the **moments** of the random variable X .

Definition 9.4. *The n th moment of the random variable X is the expectation $\mathbb{E}(X^n)$.*

Such expectations can easily be calculated for discrete random variables using Proposition 9.3.² Their values give information about the “shape” of the probability mass function. In particular, the second moment is related to the **variance** which quantifies the spread of the distribution.

Definition 9.5. *If X is a discrete random variable which takes values $x_1, x_2, x_3, \dots$, then the **variance** of X is defined by*

$$\begin{aligned}\text{Var}(X) &= \sum_k [x_k - \mathbb{E}(X)]^2 \mathbb{P}(X = x_k) \\ &= [x_1 - \mathbb{E}(X)]^2 \mathbb{P}(X = x_1) + [x_2 - \mathbb{E}(X)]^2 \mathbb{P}(X = x_2) \\ &\quad + [x_3 - \mathbb{E}(X)]^2 \mathbb{P}(X = x_3) + \dots\end{aligned}$$

Remarks:

- Armed with Proposition 9.3, we see that $\text{Var}(X) = \mathbb{E}([X - \mathbb{E}(X)]^2)$, i.e., it is the expectation of the square of the difference between X and $\mathbb{E}(X)$.
- The variance measures how sharply concentrated X is about $\mathbb{E}(X)$, with a small variance meaning sharply concentrated and a large variance meaning spread out.

²For random variables taking infinitely many values, the moments may again be infinite or not well defined; that subtlety does not concern us here.

- Since the square of any real number is non-negative and the values of the p.m.f. are also non-negative [from Definition 2.1(a) of course], it is clear that $\text{Var}(X) \geq 0$.
- The square root of the variance is called the **standard deviation**. Mathematically it is usually more convenient to work with the variance than the standard deviation.

The concept of the variance as a measure of spread is illustrated by the following example.

Example 9.7: Competing investments

Let X be the amount (in pounds) you get from one investment and Y be the amount (in pounds) you get from a second investment. Suppose that X takes value 99 with probability $1/2$ and value 101 with probability $1/2$ while Y takes value 90 with probability $1/2$ and value 110 with probability $1/2$.

- Compare $\mathbb{E}(X)$ and $\mathbb{E}(Y)$.
- Compare $\text{Var}(X)$ and $\text{Var}(Y)$.

Solution:

- From Definition 9.1 we have

$$\mathbb{E}(X) = 99 \times \frac{1}{2} + 101 \times \frac{1}{2} = 100,$$

and

$$\mathbb{E}(Y) = 90 \times \frac{1}{2} + 110 \times \frac{1}{2} = 100.$$

[These results are also obvious from a symmetry argument.] Hence the expectations of the amounts gained from the two investments are the same.

- From Definition 9.5 we have

$$\text{Var}(X) = (99 - 100)^2 \times \frac{1}{2} + (101 - 100)^2 \times \frac{1}{2} = (1)^2 = 1,$$

and

$$\text{Var}(Y) = (90 - 100)^2 \times \frac{1}{2} + (110 - 100)^2 \times \frac{1}{2} = (10)^2 = 100.$$

So the variance of Y is much bigger than that of X ; we can interpret this as the second investment being, in some sense, riskier.

There is also a useful alternative formula expressing the variance in terms of the first two moments.

Proposition 9.6. *If X is a discrete random variable then*

$$\text{Var}(X) = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2.$$

Proof:

As usual we write $x_1, x_2, x_3, \dots$ for the possible values of X . Starting from Definition 9.5 and the fact that $[x_k - \mathbb{E}(X)]^2 = (x_k)^2 - 2\mathbb{E}(X)x_k + [\mathbb{E}(X)]^2$, we have

$$\begin{aligned}
 \text{Var}(X) &= \sum_k [x_k - \mathbb{E}(X)]^2 \mathbb{P}(X = x_k) \\
 &= \sum_k (x_k)^2 \mathbb{P}(X = x_k) - \sum_k 2\mathbb{E}(X)x_k \mathbb{P}(X = x_k) + \sum_k [\mathbb{E}(X)]^2 \mathbb{P}(X = x_k) \\
 &= \mathbb{E}(X^2) - 2\mathbb{E}(X) \sum_k x_k \mathbb{P}(X = x_k) + [\mathbb{E}(X)]^2 \sum_k \mathbb{P}(X = x_k) \quad [\text{from Proposition 9.3}] \\
 &= \mathbb{E}(X^2) - 2\mathbb{E}(X) \times \mathbb{E}(X) + [\mathbb{E}(X)]^2 \times 1 \quad [\text{using Definition 9.1 and Proposition 8.4}] \\
 &= \mathbb{E}(X^2) - [\mathbb{E}(X)]^2. \quad \square
 \end{aligned}$$

Remarks:

- You can remember this expression for the variance as “the mean of the square minus the square of the mean”.
- Since $\text{Var}(X) \geq 0$, we must have $\mathbb{E}(X^2) \geq [\mathbb{E}(X)]^2$.

Example 9.8: Variance of a die

Let W be the number shown on rolling a fair six-sided die (as in Example 9.1). Find $\text{Var}(W)$.

Solution:

Using the formula in Proposition 9.6 we have

$$\begin{aligned}
 \text{Var}(W) &= \mathbb{E}(W^2) - [\mathbb{E}(W)]^2 \\
 &= \sum_{w=1}^6 w^2 \mathbb{P}(W = w) - \left(\frac{7}{2}\right)^2 \quad [\text{using result of Example 9.1}] \\
 &= 1^2 \times \frac{1}{6} + 2^2 \times \frac{1}{6} + 3^2 \times \frac{1}{6} + 4^2 \times \frac{1}{6} + 5^2 \times \frac{1}{6} + 6^2 \times \frac{1}{6} - \left(\frac{7}{2}\right)^2 \\
 &= \frac{91}{6} - \frac{49}{4} \\
 &= \frac{35}{12}.
 \end{aligned}$$

Exercise 9.9: Variance of the sum of two dice

Let X be the sum of the numbers when two fair six-sided dice are rolled. Use the p.m.f. calculated in Example 8.5 to calculate $\text{Var}(X)$ using both Definition 9.5 and Proposition 9.6. Which method do you find easiest? [Answer (to first part): $35/6$]

Exercise 9.10: Why oh why?

Let Y be a discrete random variable with $\mathbb{E}(Y) = 2$ and $\text{Var}(Y) = 6$. Find $\mathbb{E}(2Y^2)$.

9.4 Useful Properties of Expectation and Variance

Linear functions of random variables are often found in applications. In this section we summarize the particular properties of the expectation and variance in such cases.

Proposition 9.7. *If $a, b \in \mathbb{R}$ and X is a discrete random variable, then*

$$\mathbb{E}(aX + b) = a\mathbb{E}(X) + b.$$

Remarks:

- The proof of Proposition 9.7 is left as an exercise; it essentially just combines the proofs of Example 9.5.
- Setting $a = 0$ yields the special case $\mathbb{E}(b) = b$. This corresponds to the expectation of a so-called degenerate random variable which takes value b with probability one.
- The property of “linearity of expectation” also extends to sums of higher moments, for example, $\mathbb{E}(3X^2 - 4X + 5) = 3\mathbb{E}(X^2) - 4\mathbb{E}(X) + 5$. We shall see further generalizations in Chapter 11.

Proposition 9.8. *If $a, b \in \mathbb{R}$ and X is a discrete random variable, then*

$$\text{Var}(aX + b) = a^2\text{Var}(X).$$

Proof:

Starting from Definition 9.5, we have

$$\begin{aligned} \text{Var}(aX + b) &= \mathbb{E}([aX + b - \mathbb{E}(aX + b)]^2) \\ &= \mathbb{E}([aX + b - a\mathbb{E}(X) - b]^2) \quad [\text{using Proposition 9.7}] \\ &= \mathbb{E}(a^2[X - \mathbb{E}(X)]^2) \\ &= a^2\mathbb{E}([X - \mathbb{E}(X)]^2) \quad [\text{using Proposition 9.7}] \\ &= a^2\text{Var}(X). \end{aligned} \quad \square$$

Remarks:

- Note that adding a constant to a random variable does not change the variance; this is intuitively reasonable as the spread of the distribution is unchanged by a shift.
- An important special case is $\text{Var}(b) = 0$.

Example 9.11: Empirical mean

Let X once again be the sum of the numbers when two fair six-sided dice are rolled. Now define Y as the “empirical mean value” of the rolls, i.e., $Y = X/2$. Find the expectation and variance of Y .

Solution:

Using the results of Exercises 9.2 and 9.9, together with Propositions 9.7 and 9.8, we have

$$\mathbb{E}(Y) = \mathbb{E}\left(\frac{X}{2}\right) = \frac{1}{2}\mathbb{E}(X) = \frac{7}{2},$$

and

$$\text{Var}(Y) = \text{Var}\left(\frac{X}{2}\right) = \left(\frac{1}{2}\right)^2 \text{Var}(X) = \frac{35/6}{4} = \frac{35}{24}.$$

Notice that $\mathbb{E}(Y)$ is the same as the expectation of the number on a single die (Example 9.1) but $\text{Var}(Y)$ is *smaller* than the variance of the number on a single die (Example 9.8). Can you understand why?

*Exercise 9.12: *Puzzling p.m.f.*

Suppose that the discrete random variable X has $\mathbb{E}(3X + 7) = 10$ and $\text{Var}(3X + 7) = 36$. Give one example of a possible p.m.f. for X .

In the next chapter we will derive formulae for the expectation and variance of various distributions which appear so frequently that they are given special names.

9.5 Further Exercises

Exercise 9.13: Calculating expectations and variances

Let X be a discrete random variable with $\mathbb{E}(X) = 5$ and $\text{Var}(X) = 2/3$. Find the following:

- (a) $\mathbb{E}(3X)$
- (b) $\text{Var}(3X)$,
- (c) $\mathbb{E}(4 - 3X)$,
- (d) $\text{Var}(4 - 3X)$,
- (e) $\mathbb{E}(4 - 3X^2)$.

[Answers: 15, 6, -11, 6, -73]

Exercise 9.14: Unfair tossing (revisited)

A coin which has probability p of coming up Heads is tossed three times. Let X be the number of Heads observed (see Exercise 8.13).

- (a) Compute the expectation of X .
- (b) Compute the variance of X .

*Exercise 9.15: **More series*

Suppose $|z| < 1$. By differentiating the geometric series, or otherwise, show that:

(a)

$$1 + 2z + 3z^2 + 4z^3 + \dots = \sum_{k=1}^{\infty} kz^{k-1} = \frac{1}{(1-z)^2},$$

(b)

$$1 + 4z + 9z^2 + 16z^3 + \dots = \sum_{k=1}^{\infty} k^2 z^{k-1} = \frac{2}{(1-z)^3} - \frac{1}{(1-z)^2}.$$

[These results are very useful but you would not be expected to prove them in an exam for this course.]

*Exercise 9.16: *An e-zee proof?*

Let Z be a random variable for which all the possible values are in the set $\{0, 1, 2, 3, \dots, n\}$.

(a) Show that

$$\mathbb{E}(Z) = \sum_{i=1}^n \mathbb{P}(Z \geq i).$$

(b) Deduce that if $\mathbb{E}(Z) < 1$, then Z takes the value 0 with non-zero probability.(c) Use part (a) to prove that for all $1 \leq t \leq n$ we have

$$\mathbb{P}(Z \geq t) \leq \frac{\mathbb{E}(Z)}{t}.$$

*Exercise 9.17: **European coin games*

(a) Consider a game where you flip a fair coin until it comes up Heads, starting with £2 and doubling the prize fund with every appearance of Tails. Let the random variable N be the number of flips; if N takes value n , you win £ 2^n . Let X denote the prize you win (in pounds). Find $\mathbb{E}(X)$ and discuss how much you would be prepared to pay to enter such a game.

(b) Now consider the following two-player game. Angela and Boris flip a fair coin until it comes up Tails. The number of flips needed is again a random variable N taking values $n = 1, 2, 3, \dots$. If n is odd then Boris pays Angela 2^n €; if n is even then Angela pays Boris 2^n €. Let Y denote Boris' net reward (in euros). Show that $\mathbb{E}(Y)$ does not exist.

CHAPTER 10

Special Discrete Random Variables

10.1 Bernoulli Distribution

We begin our survey of special distributions with a very easy case which will nevertheless introduce some important concepts.

Consider an experiment where there are only two possible outcomes labelled “success” and “failure”. Note that no moral judgement is implied by these labels; “success” could be something as mundane as a coin landing on Heads. This set-up is called a **Bernoulli trial**. Now suppose that the probability of success is p , i.e., $\mathbb{P}(\{\text{success}\}) = p$. Then the random variable defined by $X(\text{success}) = 1$ and $X(\text{failure}) = 0$, has the probability mass function

$$\begin{array}{c|cc} k & 0 & 1 \\ \hline \mathbb{P}(X = k) & 1 - p & p \end{array}.$$

This p.m.f. is called the **Bernoulli distribution**, with parameter p , and we write $X \sim \text{Bernoulli}(p)$, where the symbol “ $\sim$ ” loosely means “has the distribution of”.

As with all the distributions in this chapter, we are interested in the expectation and variance. In this case they are very easily calculated; from Definitions 9.1 and 9.5, we have

$$\mathbb{E}(X) = 0 \times (1 - p) + 1 \times p = p, \tag{10.1}$$

$$\text{Var}(X) = (0 - p)^2 \times (1 - p) + (1 - p)^2 \times p = p(1 - p)[p + (1 - p)] = p(1 - p). \tag{10.2}$$

Note that the Bernoulli distribution also applies to experiments with more than two possible outcomes as long as the sample space is partitioned into events corresponding to “success” and “failure” and we are interested in whether or not a “success” occurs.

Example 10.1: Six success

Consider once more throwing a fair six-sided die. Let the random variable Y take the value 1 when a “six” is rolled and the value 0 otherwise. Find $\mathbb{E}(Y)$ and $\text{Var}(Y)$.

Solution:

The sample space of the experiment is $\{1, 2, 3, 4, 5, 6\}$ and the question tells us that $Y(1) = Y(2) = Y(3) = Y(4) = Y(5) = 0$ while $Y(6) = 1$. We can partition the sample space into the events $A = \{6\}$ and $A^c = \{1, 2, 3, 4, 5\}$, where A is identified with “success”; the success probability is $\mathbb{P}(Y = 1) = \mathbb{P}(A) = 1/6$ and so $Y \sim \text{Bernoulli}(1/6)$. Hence, from (10.1) and (10.2) above, we obtain

$$\mathbb{E}(Y) = \frac{1}{6} \quad \text{and} \quad \text{Var}(Y) = \frac{1}{6} \left(1 - \frac{1}{6}\right) = \frac{5}{36}.$$

10.2 Binomial Distribution

Often we are interested in the number of “successes”, not from a single trial but from multiple repeated trials (e.g., the number of Heads when we toss a coin 100 times). Building on the previous section, we thus consider performing n independent Bernoulli trials, each with the same probability p of success. [Independent trials means that if E_i denotes the event “the i th trial is a success” then $E_1, E_2, \dots, E_n$ are mutually independent events.] Let X be the number of “successes” in these n trials. To determine the p.m.f. of X we need to evaluate the probabilities $\mathbb{P}(X = k)$ for $k = 0, 1, 2, \dots, n$. We can do this by the following argument.

- First consider the special outcome that the first k trials are successes and the remaining $n-k$ trials are failures. Using mutual independence the probability of this simple event is $p^k(1-p)^{n-k}$.
- The event “ $X = k$ ” contains $\binom{n}{k}$ different outcomes with k successes and $n-k$ failures.¹ Since the trials are identical, each outcome occurs with the same probability $p^k(1-p)^{n-k}$.

Hence we obtain the p.m.f.

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k} \quad \text{for } k = 0, 1, 2, \dots, n.$$

We call this the **binomial distribution**, with parameters n and p , and write $X \sim \text{Bin}(n, p)$. Note that the Bernoulli(p) distribution is just Binomial(1, p).

¹You can think of this as unordered sampling of k trials from n without replacement: the order doesn’t matter but once a trial has been picked to be a success, it can’t be picked again.

In dealing with the binomial distribution, the following identities are often useful:

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k} \quad [\text{binomial theorem}], \quad (10.3)$$

$$\binom{n}{k} = \frac{n}{k} \binom{n-1}{k-1}. \quad (10.4)$$

Exercise 10.2: Identity check

- (a) Use (10.3) to verify Proposition 8.4 for the binomial distribution.
- (b) Starting from the definition of the binomial coefficient, prove (10.4).

Armed with (10.3) and (10.4), we now turn to the expectation and variance. The expectation is calculated as

$$\begin{aligned} \mathbb{E}(X) &= \sum_{k=0}^n k \mathbb{P}(X = k) \\ &= \sum_{k=1}^n k \binom{n}{k} p^k (1-p)^{n-k} \\ &= \sum_{k=1}^n n \binom{n-1}{k-1} p^k (1-p)^{n-k} \quad [\text{using (10.4)}] \\ &= np \sum_{k=1}^n \binom{n-1}{k-1} p^{k-1} (1-p)^{n-k} \\ &= np \sum_{\ell=0}^{n-1} \binom{n-1}{\ell} p^{\ell} (1-p)^{n-1-\ell} \quad [\text{setting } \ell = k-1] \\ &= np[p + (1-p)]^{n-1} \quad [\text{using (10.3) with } a = p \text{ and } b = 1-p] \\ &= np. \end{aligned} \quad (10.5)$$

Similarly, one can show that the variance is

$$\text{Var}(X) = np(1-p). \quad (10.6)$$

*Exercise 10.3: *Variance of binomial distribution*

Prove the expression (10.6) for the variance of the binomial distribution by using (10.4) twice. [Hint: First calculate $\mathbb{E}(X^2) - \mathbb{E}(X)$.]

In fact, we will see a much easier argument for (10.5) and (10.6) in the next chapter!

Example 10.4: Ten tosses

Suppose you toss a fair coin ten times. Determine the expectation and variance of the number of Heads seen.

Solution:

Let the number of Heads seen in ten tosses be Z . For a fair coin the probability of a Head on each toss (i.e., the success probability in each trial) is $1/2$. Hence $Z \sim \text{Bin}(10, 1/2)$ and, from (10.5) and (10.6),

$$\mathbb{E}(Z) = 10 \times \frac{1}{2} = 5 \quad \text{and} \quad \text{Var}(Z) = 10 \times \frac{1}{2} \times \left(1 - \frac{1}{2}\right) = \frac{5}{2}.$$

10.3 Geometric Distribution

Suppose we make an *unlimited* number of independent Bernoulli trials, each with (non-zero) success probability p , and let T be the number of trials up to and including the first success. We already saw a demonstration of this in Example 8.7: tossing a coin repeatedly until a Tail appears. To find the probability mass function for the general situation, we note that “ $T = k$ ” is a simple event consisting of a single outcome: $k - 1$ failures, each with probability $1 - p$, followed by one success with probability p . Since the trials are independent, we can multiply probabilities to obtain

$$\mathbb{P}(T = k) = (1 - p)^{k-1}p \quad \text{for } k = 1, 2, 3, \dots$$

We say that T has the **geometric distribution** with parameter p and write $T \sim \text{Geom}(p)$. A word of warning is in order here: there is an alternative definition of the geometric distribution which involves counting the number of failures $(0, 1, 2, \dots)$ *before* the first success; in this course, we always use the definition above but you should check carefully if consulting other books or websites.

In order to derive the expectation and variance of the geometric distribution we will need the following two results for the sums of infinite series with $|z| < 1$:

$$\sum_{k=1}^{\infty} kz^{k-1} = 1 + 2z + 3z^2 + 4z^3 + \dots = \frac{1}{(1-z)^2}, \quad (10.7)$$

$$\sum_{k=1}^{\infty} k^2 z^{k-1} = 1 + 4z + 9z^2 + 16z^3 + \dots = \frac{2}{(1-z)^3} - \frac{1}{(1-z)^2}. \quad (10.8)$$

[One way to prove these is to start from the formula for the sum of a geometric series and differentiate both sides; the derivations belong more properly in a calculus or analysis course but see Exercise 9.15 if you want to have a try.] Armed with this knowledge, the expectation of the geometric distribution is straightforwardly

given by

$$\begin{aligned}
 \mathbb{E}(T) &= \sum_{k=1}^{\infty} k \mathbb{P}(T = k) \\
 &= \sum_{k=1}^{\infty} k(1-p)^{k-1}p \\
 &= p \sum_{k=1}^{\infty} k(1-p)^{k-1} \\
 &= p \times \frac{1}{[1 - (1-p)]^2} \quad [\text{using (10.7) with } z = 1-p] \\
 &= \frac{1}{p},
 \end{aligned} \tag{10.9}$$

while for the second moment we have

$$\begin{aligned}
 \mathbb{E}(T^2) &= \sum_{k=1}^{\infty} k^2 \mathbb{P}(T = k) \\
 &= \sum_{k=1}^{\infty} k^2 (1-p)^{k-1}p \\
 &= p \left\{ \frac{2}{[1 - (1-p)]^3} - \frac{1}{[1 - (1-p)]^2} \right\} \quad [\text{using 10.8 with } z = 1-p] \\
 &= \frac{2-p}{p^2}.
 \end{aligned} \tag{10.10}$$

Substituting from (10.9) and (10.10) into Proposition 9.6, we have for the variance

$$\begin{aligned}
 \text{Var}(T) &= \mathbb{E}(T^2) - [\mathbb{E}(T)]^2 \\
 &= \frac{2-p}{p^2} - \left(\frac{1}{p}\right)^2 \\
 &= \frac{1-p}{p^2}.
 \end{aligned}$$

*Exercise 10.5: *Tail c.d.f.*

Let T be the number of tosses of a coin up to (and including) the first time you see a Tail. Find a formula for the cumulative distribution function $\mathbb{P}(T \leq k)$. [Hint: First consider $\mathbb{P}(T > k)$.]

10.4 Poisson Distribution

Suppose we consider the binomial distribution, with non-zero success probability, and make the number of trials n larger and larger whilst keeping the expectation

the same – this means that the success probability must be of the form λ/n with λ a strictly positive constant. In the $n \rightarrow \infty$ limit the p.m.f. becomes

$$\mathbb{P}(X = k) = \frac{\lambda^k}{k!} e^{-\lambda} \quad \text{for } k = 0, 1, 2, \dots$$

In this case we say that X has the **Poisson distribution** with parameter λ and we write $X \sim \text{Poisson}(\lambda)$. If $0 < \lambda \leq 1$, then the p.m.f. $\mathbb{P}(X = k)$ is non-increasing; if $\lambda > 1$, then the p.m.f. increases for small k before decreasing for large k .

*Exercise 10.6: **Limit of Binomial*

Write down the p.m.f. for a binomial distribution with n trials and success probability λ/n . Carefully take the limit $n \rightarrow \infty$ and show that the result is the above expression for the p.m.f. of a Poisson distribution.

In proofs involving the Poisson distribution, a crucial ingredient is the Taylor series of the exponential function:

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \dots \quad (10.11)$$

Exercise 10.7: Checking normalization

Use the identity (10.11) to show that Proposition 8.4 holds for the Poisson distribution.

With knowledge of (10.11), we can easily calculate the expectation of the Poisson distribution:

$$\begin{aligned} \mathbb{E}(X) &= \sum_{k=0}^{\infty} k \mathbb{P}(X = k) \\ &= \sum_{k=1}^{\infty} k \frac{\lambda^k}{k!} e^{-\lambda} \\ &= e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda \times \lambda^{k-1}}{(k-1)!} \\ &= \lambda e^{-\lambda} \sum_{\ell=0}^{\infty} \frac{\lambda^{\ell}}{\ell!} \quad [\text{setting } \ell = k - 1] \\ &= \lambda e^{-\lambda} \times e^{\lambda} \quad [\text{using (10.11)}] \\ &= \lambda. \end{aligned} \quad (10.12)$$

For the variance, we use the trick of first considering $\mathbb{E}(X^2) - \mathbb{E}(X)$ which can be

obtained as

$$\begin{aligned}
 \mathbb{E}(X^2) - \mathbb{E}(X) &= \mathbb{E}(X^2 - X) \\
 &= \sum_{k=0}^{\infty} (k^2 - k) \mathbb{P}(X = k) \\
 &= \sum_{k=2}^{\infty} k(k-1) \frac{\lambda^k}{k!} e^{-\lambda} \\
 &= e^{-\lambda} \sum_{k=2}^{\infty} \frac{\lambda^2 \times \lambda^{k-2}}{(k-2)!} \\
 &= \lambda^2 e^{-\lambda} \sum_{\ell=0}^{\infty} \frac{\lambda^{\ell}}{\ell!} \quad [\text{setting } \ell = k - 2] \\
 &= \lambda^2 \quad [\text{using (10.11)}].
 \end{aligned} \tag{10.13}$$

Combining (10.12) and (10.13) with Proposition 9.6 we thus have

$$\begin{aligned}
 \text{Var}(X) &= \mathbb{E}(X^2) - [\mathbb{E}(X)]^2 \\
 &= \mathbb{E}(X^2) - \mathbb{E}(X) + \mathbb{E}(X) - [\mathbb{E}(X)]^2 \\
 &= \lambda^2 + \lambda - (\lambda)^2 \\
 &= \lambda.
 \end{aligned}$$

The Poisson distribution can be used as an approximation in modelling situations with many trials and a small probability of success. It also frequently appears for counting events happening in continuous time, e.g., the number of clicks of a Geiger counter (monitoring radioactive decay) in a minute.

*Exercise 10.8: *Poisson presents*

Let W_S be the number of presents per hour produced by a Senior Elf in Santa's workshop. W_S is a Poisson random variable with probability mass function

$$\mathbb{P}(W_S = k) = \frac{2^k e^{-2}}{k!} \quad \text{for } k \in \{0, 1, 2, 3, \dots\}.$$

A Junior Elf produces W_J presents per hour, also according to a Poisson distribution but with parameter $\lambda = 1$. One morning, a snap inspection is organised to check whether the elves are meeting the minimum performance criterion of each producing at least one present per hour.

- Determine the probability that a Senior Elf fails to meet the criterion.
- For a team of one Senior Elf and two Junior Elves (all working independently), find the probability that at least one elf fails to meet the criterion.

[You may leave powers of e in your answers.]

10.5 Distributions in Practice

We can summarize the content of this chapter in the following table.

Distribution	Values	$\mathbb{P}(X = k)$	$\mathbb{E}(X)$	$\text{Var}(X)$
$X \sim \text{Bernoulli}(p)$	$X = 0, 1$	$\begin{cases} 1 - p & \text{for } k = 0 \\ p & \text{for } k = 1 \end{cases}$	p	$p(1 - p)$
$X \sim \text{Bin}(n, p)$	$X = 0, 1, \dots, n$	$\binom{n}{k} p^k (1 - p)^{n-k}$	np	$np(1 - p)$
$X \sim \text{Geom}(p)$	$X = 1, 2, 3, \dots$	$(1 - p)^{k-1} p$	$\frac{1}{p}$	$\frac{1 - p}{p^2}$
$X \sim \text{Poisson}(\lambda)$	$X = 0, 1, 2, \dots$	$\frac{\lambda^k}{k!} e^{-\lambda}$	λ	λ

Note, however, that the first row is not really necessary since $\text{Bernoulli}(p)$ is just $\text{Bin}(1, p)$.

An oft-asked question is how to determine which distribution applies in a particular situation. Of course, sometimes an exercise or exam problem will explicitly give a distribution (especially in the Poissonian case) but, if not, a very good clue is the range of the random variable in question – note the differences in the second column of the table above. Loosely speaking, we can say the following.

- If a random variable takes only two possible values, it can always be related to a Bernoulli random variable.
- If a random variable is counting the number of times something happens in a fixed number of independent trials, it has a binomial distribution.
- If a random variable is counting the number of independent trials until something happens, it has a geometric distribution.
- If a random variable is counting the number of times something happens in a fixed interval (continuous time), it probably has a Poisson distribution.

In real life of course, a random variable may not have any of the above distributions but one of them may serve as a good approximation. You will see more of this in future courses but, for now, note that it is always a good idea to state clearly any assumptions you are making in modelling a situation.

Exercise 10.9: Real-life distributions

Look back at the discrete random variables you thought of in Exercise 8.1. Can you suggest what distributions any of them might have? What assumptions might be needed?

Example 10.10: Shifted Bernoulli

Show that if W is a random variable taking the value a with probability $1 - p$ and the value b with probability p , then $X = (W - a)/(b - a)$ has a $\text{Bernoulli}(p)$ distribution.

Solution:

If $W = a$, then $X = (a - a)/(b - a) = 0$; if $W = b$ then $X = (b - a)/(b - a) = 1$. Hence $\mathbb{P}(X = 0) = \mathbb{P}(W = a) = 1 - p$ and $\mathbb{P}(X = 1) = \mathbb{P}(W = b) = p$, i.e., $X \sim \text{Bernoulli}(p)$. [Note that one can rearrange to get $W = a + (b - a)X$ and obtain the expectation and variance of W from the known Bernoulli results for $\mathbb{E}(X)$ and $\text{Var}(X)$; in most such cases, however, it is probably easier to calculate $\mathbb{E}(W)$ and $\text{Var}(W)$ directly.]

Example 10.11: Random red balls

Suppose you choose n balls at random from a bag containing N balls of which M are red. [These numbers are fixed.] Let the random variable R denote the number of red balls picked.

- (a) If you pick the balls *with* replacement, what is the p.m.f. of R ? State the expectation and variance in this case.
- (b) If you pick the balls *without* replacement, what is the p.m.f. of R ?

Solution:

- (a) Each random pick results in a red ball with probability M/N and the outcome of each pick is independent of all the others (i.e., we perform n independent Bernoulli trials with success probability M/N). Hence $R \sim \text{Bin}(n, M/N)$ and we have

$$\begin{aligned}\mathbb{P}(R = k) &= \binom{n}{k} \left(\frac{M}{N}\right)^k \left(1 - \frac{M}{N}\right)^{n-k} \quad \text{for } k = 0, 1, 2, \dots, n, \\ \mathbb{E}(R) &= n \frac{M}{N}, \\ \text{Var}(R) &= n \frac{M}{N} \left(1 - \frac{M}{N}\right).\end{aligned}$$

- (b) Treating the situation as unordered sampling without replacement, the sample space has cardinality $\binom{N}{n}$ since we choose n balls from N . The event “ $R = k$ ” corresponds to choosing k red balls from M red balls, which can be done in $\binom{M}{k}$ ways, and choosing $n - k$ non-red balls from $N - M$ non-red balls, which can be done in $\binom{N-M}{n-k}$ ways. Hence, since the outcomes are all equally likely,

$$\mathbb{P}(R = k) = \frac{\binom{M}{k} \binom{N-M}{n-k}}{\binom{N}{n}}. \quad (10.14)$$

Obviously, R cannot be larger than M and $n - R$ cannot be larger than $N - M$. Hence, the possible values k of R satisfy $\max(0, n - (N - M)) \leq k \leq \min(n, M)$. However, with the convention that $\binom{a}{k} = 0$ for integers $k > a \geq 0$, then the above formula for the p.m.f. is valid for $k = 0, 1, 2, \dots, n$ and gives zero probability to any impossible values of R . [You can check that one gets the same p.m.f. by treating the situation as ordered sampling without replacement.]

In fact, (10.14) gives the probability mass function of the so-called **hypergeometric distribution**; you don’t need to know formulae for its expectation and variance but

you can try to derive them for a challenge (Exercise 10.17).

Exercise 10.12: Typographical trials

Assume that, on average, a typographical error is found every 1000 typeset characters. Compute the probability that a 600-character page contains fewer than two errors.

[2016 exam question (part)]

10.6 Further Exercises

Exercise 10.13: Probabilities from distributions

Suppose that $A \sim \text{Poisson}(3)$, $B \sim \text{Geom}(1/3)$, and $C \sim \text{Bin}(4, 1/6)$ are random variables. Find the following probabilities:

- (a) $\mathbb{P}(A = 2)$,
- (b) $\mathbb{P}(A > 2)$,
- (c) $\mathbb{P}(B = 3)$,
- (d) $\mathbb{P}(B \leq 3)$,
- (e) $\mathbb{P}(C = 2)$.

You may leave any powers of e in your answers but you should simplify all factorials and other powers.

[Answers: $9e^{-3}/2$, $1 - 17e^{-3}/2$, $4/27$, $19/27$, $25/216$]

Exercise 10.14: Lifetime of a component

An electrical component is installed on a certain day and is inspected on each subsequent day. Let G be the number of days until inspection reveals that the component is broken.

- (a) Suppose that $G \sim \text{Geom}(p)$. Show that for any non-negative integers k and ℓ ,

$$\mathbb{P}(G > k + \ell | G > k) = \mathbb{P}(G > \ell).$$

- (b) Say in words what the conclusion of part (a) means for the lifetime of the component. Why do you think this is sometimes called the “memoryless property” of the geometric distribution?

*Exercise 10.15: *A fishy problem*

Let X be the number of fish caught by a fisherman in one afternoon. Suppose that X is distributed $\text{Poisson}(\lambda)$. Each fish has probability p of being a salmon independently of all other fish caught. Let Y be the number of salmon caught.

- (a) Suppose that the fisherman catches m fish. What is the probability that k of them are salmon?

(b) Show that

$$\mathbb{P}(Y = k) = \sum_{m=k}^{\infty} \mathbb{P}(Y = k | X = m) \mathbb{P}(X = m).$$

(c) Using part (b), find the probability mass function of Y . What is the name of the distribution of Y ?

Exercise 10.16: An argument about money

A fair coin is tossed four times. Let N be the number of instances of a Head followed by another Head in the sequence of tosses.

- (a) Your friend proposes the following analysis: There are three possible ways in which we could have a Head followed by another Head (at the first and second, the second and third, or third and fourth toss). We have a probability $1/2 \times 1/2 = 1/4$ of getting a Head followed by another Head at each of these positions. Hence N is the number of successes in three Bernoulli trials each with success probability $1/4$ and so $N \sim \text{Bin}(3, 1/4)$. Explain carefully what is wrong with this argument.
- (b) Determine the correct probability mass function and then the expectation and variance of N .

*Exercise 10.17: **Hypergeometric distribution*

Consider the p.m.f. of R in Example 10.11(b). Derive expressions for $\mathbb{E}(R)$ and $\text{Var}(R)$, and compare your results to the situation of Example 10.11(a) where the balls are picked with replacement.

CHAPTER 11

Several Random Variables

11.1 Joint and Marginal Distributions

In real-life situations, and exam questions, we are often interested in probabilities of events which involve two (or more) random variables. For instance, you might want to know the probability that the number of votes for Biden is greater than the number of votes for Trump, or the probability that two different stocks both go up in value. You may also wonder about the relation between two different random variables (e.g., the number of storks in a town and the number of babies born there); we will explore the concepts of independence and correlation at the end of this chapter and in the next one. First, however, we need to establish some basics.

Definition 11.1. *Let X and Y be two discrete random variables defined on the same sample space and taking values $x_1, x_2, \dots$ and $y_1, y_2, \dots$ respectively. The function*

$$(x_k, y_\ell) \mapsto \mathbb{P}((X = x_k) \cap (Y = y_\ell))$$

*is called the **joint probability mass function** of X and Y .*

Remarks:

- Usually we write $\mathbb{P}(X = x_k, Y = y_\ell)$ instead of $\mathbb{P}((X = x_k) \cap (Y = y_\ell))$.
- The values of the joint p.m.f. must be non-negative and sum up to one: $\sum_k \sum_\ell \mathbb{P}(X = x_k, Y = y_\ell) = 1$ (also written as $\sum_{k,\ell} \mathbb{P}(X = x_k, Y = y_\ell) = 1$).
- Often we write the joint p.m.f. in the form of a table.
- The definition can be easily extended to joint distributions of three (or more) random variables but three dimensional tables are more difficult to construct!

Example 11.1: Coloured balls

A bag contains three red balls, two yellow balls, and two green balls. Suppose that we pick three balls at random (without replacement). Let R denote the number of red balls we pick, and Y the number of yellow balls we pick. Find $\mathbb{P}(R = 1, Y = 1)$ and $\mathbb{P}(R = 3, Y = 1)$.

Solution:

We can treat this situation as unordered sampling without replacement (see Section 3.4). As we are picking three balls from a set of seven balls, we have

$$|\mathcal{S}| = \binom{7}{3}.$$

The event $(R = 1) \cap (Y = 1)$ is the event that we pick one red ball (from three red balls), one yellow ball (from two yellow balls), and one green ball (from two green balls). Since all outcomes are equally likely, we can calculate the probability of this event as:

$$\mathbb{P}(R = 1, Y = 1) = \frac{|(R = 1) \cap (Y = 1)|}{|\mathcal{S}|} = \frac{\binom{3}{1}\binom{2}{1}\binom{2}{1}}{\binom{7}{3}} = \frac{3 \times 2 \times 2}{35} = \frac{12}{35}.$$

It is even easier to calculate $\mathbb{P}(R = 3, Y = 1)$; it is impossible to draw three red balls and one yellow ball if we only draw three balls in total so $(R = 3) \cap (Y = 1) = \emptyset$ and

$$\mathbb{P}(R = 3, Y = 1) = 0.$$

Exercise 11.2: Coloured balls (revisited)

In the set-up of Example 11.1, evaluate $\mathbb{P}(R = r, Y = y)$ for all possible values of R and Y . Use your results to complete the following table for the joint p.m.f.:

$R \backslash Y$	0	1	2	3
0				
1		12/35		0
2				

The next proposition relates the joint distribution $\mathbb{P}(X = x_k, Y = y_\ell)$ and the so-called **marginals** $\mathbb{P}(X = x_k)$ and $\mathbb{P}(Y = y_\ell)$.

Proposition 11.2. *Let X and Y be two discrete random variables defined on the same sample space and taking values $x_1, x_2, \dots$ and $y_1, y_2, \dots$ respectively. The marginal distribution of X can be obtained from the joint distribution as*

$$\mathbb{P}(X = x_k) = \sum_{\ell} \mathbb{P}(X = x_k, Y = y_\ell).$$

Similarly, the marginal distribution of Y is given by

$$\mathbb{P}(Y = y_\ell) = \sum_k \mathbb{P}(X = x_k, Y = y_\ell).$$

Loosely speaking, the idea is that if we only care about the probability of X taking a particular value, we need to sum over all possible values of Y (and vice versa, of course). The values of the marginals are the column sums and the row sums in a table of the joint p.m.f.; they can be written in the margins, hence the name.

*Exercise 11.3: *Marginal proof*

Give a proof of Proposition 11.2. [Hint: Use Definition 2.1(c) together with the fact that the events “ $Y = y_k$ ” ($k = 1, 2, \dots$) partition the sample space.]

Exercise 11.4: Coloured balls (re-revisited)

Find the marginal distributions of R and Y from Example 11.1 and hence calculate $\mathbb{E}(R)$ and $\mathbb{E}(Y)$.

11.2 Expectations in the Multivariate Context

After doing Exercise 11.4, you may wonder if it is possible to calculate say $\mathbb{E}(Y)$ from a joint p.m.f. without first calculating the marginal distribution $\mathbb{P}(Y = y_k)$. You may also wonder if one can define/calculate expectations of functions of two random variables in a similar way to we did for functions of a single random variable in Proposition 9.3. Both these questions are answered in the affirmative by the following proposition.

Proposition 11.3. *If $g(X, Y)$ is a real-valued function of the two discrete random variables X and Y then the expectation of $g(X, Y)$ is obtained as*

$$\mathbb{E}(g(X, Y)) = \sum_k \sum_\ell g(x_k, y_\ell) \mathbb{P}(X = x_k, Y = y_\ell)$$

where the sum ranges over all possible values x_k, y_ℓ of the two random variables.

Remarks:

- Again we implicitly assume (in this course) that such expectations are well defined, even when they involve infinite sums.
- Setting $g(X, Y) = 1$ recovers the result that the values of the joint p.m.f. sum to one; setting $g(X, Y) = Y$ gives an expression for $\mathbb{E}(Y)$, and so on.

Example 11.5: Easy expectation?

Find $\mathbb{E}(UV + U)$ if the random variables U and V have joint p.m.f. given by the table:

$V \backslash U$	1	2
	$1/2$	$1/6$
3	$1/3$	0

Solution:

From Proposition 11.3, the required expectation can be written as a sum of four terms:

$$\begin{aligned}
 \mathbb{E}(UV + U) &= (1 \times 1 + 1) \times \frac{1}{2} + (2 \times 1 + 2) \times \frac{1}{6} + (1 \times 3 + 1) \times \frac{1}{3} + (2 \times 3 + 2) \times 0 \\
 &= 2 \times \frac{1}{2} + 4 \times \left(\frac{1}{6} + \frac{1}{3} \right) \\
 &= 3.
 \end{aligned}$$

Proposition 11.3 leads to the following important, if unsurprising, theorem.

Theorem 11.4. *If X and Y are discrete random variables then*

$$\mathbb{E}(X + Y) = \mathbb{E}(X) + \mathbb{E}(Y).$$

Proof:

Starting from Proposition 11.3, and using series properties, we have

$$\begin{aligned}
 \mathbb{E}(X + Y) &= \sum_k \sum_\ell (x_k + y_\ell) \mathbb{P}(X = x_k, Y = y_\ell) \\
 &= \sum_k \sum_\ell x_k \mathbb{P}(X = x_k, Y = y_\ell) + \sum_k \sum_\ell y_\ell \mathbb{P}(X = x_k, Y = y_\ell) \\
 &= \sum_k x_k \left(\sum_\ell \mathbb{P}(X = x_k, Y = y_\ell) \right) + \sum_\ell y_\ell \left(\sum_k \mathbb{P}(X = x_k, Y = y_\ell) \right) \\
 &= \sum_k x_k \mathbb{P}(X = x_k) + \sum_\ell y_\ell \mathbb{P}(Y = y_\ell) \quad [\text{from Proposition 11.2}] \\
 &= \mathbb{E}(X) + \mathbb{E}(Y) \quad [\text{from Definition 9.1}]. \quad \square
 \end{aligned}$$

We can apply Theorem 11.4 repeatedly to obtain, for example,

$$\mathbb{E}(X + Y + Z) = \mathbb{E}(X + Y) + \mathbb{E}(Z) = \mathbb{E}(X) + \mathbb{E}(Y) + \mathbb{E}(Z).$$

In fact, using the properties of expectations (Proposition 9.7) one can show a useful general result.

Corollary 11.5 (Linearity of expectation). *If $X_1, X_2, \dots, X_n$ are discrete random variables and $c_1, c_2, \dots, c_n$ real-valued constants, then*

$$\mathbb{E}(c_1X_1 + c_2X_2 + \dots + c_nX_n) = c_1\mathbb{E}(X_1) + c_2\mathbb{E}(X_2) + \dots + c_n\mathbb{E}(X_n).$$

Remark: This is a very general statement with no assumptions needed on X_1, X_2 , etc.; it includes the case where the random variables are related in some way, e.g., we could have $X_2 = (X_1 + 3)^2$.

Exercise 11.6: Ball expectations

For the set-up of Example 11.1, find $\mathbb{E}(R + Y)$ and $\mathbb{E}(RY)$ using Proposition 11.3. Check that $\mathbb{E}(R + Y) = \mathbb{E}(R) + \mathbb{E}(Y)$. Is it also true that $\mathbb{E}(RY) = \mathbb{E}(R) \times \mathbb{E}(Y)$?

11.3 Independence for Random Variables

We now turn to the question of independence for random variables. This builds on the idea of independence for events.

Definition 11.6. *Two discrete random variables X and Y are **independent** if the events “ $X = x_k$ ” and “ $Y = y_\ell$ ” are independent for all possible values x_k, y_ℓ , i.e., if*

$$\mathbb{P}(X = x_k, Y = y_\ell) = \mathbb{P}(X = x_k)\mathbb{P}(Y = y_\ell)$$

for all x_k and y_ℓ .

This generalizes to more than two random variables; $X_1, X_2, \dots, X_n$ are independent if their joint probability mass function factorizes into the product of the marginal probability mass functions for all possible values of the random variables.

*Exercise 11.7: *Three independent random variables*

If the discrete random variables X, Y , and Z are independent then

$$\mathbb{P}(X = x_k, Y = y_\ell, Z = z_m) = \mathbb{P}(X = x_k)\mathbb{P}(Y = y_\ell)\mathbb{P}(Z = z_m)$$

for all x_k, y_ℓ and z_m . Show that this implies

$$\mathbb{P}(Y = y_\ell, Z = z_m) = \mathbb{P}(Y = y_\ell)\mathbb{P}(Z = z_m)$$

for all y_ℓ and z_m . [In this sense, the definition of independence for three random variables is simpler than the definition of (mutual) independence for three events.]

Example 11.8: Easy independence?

Determine whether U and V in Example 11.5 are independent.

Solution:

We clearly have $\mathbb{P}(U = 2) = 1/6$ and $\mathbb{P}(V = 3) = 1/3$ but $\mathbb{P}(U = 2, V = 3) = 0$ so $\mathbb{P}(U = 2, V = 3) \neq \mathbb{P}(U = 2)\mathbb{P}(V = 3)$ and hence U and V are *not* independent.

Independence of random variables has important consequences.

Theorem 11.7. *If X and Y are independent discrete random variables then:*

- (a) $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$,
- (b) $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$.

Proof:

- (a) The proof relies on the fact that

$$\sum_k \sum_\ell a_k b_\ell = \left(\sum_k a_k \right) \left(\sum_\ell b_\ell \right),$$

which can be easily checked for sums over small numbers of values. Using this, we have

$$\begin{aligned} \mathbb{E}(XY) &= \sum_k \sum_\ell x_k y_\ell \mathbb{P}(X = x_k, Y = y_\ell) && [\text{from Proposition 11.3}] \\ &= \sum_k \sum_\ell x_k y_\ell \mathbb{P}(X = x_k) \mathbb{P}(Y = y_\ell) && [\text{by independence}] \\ &= \left(\sum_k x_k \mathbb{P}(X = x_k) \right) \left(\sum_\ell y_\ell \mathbb{P}(Y = y_\ell) \right) \\ &= \mathbb{E}(X)\mathbb{E}(Y). \end{aligned}$$

- (b) Starting from Proposition 9.6, and employing Theorem 11.4/Corollary 11.5, we have

$$\begin{aligned} \text{Var}(X + Y) &= \mathbb{E}((X + Y)^2) - [\mathbb{E}(X + Y)]^2 \\ &= \mathbb{E}(X^2 + 2XY + Y^2) - [\mathbb{E}(X) + \mathbb{E}(Y)]^2 \\ &= \mathbb{E}(X^2) + 2\mathbb{E}(XY) + \mathbb{E}(Y^2) - [\mathbb{E}(X)]^2 - 2\mathbb{E}(X)\mathbb{E}(Y) - [\mathbb{E}(Y)]^2 \\ &= \mathbb{E}(X^2) - [\mathbb{E}(X)]^2 + \mathbb{E}(Y^2) - [\mathbb{E}(Y)]^2 + 2[\mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)] \\ &= \text{Var}(X) + \text{Var}(Y) && [\text{by part (a)}]. \quad \square \end{aligned}$$

Theorem 11.7 generalizes to three and more random variables. For instance, if X , Y and Z are independent random variables, then,

$$\text{Var}(X + Y + Z) = \text{Var}(X) + \text{Var}(Y + Z) = \text{Var}(X) + \text{Var}(Y) + \text{Var}(Z).$$

Indeed, using Theorem 11.7 repeatedly together with properties of the variance (Proposition 9.8), one arrives at the following corollary.

Corollary 11.8. *If $X_1, X_2, \dots, X_n$ are independent discrete random variables and $c_1, c_2, \dots, c_n$ real-valued constants, then*

$$\text{Var}(c_1X_1 + c_2X_2 + \dots + c_nX_n) = c_1^2\text{Var}(X_1) + c_2^2\text{Var}(X_2) + \dots + c_n^2\text{Var}(X_n).$$

Remarks:

- Note that while Corollary 11.5 applies for *all* random variables, Corollary 11.8 applies for *independent* random variables.
- Independence of X and Y implies $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$ but the converse does *not* hold.

*Exercise 11.9: *Converse counterexample*

Find an example where $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$ but X and Y are not independent.

Exercise 11.10: Ball independence

Determine whether the random variables R and Y of Example 11.1 are independent.

Exercise 11.11: Five dice

You roll five fair six-sided dice. Let X denote the sum of the numbers shown. Compute the expectation and the variance of X .

11.4 Binomial Distribution Revisited

We now demonstrate how the results of this chapter can be leveraged to re-derive the expressions for the expectation and variance of the binomial distribution which we saw in Section 10.2. We start by considering n Bernoulli trials each with probability p of success. We count the number of successes in each of these trials with the random variables $X_1, X_2, \dots, X_n$; the event “ $X_k = 1$ ” corresponds to success in the k th trial while “ $X_k = 0$ ” corresponds to failure in the k th trial. Since the trials are identical, we obviously have $\mathbb{E}(X_k) = p$ and $\text{Var}(X_k) = p(1 - p)$ for $k = 1, 2, \dots, n$.

Denoting the total number of successes by the random variable X , we have

$$X = X_1 + X_2 + \dots + X_n$$

and, by Corollary 11.5,

$$\begin{aligned}\mathbb{E}(X) &= \mathbb{E}(X_1 + X_2 + \cdots + X_n) \\ &= \mathbb{E}(X_1) + \mathbb{E}(X_2) + \cdots + \mathbb{E}(X_n) \\ &= np.\end{aligned}$$

This conclusion does not require the trials to be independent. However, if they *are* independent, we can also employ Corollary 11.8 to obtain

$$\begin{aligned}\text{Var}(X) &= \text{Var}(X_1 + X_2 + \cdots + X_n) \\ &= \text{Var}(X_1) + \text{Var}(X_2) + \cdots + \text{Var}(X_n) \\ &= np(1 - p).\end{aligned}$$

Hence we easily recover (10.5) and (10.6) for the expectation and variance of the number of successes in n independent Bernoulli trials, i.e., the expectation and variance of the binomial distribution.

11.5 Further Exercises

Exercise 11.12: Expectations and variances

Suppose that X , Y , and Z are random variables with $X \sim \text{Bin}(7, 1/6)$, $Y \sim \text{Geom}(1/2)$, and $Z \sim \text{Poisson}(6)$. Suppose further that X and Y are independent but that X and Z are not independent. Which of the following can be determined from this information? Find the value of those which can be determined.

- (a) $\mathbb{E}(X + Y)$,
- (b) $\mathbb{E}(X + Z)$,
- (c) $\mathbb{E}(X + 2Y + 3Z)$,
- (d) $\mathbb{E}(X^2 + Y^2 + Z^2)$,
- (e) $\text{Var}(X + Y)$,
- (f) $\text{Var}(X + Z)$,
- (g) $\text{Var}(X + 2Y + 3Z)$.

[Answers (in jumbled order): cannot be determined, $151/3$, $19/6$, $107/36$, $43/6$, cannot be determined, $139/6$]

Exercise 11.13: Ones and twos

Two fair six-sided dice are rolled. Let V be the number of “one”s seen in the outcome and W be the number of “two”s seen. Find the joint distribution of V and W and the two marginal distributions. Are V and W independent random variables?

*Exercise 11.14: *Joint deductions*

Let X and Y be discrete random variables with probability mass functions given by

$$\begin{array}{c|c|c} x_k & 0 & 1 \\ \hline \mathbb{P}(X = x_k) & 1/2 & 1/2 \end{array},$$

and

$$\begin{array}{c|c|c|c} y_\ell & 0 & 1 & 2 \\ \hline \mathbb{P}(Y = y_\ell) & 1/3 & 1/3 & 1/3 \end{array}.$$

Furthermore assume that

$$\mathbb{P}(X = 0, Y = 0) = \mathbb{P}(X = 1, Y = 2) = 0.$$

Find the joint probability mass function of X and Y .

CHAPTER 12

Covariance and Conditional Expectation

12.1 Covariance and Correlation

We know from the last chapter that if X and Y are independent then we always have $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$, while if X and Y are not independent we usually (but *not* always) have $\mathbb{E}(XY) \neq \mathbb{E}(X)\mathbb{E}(Y)$. You may wonder if we can say any more about the strength and type of relationship between two random variables. One way to do this is by defining two new quantities.¹

Definition 12.1. *Let X and Y be discrete random variables defined on the same sample space. The **covariance of X and Y** is defined by*

$$\text{Cov}(X, Y) = \mathbb{E}([X - \mathbb{E}(X)][Y - \mathbb{E}(Y)]).$$

*If $\text{Var}(X) > 0$ and $\text{Var}(Y) > 0$, then the **correlation coefficient of X and Y** is defined by*

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}.$$

We shall consider how to interpret these quantities shortly. However, we can immediately see by comparison with Definition 9.5, that $\text{Cov}(X, X) = \text{Var}(X)$. Just as there is an alternative formula for the variance (Proposition 9.6), there is an alternative formula for the covariance which, in practice, is often easier for calculations.

Proposition 12.2. *If X and Y are discrete random variables defined on the same sample space, then*

$$\text{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y).$$

¹As in previous chapters, we assume here that all expectations are finite, even if they involve infinite sums.

Proof:

From Definition 12.1, we have

$$\begin{aligned}
 \text{Cov}(X, Y) &= \mathbb{E}([X - \mathbb{E}(X)][Y - \mathbb{E}(Y)]) \\
 &= \mathbb{E}(XY - X\mathbb{E}(Y) - Y\mathbb{E}(X) + \mathbb{E}(X)\mathbb{E}(Y)) \\
 &= \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y) - \mathbb{E}(X)\mathbb{E}(Y) + \mathbb{E}(X)\mathbb{E}(Y) \\
 &= \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y),
 \end{aligned}$$

where we use the properties of expectation (Proposition 9.7 and Corollary 11.5) in going from the second to the third line. $\square$

Remarks:

- When $\text{Cov}(X, Y) = 0$, we say that X and Y are **uncorrelated**. If X and Y are independent then, from Theorem 11.7(a) and Proposition 12.2, we always have $\text{Cov}(X, Y) = 0$. Conversely, however, $\text{Cov}(X, Y) = 0$ does not necessarily mean X and Y are independent.
- $\text{Cov}(X, Y) > 0$ if, on average, $[X - \mathbb{E}(X)]$ and $[Y - \mathbb{E}(Y)]$ have the same sign. In other words, when $[X - \mathbb{E}(X)]$ is positive, $[Y - \mathbb{E}(Y)]$ tends to be positive too; when $[X - \mathbb{E}(X)]$ is negative, $[Y - \mathbb{E}(Y)]$ tends to be negative too. Loosely, we can say in this case that X and Y tend to deviate together above or below their expectation. An example of such **positive correlation** might be the number of hours of revision and the final exam score.
- $\text{Cov}(X, Y) < 0$ if, on average, $[X - \mathbb{E}(X)]$ and $[Y - \mathbb{E}(Y)]$ have opposite signs. Loosely, we can say in this case that X and Y tend to deviate in opposite directions from their expectations. An example of such **negative correlation** might be temperature (recorded to the nearest degree, say) and sales of hot chocolate.
- Correlation does not necessarily indicate causation. An oft-discussed example in this respect is the positive correlation between the number of storks and the number of babies in Germany [Sie88]. The “Spurious Correlations” website, and accompanying book [Vig15], also contains many entertaining graphs (yes, really!) and bizarre facts.

Example 12.1: Correlated balls

Find the covariance of the random variables R and Y of Example 11.1. Discuss the interpretation of the sign of the covariance in this case.

Solution:

From Proposition 12.2, we have

$$\begin{aligned}\text{Cov}(R, Y) &= \mathbb{E}(RY) - \mathbb{E}(R)\mathbb{E}(Y) \\ &= \frac{6}{7} - \frac{9}{7} \times \frac{6}{7} \quad [\text{see Exercises 11.4 and 11.6}] \\ &= -\frac{12}{49}.\end{aligned}$$

R and Y are negatively correlated so having more red balls means having fewer yellow ball *on average*, and vice versa.

The correlation coefficient is a kind of scaled version of the covariance which is insensitive, for example, to the choice of units in a measurement. In particular, we have the following nice properties.

Proposition 12.3. *Let X and Y be discrete random variables defined on the same sample space and with $\text{Var}(X) > 0$ and $\text{Var}(Y) > 0$.*

(a) *If a, b, c, d are real-valued constants with $a > 0$ and $c > 0$, then*

$$\text{Corr}(aX + b, cY + d) = \text{Corr}(X, Y).$$

(b) *$\text{Corr}(X, Y)$ lies in $[-1, 1]$, i.e., $-1 \leq \text{Corr}(X, Y) \leq 1$.*

Exercise 12.2: Proving properties

Suppose that X and Y are discrete random variables defined on the same sample space (with $\text{Var}(X) > 0$ and $\text{Var}(Y) > 0$) and that a, b, c, d are real-valued constants.

(a) Show that

$$\text{Cov}(aX + b, cY + d) = ac\text{Cov}(X, Y).$$

(b) If $a > 0$ and $c > 0$, use the result of part (a) to show that

$$\text{Corr}(aX + b, cY + d) = \text{Corr}(X, Y).$$

(c) Show that if $Y = aX + b$ with $a > 0$, then $\text{Corr}(X, Y) = 1$ (perfect positive correlation). What relationship between X and Y would give $\text{Corr}(X, Y) = -1$ (perfect negative correlation)?

12.2 Conditional Expectation

Example 12.3: Conditional balls

Consider again the experiment of Example 11.1. Assuming that we pick one yellow ball (i.e., that $Y = 1$), compute the probability that we pick r red balls for $r = 0, 1, 2, 3$.

Solution:

By Definition 4.1 for conditional probability, we have

$$\mathbb{P}(R = r|Y = 1) = \frac{\mathbb{P}((R = r) \cap (Y = 1))}{\mathbb{P}(Y = 1)} = \frac{\mathbb{P}(R = r, Y = 1)}{\mathbb{P}(Y = 1)}.$$

Hence the values of $\mathbb{P}(R = r|Y = 1)$ for different r can be obtained by dividing the values in the second row of the joint probability mass function (Exercise 11.2) by the marginal $\mathbb{P}(Y = 1)$. We can calculate $\mathbb{P}(R = 1|Y = 1)$, for example, as

$$\begin{aligned} \mathbb{P}(R = 1|Y = 1) &= \frac{\mathbb{P}(R = 1, Y = 1)}{\mathbb{P}(Y = 1)} \\ &= \frac{12/35}{4/7} \quad [\text{see Exercise 11.4}] \\ &= \frac{3}{5}. \end{aligned}$$

The resulting conditional p.m.f. is given by the following table:

r	0	1	2	3
$\mathbb{P}(R = r Y = 1)$	1/10	3/5	3/10	0

The above example illustrates the general idea of conditioning an event involving a random variable on another event.

Definition 12.4. Let X be a discrete random variable and A be an event with $\mathbb{P}(A) > 0$. The conditional probability

$$\mathbb{P}(X = x_k|A) = \frac{\mathbb{P}((X = x_k) \cap A)}{\mathbb{P}(A)}$$

defines the **probability mass function of X given A** . The corresponding expectation

$$\mathbb{E}(X|A) = \sum_k x_k \mathbb{P}(X = x_k|A)$$

is called the **conditional expectation**.

Remarks:

- One may consider $X|A$ as a random variable in its own right: “the random variable X given event A ”.
- The event A being conditioned on is often an event involving another random variable, e.g., the event “ $Y = 1$ ” in Example 12.3.
- If X and Y are independent random variables, then $\mathbb{P}(X = x_k|Y = y_l) = \mathbb{P}(X = x_k)$ and we *always* have $\mathbb{E}(X|Y = y_l) = \mathbb{E}(X)$. [If X and Y are not independent, then we *usually* have $\mathbb{E}(X|Y = y_l) \neq \mathbb{E}(X)$.]

Example 12.4: Conditional expectation of balls

Consider once again the experiment of Example 11.1. Assuming that we pick one yellow ball, determine the expectation of the number of red balls.

Solution:

The conditional expectation $\mathbb{E}(R|Y = 1)$ is easily calculated from the table of the conditional p.m.f. in Example 12.3:

$$\begin{aligned}\mathbb{E}(R|Y = 1) &= \sum_{r=0}^3 r\mathbb{P}(R = r|Y = 1) \\ &= 0 \times \frac{1}{10} + 1 \times \frac{3}{5} + 2 \times \frac{3}{10} + 3 \times 0 \\ &= \frac{6}{5}.\end{aligned}$$

Recalling from Exercise 11.4 that $\mathbb{E}(R) = 9/7$, we note that $\mathbb{E}(R|Y = 1) < \mathbb{E}(R)$ which makes sense on thinking about the physical set-up: if we pick one yellow ball, we have more than the expected number of yellow balls (again from Exercise 11.4, $\mathbb{E}(Y) = 6/7$) and hence, since the total number of balls picked is always three, the expected number of red balls is reduced. [R and Y are not independent, see Exercise 11.10.]

Exercise 12.5: Geometric and binomial

Let X be a random variable with $X \sim \text{Geom}(1/3)$. Let Y be another random variable with $Y \sim \text{Bin}(n, 1/4)$ where n is the value taken by the random variable X .

- (a) State the expectation of the random variable X .
- (b) Assuming X takes the value n , compute the conditional expectation of the random variable Y .
[based on 2018 exam question (part)]

12.3 Law of Total Probability for Expectations

Recall that Theorem 6.2 enables us to write (total) probabilities in terms of conditional probabilities. Analogously, the final theorem of our course enables us to write (total) expectations in terms of conditional expectations.

Theorem 12.5. *Suppose that $E_1, E_2, \dots, E_n$ partition $\mathcal{S}$ with $\mathbb{P}(E_i) > 0$ for $i = 1, 2, \dots, n$. Then for any discrete random variable X we have*

$$\mathbb{E}(X) = \sum_{i=1}^n \mathbb{E}(X|E_i)\mathbb{P}(E_i).$$

Proof:

Using the law of total probability (Theorem 6.2) for the event “ $X = x_k$ ” with

partition formed by $E_1, E_2, \dots, E_n$, we have

$$\mathbb{P}(X = x_k) = \sum_{i=1}^n \mathbb{P}(X = x_k | E_i) \mathbb{P}(E_i).$$

Then, using the definition of expectation (Definition 9.1), we get

$$\begin{aligned} \mathbb{E}(X) &= \sum_k x_k \mathbb{P}(X = x_k) \\ &= \sum_k x_k \sum_{i=1}^n \mathbb{P}(X = x_k | E_i) \mathbb{P}(E_i) \\ &= \sum_{i=1}^n \sum_k x_k \mathbb{P}(X = x_k | E_i) \mathbb{P}(E_i) \\ &= \sum_{i=1}^n \mathbb{P}(E_i) \sum_k x_k \mathbb{P}(X = x_k | E_i) \\ &= \sum_{i=1}^n \mathbb{P}(E_i) \mathbb{E}(X | E_i) \quad [\text{using Definition 12.4}]. \quad \square \end{aligned}$$

Example 12.6: Last throw of the dice

We toss a fair coin. If the coin comes up Heads we roll a fair six-sided die. If it comes up Tails we roll a fair four-sided die (with sides numbered 1, 2, 3, 4). Let X be the number we roll. Find $\mathbb{E}(X)$.

Solution:

Let H be the event “the coin shows Heads”. Clearly $\{H, H^c\}$ is a partition of $\mathcal{S}$ and, since the coin is fair, $\mathbb{P}(H) = \mathbb{P}(H^c) = 1/2$.

Now, the random variable $X|H$ is the number shown by the die given that the coin shows Heads, i.e., the number shown given we roll the six-sided die. Clearly its expectation is

$$\mathbb{E}(X|H) = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = \frac{7}{2}.$$

Similarly, $X|H^c$ is the number shown by the die given that the coin shows Tails, i.e., the number shown given we roll the four-sided die, and so

$$\mathbb{E}(X|H^c) = \frac{1 + 2 + 3 + 4}{4} = \frac{5}{2}.$$

Finally, Theorem 12.5 with the partition $\{H, H^c\}$ gives

$$\mathbb{E}(X) = \mathbb{E}(X|H)\mathbb{P}(H) + \mathbb{E}(X|H^c)\mathbb{P}(H^c) = \frac{7}{2} \times \frac{1}{2} + \frac{5}{2} \times \frac{1}{2} = 3.$$

*Exercise 12.7: *Geometric and binomial (continued)*

Continue Exercise 12.5 with the following.

- (c) Using the law of total probability, or otherwise, compute the expectation of Y .
- (d) Using the law of total probability, or otherwise, compute the expectation of the product XY and hence compute the covariance of the random variables X and Y .

[based on 2018 exam question (part)]

12.4 Further Exercises

Exercise 12.8: Variance and covariance

Suppose that X and Y are discrete random variables defined on the same sample space. Show that

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y).$$

*Exercise 12.9: *An odd question*

A fair six-sided die is rolled twice. Let X be the number showing on the first roll and A be the event “at least one odd number is rolled”. Find the probability mass function of $X|A$ and hence calculate $\mathbb{E}(X|A)$. [Answer: 10/3]

*Exercise 12.10: *A many-headed puzzle*

A fair six-sided die is rolled. Let N be the number showing on the die. Now a fair coin is tossed N times. Let X be the number of Heads seen. Find the conditional distribution of X given the event “ $N = n$ ”, and hence find the expectation of X . [Answer: 7/4]

*Exercise 12.11: *Coins and dice again*

You have two fair six-sided dice and two coins, one of which is fair while the other *always* shows Tails. First you roll the two dice, then you pick a coin at random and toss it. If it shows Heads, your score is the product of the numbers on the two dice; if it shows Tails, your score is the sum of the two numbers. Determine the expectation of your score. [Answer: 133/16]

*Exercise 12.12: **Random walk*

Consider a “random walker” on an infinite one-dimensional lattice (with sites labelled by integers). Let X_t denote the position of the walker at discrete time t where $t = 0, 1, 2, 3, \dots$

At time $t = 0$ the walker is at site 0, i.e., $X_0 = 0$. At each subsequent time step the walker moves at random to one of the two neighbouring sites, i.e., if $X_{t-1} = \ell$ then $X_t = \ell + 1$ with probability $1/2$ and $X_t = \ell - 1$ with probability $1/2$. In terms of conditional probabilities, this means $\mathbb{P}(X_t = \ell + 1 | X_{t-1} = \ell) = 1/2$, $\mathbb{P}(X_t = \ell - 1 | X_{t-1} = \ell) = 1/2$, and $\mathbb{P}(X_t = k | X_{t-1} = \ell) = 0$ if $k \neq \ell \pm 1$.

- (a) Use Theorem 12.5 to show that $\mathbb{E}(X_t) = 0$ for all t .
- (b) Show further that $\text{Var}(X_t) = t$ for all t .

CHAPTER 13

Epilogue (and Exams etc.)

13.1 Tips for Revision and Exams

There is no silver bullet for exam success and effective revision will depend on your own learning style. However, the earlier tips for reading the lecture notes (see Section 7.2) should apply particularly as you revise – *understanding is key*, especially as the results in later parts of the course continue to build on the earlier ones. Similarly, for tackling questions in the exam itself, the tips for doing examples and exercises (see Section 7.3) are relevant. In addition, there are some specific elements of common sense and exam strategy, including those below.

- *Read the question.* Take a few moments to be clear about what is being asked and, if necessary, work out how to convert the words to symbols. Make sure you transcribe any relevant material correctly; if you copy something down incorrectly you may even make a question impossible to solve!
- *Show that you understand what you are doing.* In a written question, you must clearly show all your working except where you are simply asked to “state” or “write down” a result. If you give a correct answer without justification, you may only be awarded partial marks. If you give an incorrect answer without justification, you are likely to get zero marks! However, if you give an incorrect answer but show signs of a correct method, you will earn partial marks.
- *Use your time wisely.* Don’t spend ages struggling with a question (or question part) worth only a few marks, when you know you can do something else. Once the easier marks are safely “in the bag”, you can always come back to the harder ones.

- *Check your answers.* Try to avoid losing marks for “silly mistakes”! Don’t just assume that you know how to do something and have done it correctly. Think about whether each answer makes sense (e.g., is a calculated probability plausible) and if there is another way you could check it.

13.2 Probability in Perspective

I hope that you have enjoyed this course and seen something of the beautiful structure of probability. The focus naturally turns to exams at this point but there should also be a rather broader perspective. The aim of this module is not merely to prepare you for a specific exam, nor even to prepare you for future exams in courses which build on this one, but to give you tools to analyse probabilistic scenarios and tackle unfamiliar problems wherever they may arise. This could ultimately be in research-level mathematics or in a more applied career (e.g., those involving finance, medical statistics, insurance, weather prediction, etc.) but just as importantly it could be in the decisions or discussions of everyday life – *probability matters!*

13.3 Bonus Further Exercise

Exercise 13.1: Seasonal exam question

- (a) Santa’s sleigh is pulled by Rudolph and eight other reindeer. Rudolph always has a red nose whereas each of the other eight reindeer has a red nose if and only if they have a cold. The events that the reindeer catch cold are mutually independent; at any time, each reindeer has a cold with probability $1/4$.
- Using the law of total probability, or otherwise, determine the probability that a randomly-chosen reindeer has a red nose. [5 marks]
 - Given that the chosen reindeer has a red nose, find the probability that it is Rudolph. [4 marks]
 - Let X denote the random variable counting the total number of red-nosed reindeer pulling Santa’s sleigh. Let $Y = X - 1$ be the number of red-nosed reindeer excluding Rudolph. State the distribution of Y and hence, or otherwise, calculate $\mathbb{E}(X)$. [5 marks]
- (b) Suppose that U and V are discrete random variables defined on the same sample space. Show that $\text{Var}(U - V) = \text{Var}(U) + \text{Var}(V) - 2\text{Cov}(U, V)$. [6 marks]

BIBLIOGRAPHY

- [ASV18] David F. Anderson, Timo Seppäläinen, and Benedek Valkó, *Introduction to Probability*, Cambridge University Press, Cambridge, 2018.
- [Bay63] Thomas Bayes, *An essay towards solving a problem in the doctrine of chances. by the late Rev. Mr. Bayes, F. R. S. communicated by Mr. Price, in a letter to John Canton, A. M. F. R. S.*, Phil. Trans. R. Soc. **53** (1763), 370–418.
- [Ros20] Sheldon Ross, *A First Course in Probability*, 10th ed., Pearson Education Limited, Harlow, 2020.
- [Rot64] Gian-Carlo Rota, *On the foundations of combinatorial theory I. Theory of Möbius Functions*, Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete **2** (1964), 340—368.
- [Sie88] Helmut Sies, *A new parameter for sex education*, Nature **332** (1988), 495.
- [Tij12] Henk Tijms, *Understanding Probability*, 3rd ed., Cambridge University Press, New York, 2012.
- [Vig15] Tyler Vigen, *Spurious correlations*, Hatchette Books, USA, 2015.

APPENDIX A

Errata

This appendix lists the points in these notes where there are non-trivial corrections/clarifications from earlier released versions.

- [Page 55] H corrected to read H_1 below (6.5).
- [Page 55] P corrected to read $\mathbb{P}$ in first line of argument for $\mathbb{P}(H_2|H_1 \cap F)$.